InnoPick Manager Operations Manual

Automated Case Picking System

Phil Cotnoir - DRL Systems

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1. InnoPick Manager Operations Manual

Automated Order Preparation Solutions

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1.1 Welcome

Welcome to the InnoPick Manager Technical Manual, designed to provide you with an in-depth understanding of the features, functionalities, and operational details of the InnoPick system.

This manual has been prepared to guide users of all levels through the setup, operation, and optimization of the InnoPick Manager software.

1.2 Documentation Sections

Learn how to access and navigate InnoPick Manager for the first time.

Standard procedures for starting up, monitoring, and shutting down the system.

Detailed reference for all screens and features in InnoPick Manager.

Alert resolution procedures and problem-solving guidance.

Quick lookup tables for alert codes and system status indicators.

1.3 User Roles

Access to certain features depends on your user role:

- **Admin**: Full access to all controls and settings
- **Manager**: Access to operational settings, but not site configuration
- **Clerk**: Access to operate the system for production
- **Guest**: View-only access to browse screens

See User Manager for more information.

1.4 System Overview

InnoPick Manager (IPM) is a web-based software interface for controlling and monitoring the InnoPick automated case picking system. The system manages:

- Automated case storage and retrieval across multiple levels
- Replenishment scheduling and tracking
- Case sequencing for order fulfillment
- Real-time inventory management
- Alert monitoring and resolution

1.5 Getting Help

- For operational questions, consult the Troubleshooting section
- For alert resolution, see the Alert Reference Page
- For system configuration, see Administration Pages

Navigation: Next: Getting Started →

2. Getting Started

2.1 Getting Started

Home > Getting Started

Overview

This section will guide you through your first interactions with InnoPick Manager, from logging in to understanding the basic interface layout.

What You'll Learn

- Introduction - Overview of InnoPick Manager and this manual
- Logging In - How to access InnoPick Manager at your terminal
- Interface Overview - Understanding the main navigation and screen layout

---

Prerequisites

Before you begin:

- You need a valid username and password for InnoPick Manager
- The system must be accessible via your facility's network

Quick Start

1. **First time logging in?** Start with Logging In
2. **Want to understand the interface?** See Interface Overview
3. **Ready to operate the system?** Move to Daily Operations

Access Levels

Your user account will have one of the following roles, which determines what you can view and control:

- **Guest**: Browse screens but cannot make changes
- **Clerk**: Operate the system for production
- **Manager**: Access operational settings
- **Admin**: Full access to all controls and configuration

If you're unsure of your access level, ask your supervisor or system administrator.

Navigation: ← Home | Introduction →

2.2 Introduction

Home > Getting Started > Introduction

About InnoPick Manager

InnoPick Manager is a web-based software interface designed to control and monitor the InnoPick automated case picking system. This powerful system manages the storage, retrieval, and sequencing of cases across multiple levels to efficiently fulfill orders.

What InnoPick Manager Does

InnoPick Manager provides operators with tools to:

- Monitor real-time system status and inventory levels
- Control operation modes and settings
- Track and manage replenishments
- View and resolve system alerts
- Monitor case sequences, inventory, and production progress
- Configure system parameters (for authorized users)

About This Manual

This manual has been prepared to guide users of all levels through the setup, operation, and optimization of the InnoPick Manager software.

Manual Organization

The manual is organized into logical sections that match how you'll actually use the system:

- **Getting Started**: Initial login and interface orientation
- **Daily Operations**: Routine startup, monitoring, and shutdown procedures
- **Main Screens**: Detailed reference for each screen and feature
- **Troubleshooting**: Alert resolution and problem-solving
- **Reference**: Quick lookup information for codes and indicators

Within each section, content is arranged by the menu structure in InnoPick Manager, making it easy to follow along as you learn the system.

Who Should Use This Manual

This manual is designed for:

- **Operators**: Daily startup, monitoring, and basic troubleshooting
- **Supervisors**: Advanced troubleshooting and system configuration
- **Maintenance Personnel**: Understanding system behavior during repairs
- **Administrators**: Complete system configuration and user management

Different sections will be relevant depending on your role and access level.

System Architecture Overview

InnoPick consists of:

- **Multiple Levels**: Independent operating levels, each with storage lanes and sequence conveyors
- **Storage Lanes**: Buffer lanes that hold cases until needed
- **Sequence Conveyors**: Conveyors that move cases from infeed to storage and from storage to outfeed
- **Infeed System**: Receives replenishment cases from upstream conveyor system and depalletizers
- **Outfeed System**: Releases sequenced cases to downstream conveyor system and palletizers
- **InnoPick Manager**: The software interface you'll use to control it all

---

Getting the Most from This Manual

For New Users:

- Read through Getting Started and Daily Operations sections first
- Practice navigating the screens in Guest mode if available
- Ask questions and take notes as you learn

For Experienced Users:

- Use this as a reference when encountering unfamiliar alerts or features
- Review the Troubleshooting section for efficient problem-solving
- Bookmark frequently-used pages in your browser

For Administrators:

- Pay special attention to the Administration section
- Review security and access control features
- Understand the implications of configuration changes

Next Steps

Now that you understand what InnoPick Manager does, proceed to Logging In to access the system.

Navigation: ← Getting Started | Logging In →

2.3 Logging In to InnoPick Manager

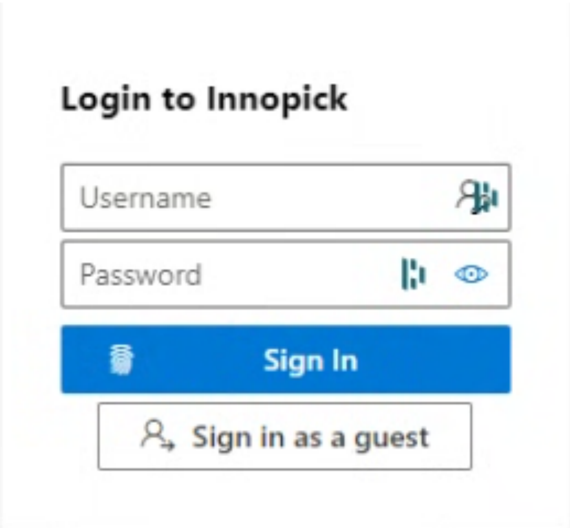
Home > Getting Started > Logging In

Accessing InnoPick Manager

InnoPick Manager is accessed through a standard web browser at your operator terminal.

Step-by-Step Login

1. **Open a web browser** (Chrome, Firefox, or Edge recommended)
2. **Navigate to the InnoPick Manager address**
    - Enter the web address provided by your system administrator
3. **The login page will appear**



1. **Enter your credentials**
    - Username: Your assigned username
    - Password: Your assigned password
2. **Click the** Sign In **button**

After Logging In

Once you successfully log in:

- You will remain logged in until the session expires or you manually log out
- The system will remember your session across page refreshes
- Your access level determines which screens and features you can use

First Login

If this is your first time logging in, you'll see the Home Page with the main interface. Take a moment to familiarize yourself with the layout described in Interface Overview.

Access Levels

Depending on your assigned user role, some menu options may not be accessible:

- **Guest**: Can browse screens but cannot make changes to configuration or equipment status
- **Clerk**: Can operate the system for production but cannot change settings
- **Manager**: Can access operational settings but not site configuration
- **Admin**: Full access to all controls and configuration options

If you need different access permissions, contact your supervisor or system administrator.

Session Management

Session Timeout

For security, your session will automatically expire after a period of inactivity. If this happens:

1. You'll be redirected to the login page
2. Simply log in again to resume your session
3. Any unsaved changes may be lost

Logging Out

To manually log out:

1. Click on your user avatar or the logout option (in the upper right corner of the page)
2. Confirm that you want to log out
3. You'll be returned to the login page

Best Practice: Always log out when leaving your workstation, especially if you have elevated access privileges.

Troubleshooting Login Issues

Cannot Access the Login Page

- Verify your network connection
- Confirm you're using the correct web address
- Contact IT support if the page won't load

Invalid Credentials

- Double-check your username and password (case-sensitive)

- Ensure Caps Lock is not enabled
- Contact your supervisor if you've forgotten your password

Account Locked

- After multiple failed login attempts, your account may be locked for security
- Contact your supervisor or system administrator to unlock it

---

Next Steps

Now that you're logged in, proceed to Interface Overview to learn about the main navigation and screen layout.

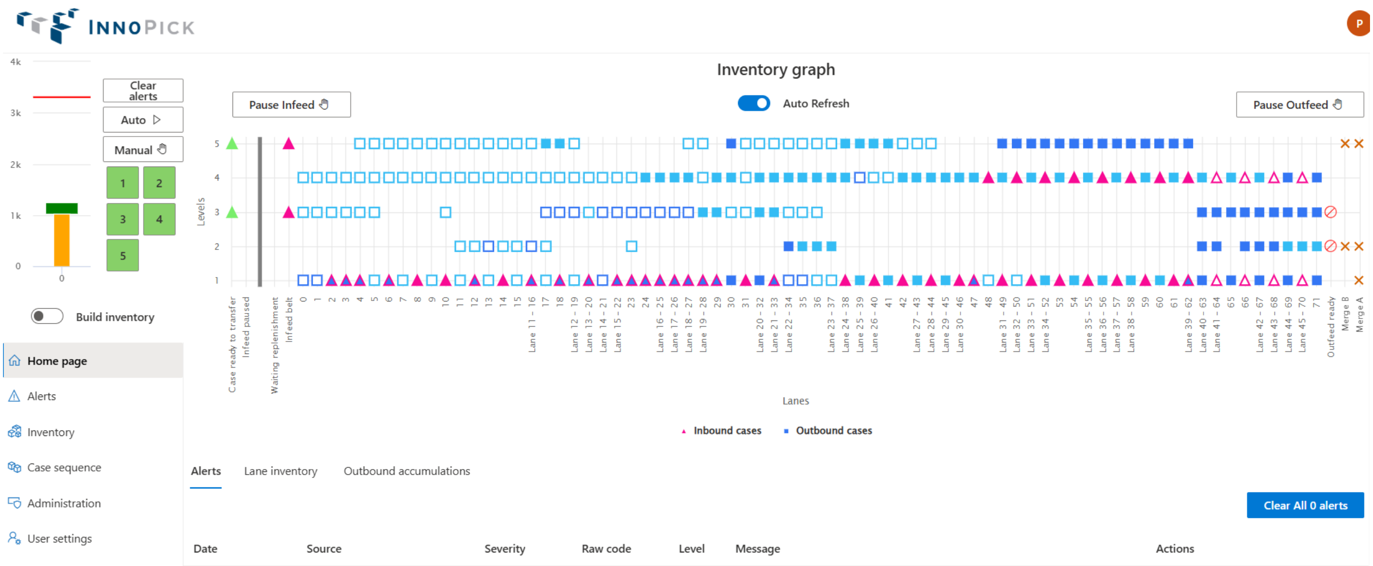
Navigation: ← Introduction | Interface Overview →

2.4 Interface Overview

Home > Getting Started > Interface Overview

Main Navigation

After logging in, you'll see the InnoPick Manager interface with a navigation menu on the left side of the screen. This menu is available from any page within the application.



Primary Navigation Buttons

Six main navigation buttons are always accessible on the left side:

1. **Home Page**
    - System overview and real-time status
    - Visual representation of cases and conveyors
    - Current alerts and production progress
2. **Alerts**
    - Current and historical alert listing
    - Alert filtering and search functionality
3. **Inventory**
    - Replenishments management
    - Product configuration
    - Lane inventory tracking
    - Accumulation monitoring
    - Product statistics
4. **Case Sequence**
    - Order of cases exiting InnoPick
    - Case tracking and filtering
5. **Administration**
    - System configuration (authorized users only)
    - Manual operations
    - User management
    - Recovery actions
    - Operations Settings
6. **User Settings**
    - Personal preferences
    - Account settings

Screen Layout

Each screen in InnoPick Manager follows a consistent layout:

Top Section

- User information and logout option
- Current system date/time
- Navigation breadcrumbs (on some pages)

Left Sidebar

- Main navigation buttons
- Quick access to all major sections

Main Content Area

- Primary information display
- Interactive controls and buttons
- Data tables and visualizations

Bottom Section

- Action buttons (Clear Alerts, Save, Cancel, etc.)
- Status indicators

Common Interface Elements

Status Indicators

Level Status Colors:

- **Green**: Automatic mode - system is running or ready
- **Yellow**: Manual mode - system is paused
- **Red**: Fault mode - system has an active alert
- **Dark Red**: E-Stop active - safety system engaged
- **Blue**: Communication Loss

Interactive Elements

Clickable Icons and Buttons:

- Colored level numbers can be clicked to change mode
- Three-dot menus (⋮) reveal additional actions
- Blue hyperlinks navigate to related information
- Hover tooltips provide additional details

Data Tables:

- Column headers can often be clicked to sort
- Search bars filter table contents
- Action columns contain buttons for specific operations

Navigation Tips

Getting Around Efficiently

1. **Use the Home button** to quickly return to the main overview
2. **Bookmark frequently-used pages** in your web browser
3. **Use browser Back/Forward** buttons to navigate your history
4. **Open links in new tabs** (right-click, "Open in new tab") to keep your place

Finding Information Quickly

- Use the search features in data tables
- Check the Alerts page for system status issues
- Use the Home Page for at-a-glance system overview
- Reference this manual's index for specific topics

Understanding Access Restrictions

Based on your user role, you may see some menu options grayed out or missing entirely:

- **Guest users**: Can view most screens but cannot modify settings or control equipment
- **Clerk users**: Can operate the system but cannot access configuration
- **Manager users**: Can access operational controls but not site setup
- **Admin users**: Have full access to all features

If you need access to a restricted feature, contact your supervisor or system administrator.

Next Steps

Now that you understand the interface layout, you're ready to:

- Learn Daily Operations procedures
- Explore the Main Screens in detail
- Review Troubleshooting procedures

For routine operation, proceed to Daily Operations: Startup.

Navigation: ← Logging In | Daily Operations →

3. Daily Operations

3.1 Daily Operations

Home > Daily Operations

Overview

This section covers the standard procedures for operating InnoPick Manager on a daily basis, including startup, monitoring, and shutdown procedures.

Startup

Complete checklist for beginning operations each day, including visual inspection, login, and enabling automatic mode.

Pre-Startup Visual Inspection:

Before powering up or enabling the system, conduct a thorough visual inspection of InnoPick:

Check for:

- Cases that are out of place or fallen
- Maintenance tools or equipment left on the machine
- Damaged cases or debris on conveyors
- Any unusual conditions or spills
- All safety guards and doors are properly closed
- E-Stop buttons are in the released (pulled out) position

Action: Remove any foreign objects and correct any unsafe conditions before proceeding.

Startup Procedure

ACCESS INNOPICK MANAGER

1. Open a web browser at your operator terminal
2. Navigate to the InnoPick Manager web address
3. Log in using your username and password

See Logging In for detailed instructions if needed.

2. REVIEW SYSTEM STATUS

Once logged in, you'll see the Home Page:

1. **Check for existing alerts**
    - Review the Alerts List on the Home Page
    - If alerts are present, see Resolving Alerts
    - Clear any alerts from previous shift
2. **Check level statuses**
    - All levels should show their current mode (Manual/Auto/Fault)
    - Verify the system is in the expected state
    - Each level must be in Manual mode (Yellow) in order for it to be put into Auto mode.
3. **Review disabled lanes**
    - Check for disabled lanes on all levels
    - Check with maintenance staff if any disabled lanes can be put re-enabled following maintenance work.

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3. PUT INNOPICK INTO AUTOMATIC MODE

Once you've verified the system is ready:

1. **Click the Auto button** to put all levels into Auto mode

1. **Verify the level(s) turn green** indicating Automatic mode

Important: Only enable Automatic mode if:

- Visual inspection is complete
- No unresolved alerts exist
- All safety devices are functional
- Nobody is inside the InnoPick safety fence
- You're ready to begin production

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4. MONITOR INITIAL OPERATION

After enabling Automatic mode:

1. **Watch the first few cycles** to ensure smooth operation
2. **Listen for unusual sounds** that might indicate problems
3. **Check the Inventory Graph** to verify cases are moving as expected on all levels
4. **Monitor for new alerts**

See Monitoring Production for ongoing monitoring procedures.

Startup After Extended Downtime

If the system has been shut down for more than 24 hours:

1. Perform a more thorough visual inspection
2. Verify all utilities are functioning (power, air pressure, network)
3. Check with maintenance that all scheduled work is complete
4. Manually jog any lanes or sequence conveyor chains that have had maintenance work done on them
5. Allow a few moments for systems to initialize
6. Test one level in Automatic mode before enabling all levels

---

Troubleshooting Startup Issues

SYSTEM WON'T ENTER AUTOMATIC MODE

Possible Causes:

- Active alerts must be cleared first
- If no alerts appear at first, go to Alerts Page and remove all filters to see *all* alerts.
- Because InnoPick automatically filters out alerts it thinks are not critical, sometimes an important alert is not immediately visible.
- E-Stop or door is open
- Upstream or downstream systems not ready

Solution: Check the Alerts Page for details

Shutdown

Proper procedures for ending the shift and securing the system.

1. Verify Production is Complete

Check that there are no:

- **In Progress Cases**: Cases currently being processed on sequence conveyors
- **Incomplete Cases**: Cases scheduled but not yet completed
- **Pending Cases**: Cases in the case sequence waiting to be processed
- **Active Replenishments**: Replenishments currently inducting or in progress

If production is not complete:

- Coordinate with supervision about completing or postponing remaining orders
- Allow system to finish current work before proceeding
- Document any incomplete work for the next shift

---

2. Put InnoPick into Manual Mode

- On the Home Page, click the **Manual** button

- Verify all level indicators turn **Yellow** (Manual mode)

Why Manual Mode?

- Prevents the system from automatically starting new moves
- Allows safe inspection and shutdown
- Ensures system won't start unexpectedly

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3. Conduct Visual Inspection

Perform a walk-around inspection of InnoPick:

Check for:

- Cases out of place or fallen
- Cases left on conveyors
- Any damage to equipment or cases
- Spills or debris that need cleanup

Document any issues:

- Report safety concerns or mechanical issues to supervision

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4. Log Out

1. Click on your username or logout button
2. Confirm logout
3. Close the browser

Security Note: Always log out when leaving your workstation to prevent unauthorized access.

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5. Suggested: Open an InnoPick Door

- It is recommended to request access to an InnoPick door, as this removes all latent electrical and pneumatic energy from the system
- This puts the system into a state equivalent to E-Stop
- Reduces wear on certain components

Essential Skills for Operators

To effectively operate InnoPick, you should be familiar with:

1. **System Status Monitoring**
    - Understanding level status colors (Green/Yellow/Red)
    - Reading the inventory graph
    - Monitoring production progress
2. **Alert Management**
    - Recognizing common alerts
    - Basic alert resolution
    - When to escalate to maintenance or support
3. **Replenishments Awareness**
    - Understanding replenishment status

Safety Reminders

- Always perform visual inspections before starting equipment
- Never bypass safety interlocks or E-stops
- Ensure all personnel are clear before resuming automatic operation
- Report any unsafe conditions to your supervisor

Navigation: ← Interface Overview | Monitoring Production →

3.2 Monitoring Production

Home > Daily Operations > Monitoring

Overview

During production, operators should actively monitor InnoPick Manager to ensure smooth operation, catch problems early, and maintain production efficiency.

Primary Monitoring Dashboard

The Home Page provides all essential information for monitoring production at a glance.

Key Elements to Monitor

1. **Level Status Indicators**
    - **Green**: Level is in Automatic mode and operating normally
    - **Yellow**: Level is in Manual mode (paused)
    - **Red**: Level has an active fault
    - **Dark Red**: E-Stop condition is active
    - **Blue**: Communication Loss
2. **Inventory Graph**
    - Visual representation of all cases on sequence conveyors
    - Squares = Outbound cases (for customer pallets)
    - Triangles = Inbound cases (replenishments being stored)
    - Filled shapes = Physical cases present
    - Hollow shapes = Scheduled case positions
3. **Alerts List**
    - Shows current system alerts
    - Alerts appear when issues require attention
    - Go to the Alerts page for more details.
4. **Production Graph**
    - Shows progress of current production run
    - Green bar = In Progress cases
    - Orange bar = Completed cases
    - Red line = Total cases in current orders
5. **Lane Inventory**
    - Select a level to view storage lane status
    - Hover over lanes for detailed inventory info
    - Monitor for abnormal low inventory levels

What to Watch During Production

Continuous Monitoring Tasks

Every 15-30 Minutes:

- Glance at level status indicators (all should be green)
- Check for new alerts
- Verify production progress is advancing

Every 1-2 Hours:

- Check replenishment status
- Monitor case sequence progress

Throughout the Shift:

- Listen for unusual mechanical sounds
- Watch for stopped or hesitating conveyors
- Be aware of downstream system status

Recognizing Normal Operation

When InnoPick is operating correctly:

- All levels show **green** status
- Cases move smoothly through the system
- **Inventory Graph** shows cases advancing position by position
- **Production Graph** shows steady progress
- **Alerts List** is empty or shows only minor warnings
- Replenishments are being inducted as needed
- Output cases are feeding downstream without backing up

Recognizing Problems Early

Warning Signs to Watch For

Immediate Attention Required:

- Any level turns **Red** (fault condition)
- E-Stop condition (all levels **Dark Red**)
- Critical alerts appear
- Cases stopped on conveyors
- Unusual sounds or movements

Monitor Closely:

- Repeated alerts of the same type
- Accumulation building up at outfeed
- Replenishments not arriving as scheduled

Responding to Alerts

When an alert occurs:

1. **Note the alert details**
    - Which level is affected
    - What type of alert (code and message)
    - Time of occurrence
2. **Assess the situation**
    - Is it a single occurrence or repeating?
    - Is production stopped or continuing?
3. **Take appropriate action**
    - For common alerts: Follow Alert Reference
    - For safety issues: Use E-Stop if necessary.
    - For unknown issues: Contact supervisor, maintenance, or remote support.

See Alert Guidelines for detailed alert resolution procedures.

Monitoring Replenishments

Replenishments ensure InnoPick has adequate inventory to fulfill orders.

What to Monitor

Access the Replenishments Page to check:

- **Pending**: Replenishments requested but not yet started
- **In Progress**: Replenishments being prepared for induction
- **Inducting**: Replenishments currently entering InnoPick
- **Completed**: Replenishments fully inducted

When to Take Action

Normal: Replenishments progress from Pending → In Progress → Inducting → Completed

Problem Indicators:

- Inducted Quantity doesn't match expected quantity
- Replenishments backing up at infeed

Action: Coordinate with upstream depalletizing operations or see Troubleshooting

Understanding System Settings That Affect Operations

Be aware of these settings and their effects:

Build Inventory Mode

- When ON: Prioritizes storing replenishment cases over outputting cases
- When OFF: Balances input and output for maximum throughput

Pause Infeed / Pause Outfeed

- Controls whether cases enter or exit InnoPick
- Check: Home Page

Shift Handoff Checklist

When ending your shift, review with the next operator:

- Current system status and any active alerts
- Any recurring issues or unusual behavior observed
- Replenishments in progress or pending
- Any lanes disabled or products having problems
- Upcoming scheduled maintenance or system changes
- Any instructions from supervision

Navigation: ← Daily Operations | Main Screens →

4. Main Screens

4.1 Main Screens

Home > Main Screens

Overview

This section provides detailed documentation for each screen and feature in InnoPick Manager. Use this as a reference guide to understand what information is available and how to use each feature.

Screen Reference Guide

Home Page

The main dashboard showing system status at a glance, including:

- Inventory graph with real-time case positions
- Level status indicators and mode controls
- Current alerts listing
- Lane inventory display
- Production progress graph

Alerts Page

Comprehensive alert monitoring and history, including:

- Current and past alerts
- Alert filtering and search
- Alert details and severity levels
- Common alert descriptions

Inventory Section

Complete inventory management tools:

- **Replenishments Page**: Track & manage incoming replenishments
- **Products Page**: Configure product settings and parameters
- **Lane Inventory Page**: View and edit storage lane contents
- **Accumulations Page**: Monitor outfeed accumulation space
- **Product Statistics Page**: Analyze product usage and performance

Case Sequence Page

View the sequence of outgoing cases:

- Current case sequence
- Case status
- Case information (HUID, level, case #, etc)
- Filtering and search tools
- Case cancellation options

Administration Pages

System configuration and advanced features (authorized users only):

- **Manual Motion Control**: Manual jog controls
- **Setup InnoPick**: System configuration parameters
- **Algorithm Parameters**: Algorithm tuning and optimization
- **User Manager**: Create and manage user accounts
- **Recovery Actions**: Advanced troubleshooting tools
- **Operations**: Level and Merge statuses / controls
- **User Settings**: Personal preferences

How to Use This Section

For Daily Operation

Focus on these screens:

- Home Page - Your primary monitoring dashboard
- Alerts Page - When issues occur
- Replenishments Page - Monitor replenishment statuses

For Troubleshooting

Refer to these screens:

- Alerts Page - Identify problems
- Lane Inventory - Adjust lane counts
- Case Sequence Page - Track specific output cases
- Manual Motion Control - Jog lanes and conveyors
- Also, see the Troubleshooting section of this manual

For Administration

Authorized users should understand:

- Setup InnoPick - System configuration
- Recovery Actions - Advanced tools

Screen Navigation Tips

Getting to Screens:

- Click the navigation buttons on the left sidebar
- Use browser bookmarks for frequently-accessed pages
- Some screens have sub-sections accessed via tabs

Working with Data Tables:

- Click column headers to sort
- Use search boxes to filter results
- Look for action menus (three dots ⋮) for additional options

Understanding Visual Elements:

- Green/Yellow/Red color coding indicates status
- Hover tooltips provide additional information
- Blue hyperlinks navigate to related pages

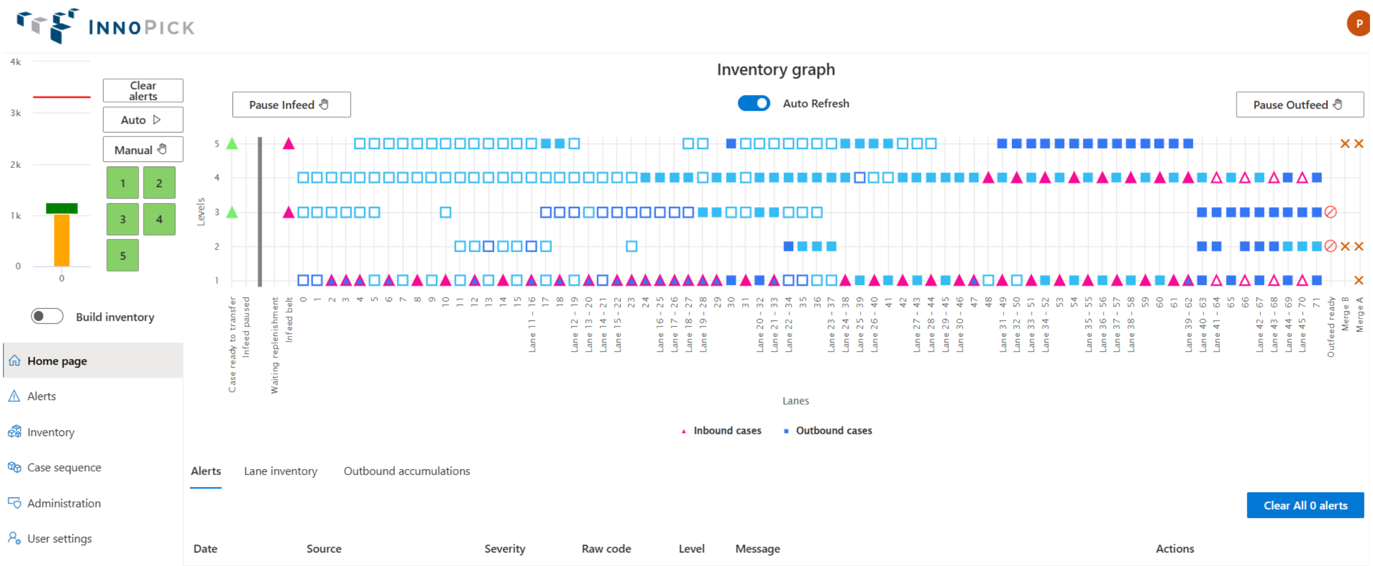
Navigation: ← Monitoring Production | Home Page →

4.2 Home Page

Home > **Main Screens** > **Home Page**

Overview

The Home Page is your primary dashboard for monitoring InnoPick operations. It provides a real-time snapshot of system status, case locations, alerts, and production progress.



Accessing the Home Page

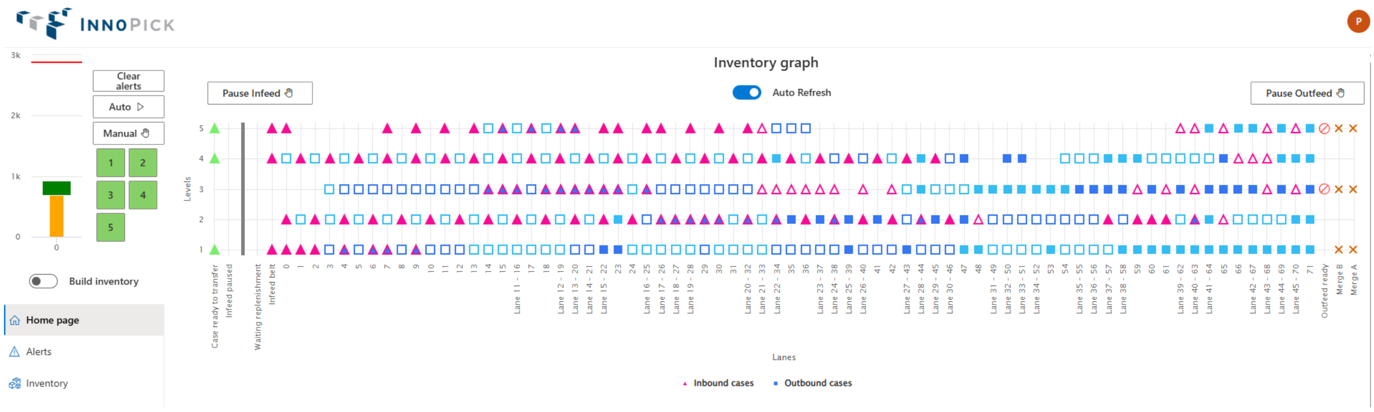
- Click the **Home Page** button in the left navigation menu
- The Home Page appears automatically after logging in
- Available from any screen in InnoPick Manager

Home Page Components

The Home Page displays the following key information:

1. Inventory Graph - Visual representation of cases on sequence conveyors
2. Level Statuses & Controls - Mode indicators and control buttons
3. Alerts List - Current system alerts
4. Lane Inventory - Storage lane contents by level
5. Production Graph - Progress of current production run

Inventory Graph



The main visual grid in the center-right of the Home Page provides a quick summary of case positions on InnoPick sequence conveyors.

Understanding the Symbols

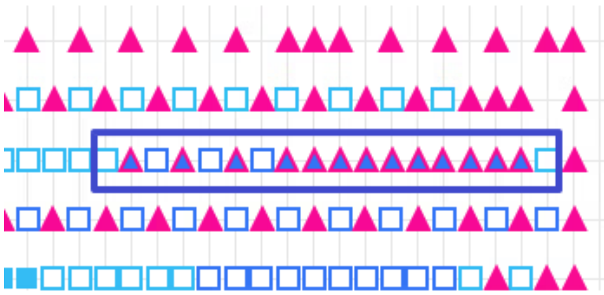
SHAPES:

- **Squares** = Outbound cases (scheduled for customer pallets)
- **Triangles** = Inbound cases (replenishments being stored in lanes)

FILL STATUS:

- **Filled shapes** = Physical case is present at that position
- **Hollow shapes** = Position is empty or case is expected
- If a triangle, it means the case has been stored
- If a square, it means the case is scheduled to be dispensed on this position

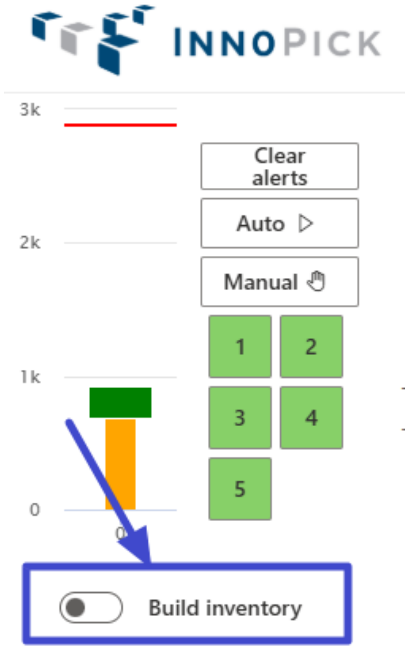
"GENDER BENDER" CASES:



- Some Inbound case positions can be re-used as Outbound case positions when the system determines that the inbound case will be stored in a lane upstream of the outbound case's lane.
- This is shown via a special border around the inbound case on the Inventory graph (see screenshot above - cases highlighted in blue rectangle)
- When this case is stored, the triangle will be replaced with a square.

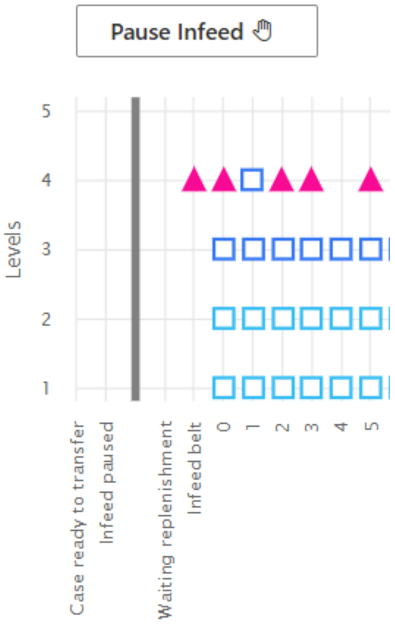
Operational Controls on the Inventory Graph

BUILD INVENTORY MODE

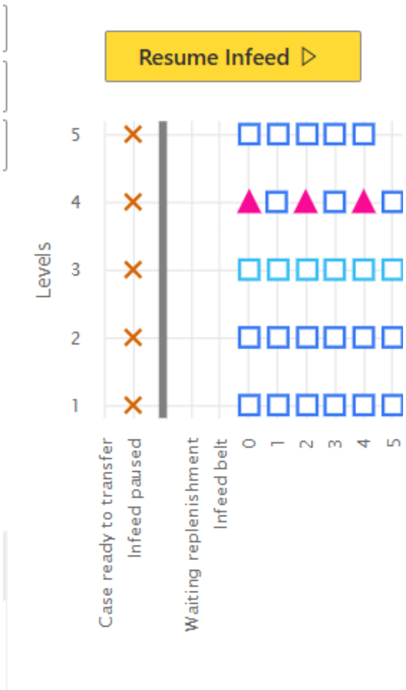


- Build Inventory mode prioritizes letting replenishment cases come into InnoPick over scheduling output cases.
- More information on Build Inventory Mode

PAUSE / RESUME INFEED

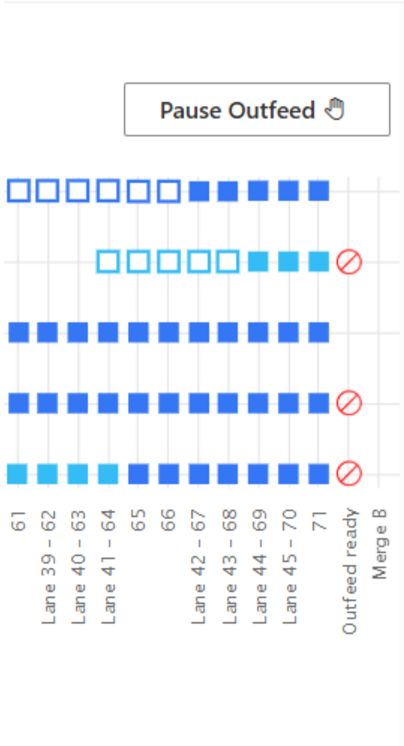


- Click the **Pause Infeed** button to pause incoming cases
- Cases will stop and wait on the first infeed position of InnoPick

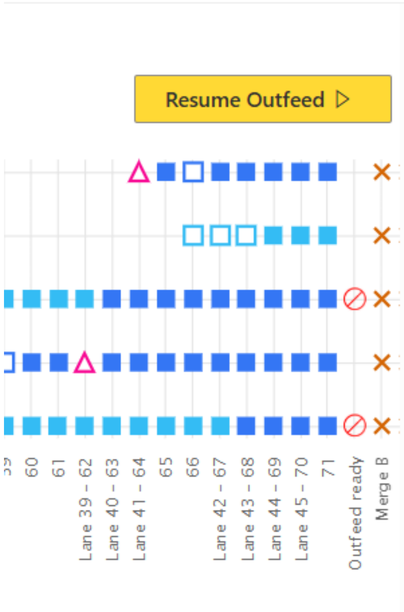


- Click the **Resume Infeed** button to resume incoming cases.

PAUSE / RESUME OUTFEED



- Click the **Pause Outfeed** button to pause outgoing cases
- Cases will stop and wait on the last position of InnoPick



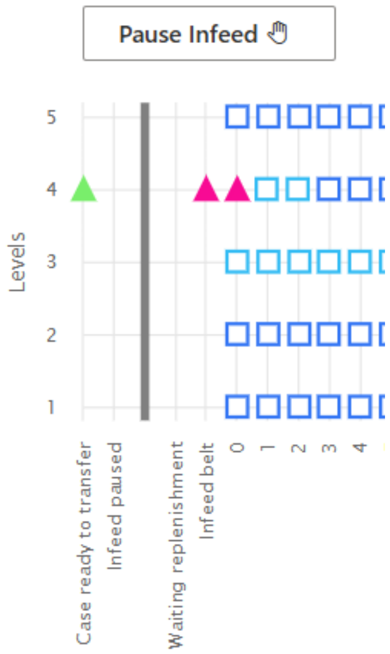
- Click the **Resume Outfeed** button to resume outgoing cases.

Interpreting the Display

The inventory graph shows:

- Cases currently on all sequence conveyor positions, with the storage lanes displayed below the graph, indicating their position vertically.
- Various statuses of the transition points at both Infeed and Outfeed:

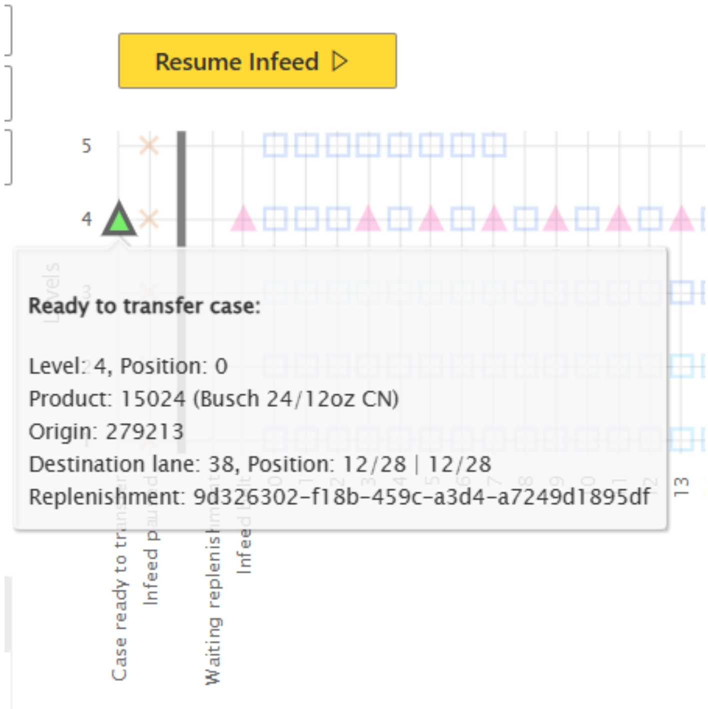
STATUSES AT THE INFEED



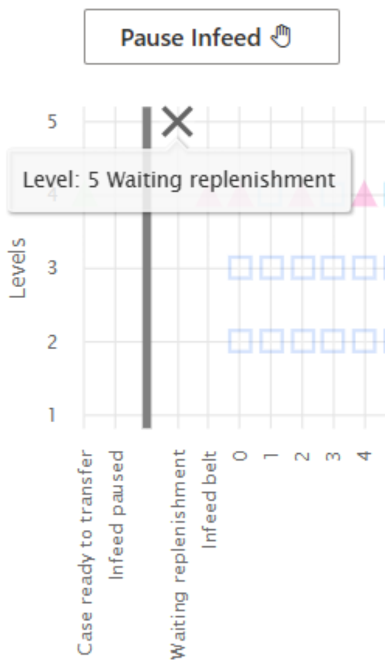
CASE READY TO TRANSFER

- When a case is to be transferred from the upstream conveyor system, it appears as a coloured triangle on that level.

• Hover over the triangle to get more information:



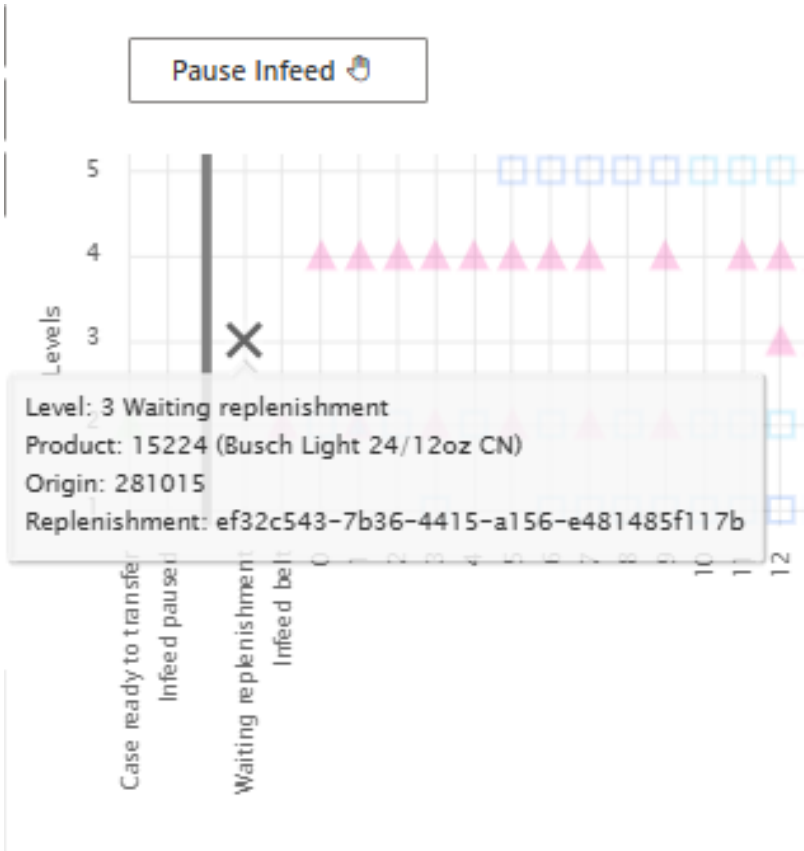
WAITING REPLENISHMENT



• When a level InnoPick is not able to schedule output cases because it is waiting for a replenishment, an **x** appears at that level.

• This allows an operator to know immediately why a given level is not scheduling output cases.

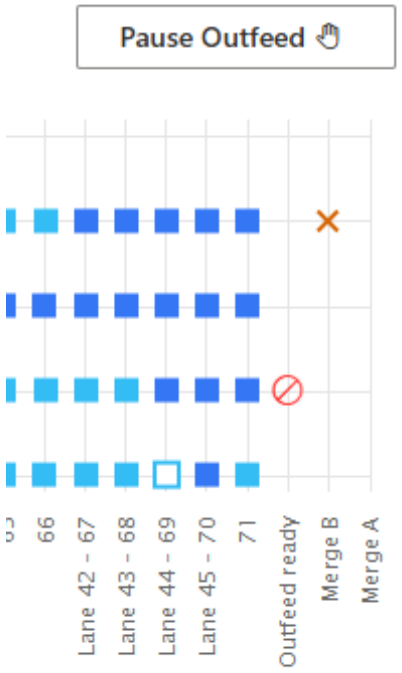
• Hover over the **x** to see which replenishment the level is waiting for.



INFEED PAUSED

- See above

STATUSES AT THE OUTFEED

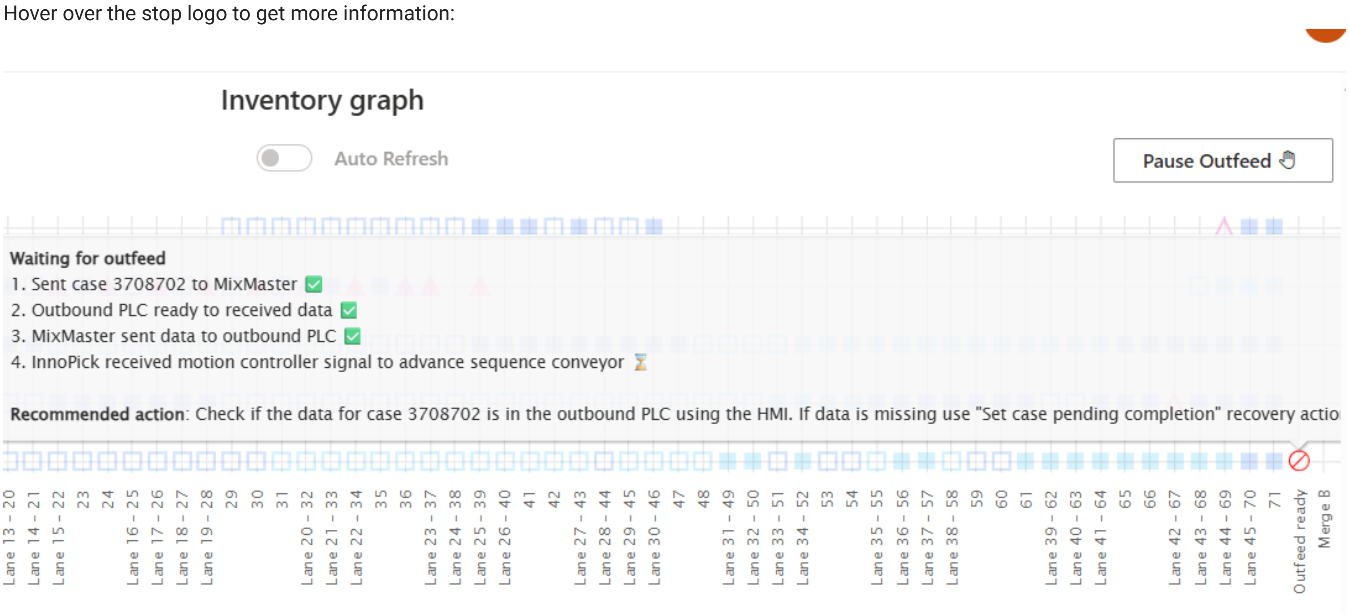


MERGE A / B

- If InnoPick calculates that a certain destination (Merge A or Merge B) cannot receive any more cases, it will stop scheduling output cases for that destination and indicate this with an **X** on the given level and over the given destination.

OUTFEED READY

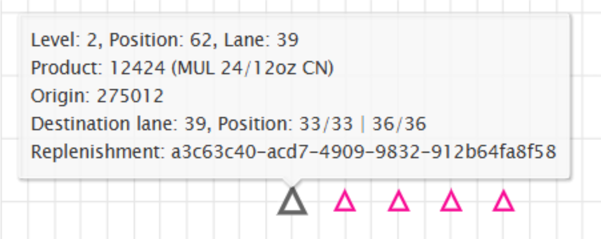
- If there is a case on the last position of InnoPick and the downstream conveyor system is not ready to receive it, a stop logo appears.
- Hover over the stop logo to get more information:



Getting More Information

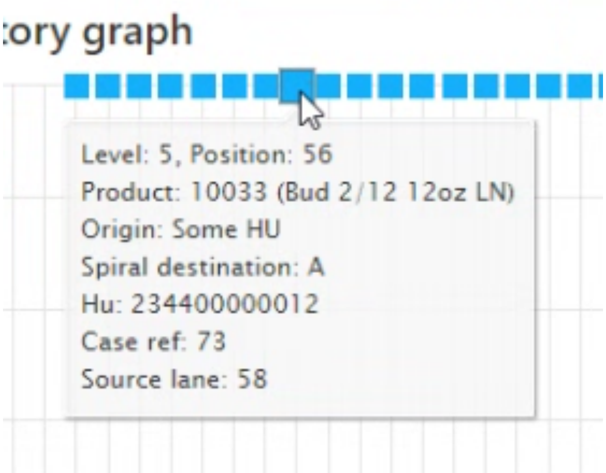
HOVER FOR DETAILS:

Hover your cursor over any case icon to see additional information:



Information includes:

- Level #, current position #, and lane # (if applicable).
- Product SKU and description
- Origin PalletID / HUID
- If Inbound case:
    - Destination lane
    - Position: the position of the case in the storage lane; followed by the position of the case in terms of the total replenishment quantity. In the example in the screenshot above, it is case 33 out of 33 in lane 39; and it was the last case of a 36-case replenishment.
    - Replenishment ID (unique GUID); this can be used to identify the replen in InnoPick and MixMaster.
- If Outbound Case:



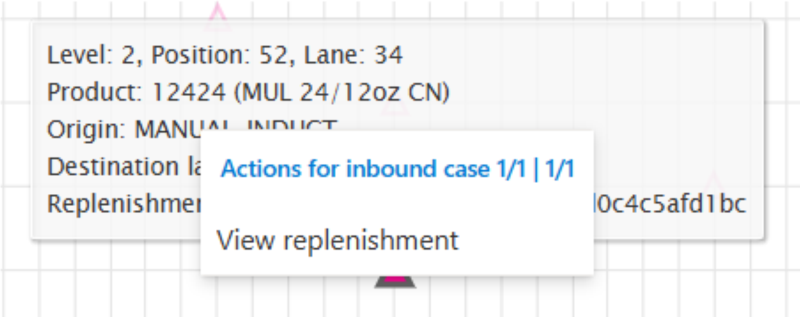
- Merge / Spiral destination
- Client Order PalletID
- Case reference number
- Source lane (where the case was dispensed from)

QUICK LINKS

Users can click on:

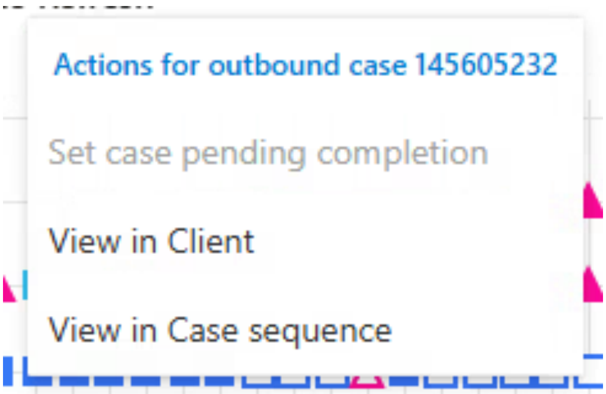
1. REPLENISHMENT LINKS:

For inbound cases (triangles), you can click on the case and see a clickable "View replenishment" link:



Clicking this link opens the Replenishments Page and displays details for that specific replenishment.

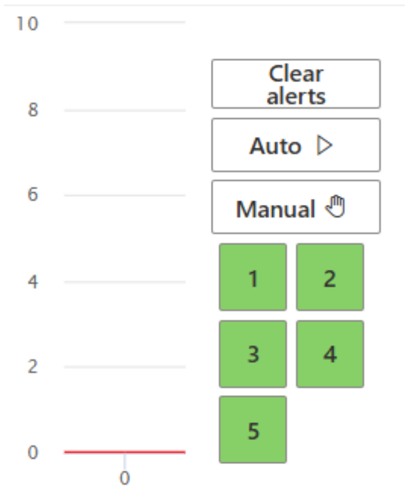
2. OUTBOUND CASE LINKS



For outbound cases (squares), you can click on the case and see between 2 and 3 clickable links:

- **Set case pending completion**: This is only clickable if the case if on the last position of InnoPick. If clicked, a second confirmation will be required in order to execute the recovery action.
- For detailed information on what this recovery action does, see the Commonly Used Recovery Actions.
- **View in Client**: This will redirect the user to the Case Sequence page in MixMaster where the case can be found.
- Tip: Right-click and "Open in New Tab" to keep this page open. You may need to login again in the new tab.
- **View in Case Sequence**: This will redirect the user to the Case Sequece page in InnoPick Manager, with a filter applied so only this case appears.
- Tip: Right-click and "Open in New Tab" to keep this page open. You may need to login again in the new tab.

Level Statuses & Auto / Manual Controls



The numbered squares in the top left corner indicate the operational status of each InnoPick level.

Status Color Meanings

- **Green**: Automatic mode - level is running or ready for production
- **Yellow**: Manual mode - level is paused
- **Red**: Fault mode - level has an active alert
- **Dark Red**: E-Stop active - safety system engaged
- **Blue**: Communication Loss

Controlling Level Modes

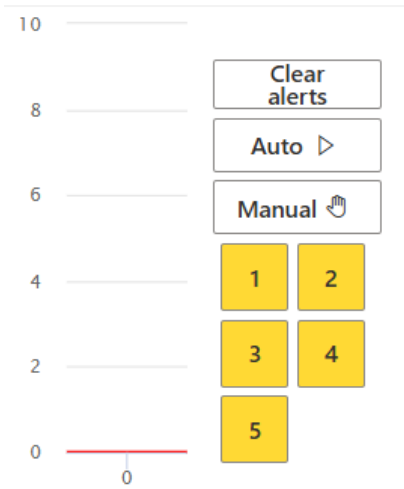
To change a level's mode:

1. Click on the colored level number
2. A pop-up may appear to confirm the mode change
3. The color will change to reflect the new mode

Mode Control Buttons:

- **Auto Button**: Puts InnoPick (all levels) into Automatic (production) mode
- Use when ready to begin or resume production
- System will autonomously manage case movements
- **Manual Button**: Puts InnoPick (all levels) into Manual mode
- System pauses automatic operation
- Used for troubleshooting and maintenance
- Required in order to jog lanes (manual control) or change certain settings, such as the inventory of a lane.

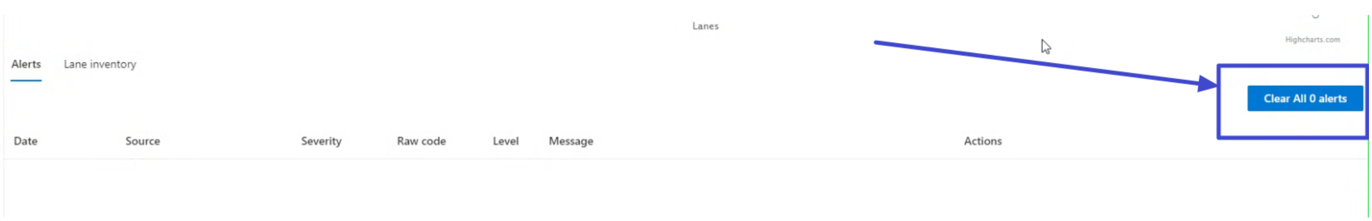
When Levels Are in Manual Mode



When InnoPick has an alarm, the affected level automatically enters Manual (Yellow) or Fault (Red) mode. This pauses production and allows the operator to:

- Assess the problem
- Resolve any issues
- Clear the alarm
- Resume production

Alerts List

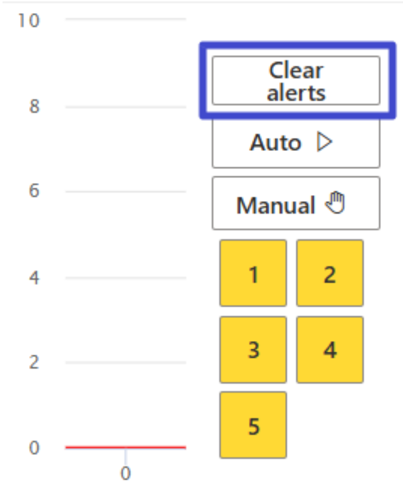


The Alerts section shows current system alerts that require attention.

Clear All Alerts Button

The **Clear All Alerts** button (bottom right of the inventory graph) attempts to clear all current alerts in the system.

There is also a **Clear Alerts** button always visible just above the Auto and Manual buttons:



Important Notes:

- If the underlying issue has not been resolved, the alert **cannot** be cleared
- Common reasons alerts won't clear:
- Photocell sensor still blocked
- E-Stop button still pressed
- Door still open
- Physical case still in wrong position

When Alerts Won't Clear:

1. Read the alert message carefully
2. Resolve the underlying issue
3. Try clearing alerts again
4. See Alert Guidelines for specific resolution steps
5. Try using the **Clear alarm on level** recovery action.
6. Open the Alerts Page and remove all filters to see if some other alert could be affecting the system.
    - Sometimes the system filters out an alert that is still active and cannot be cleared until the underlying issue has been fixed.

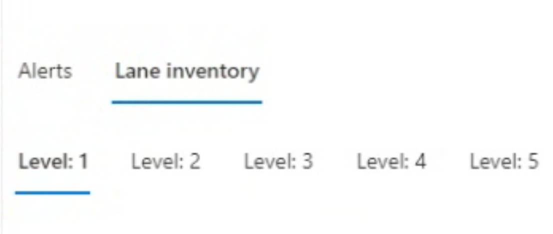
Viewing Complete Alert History

For a full list of all current and past alerts with search and filtering capabilities, see the Alerts Page.

Lane Inventory

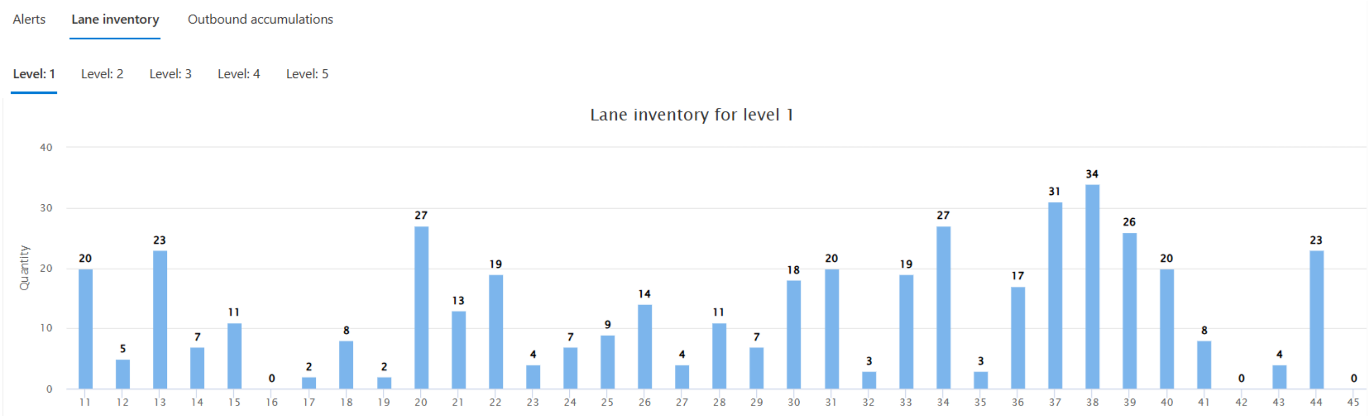
The Lane Inventory display shows the contents of storage lanes for each level.

Selecting a Level



Click on a level number to view its lane inventory.

Lane Display

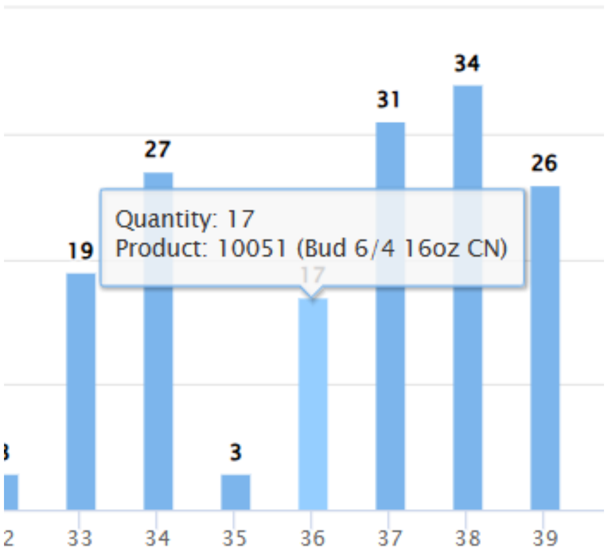


Each storage lane is shown as a horizontal bar:

- Bar length represents relative fullness
- The display adjusts automatically to show all lanes for that level

---

Getting Lane Details



Hover your cursor over any lane to see:

- Quantity of cases
- Product stored (SKU and description)

Complete Lane Management

For more detailed lane inventory information and editing capabilities, see the Lane Inventory Page.

Production Graph

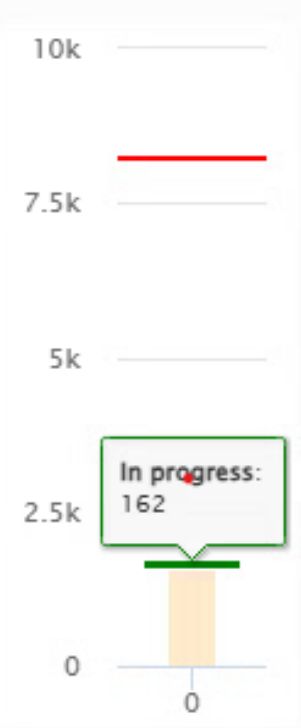
The Production Graph on the far left of the Home Page shows the current state of the production run.

Understanding the Graph

Red Line at Top:

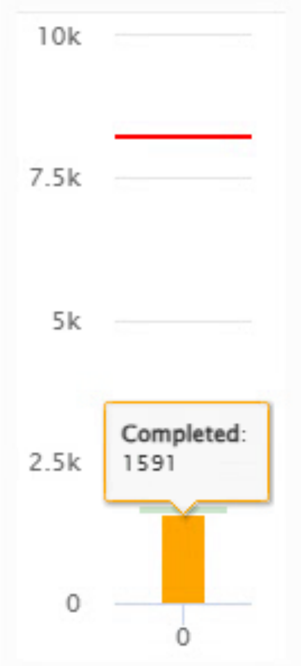
- Represents the total number of cases in currently launched orders
- This is the production target

Green Bar:



- Represents **In Progress** cases
- These are cases currently on the sequence conveyors (outbound)
- Corresponds to the squares (filled and hollow) on the inventory graph
- Hover to see exact count

Orange Bar:



- Represents **Completed** cases
- These cases have been output from InnoPick
- This counter resets periodically as routes/truckloads are completed
- Hover to see exact count

Using the Home Page Effectively

For Daily Monitoring

The Home Page should be your primary screen during production:

1. **Glance regularly at level statuses** - All should be green during production
2. **Watch for new alerts** - Address immediately when they appear
3. **Monitor production progress** - Verify steady advancement

---

Quick Troubleshooting

When an alert occurs:

1. **Check which level** has the issue (red or yellow indicator)
2. **Read the alert message** in the Alerts List
3. **Look at the inventory graph** - does it match physical reality?
4. **Address the issue** per the Troubleshooting Guidelines
5. **Clear the alert** once resolved

Navigation from Home Page

To investigate issues further:

- Click level numbers → Changes mode or shows level details
- Click "See replenishment" links → View replenishment details
- Click alerts → Navigate to Alerts page for more information

To access other features:

- Use left sidebar navigation buttons
- See Main Screens Index for all available screens

Related Topics

- Monitoring Production - How to use the Home Page during operation
- Alerts Page - Detailed alert information and history
- Alert Guidelines - How to resolve alerts
- Lane Inventory Page - Detailed lane management

Navigation: ← Main Screens | Alerts Page →

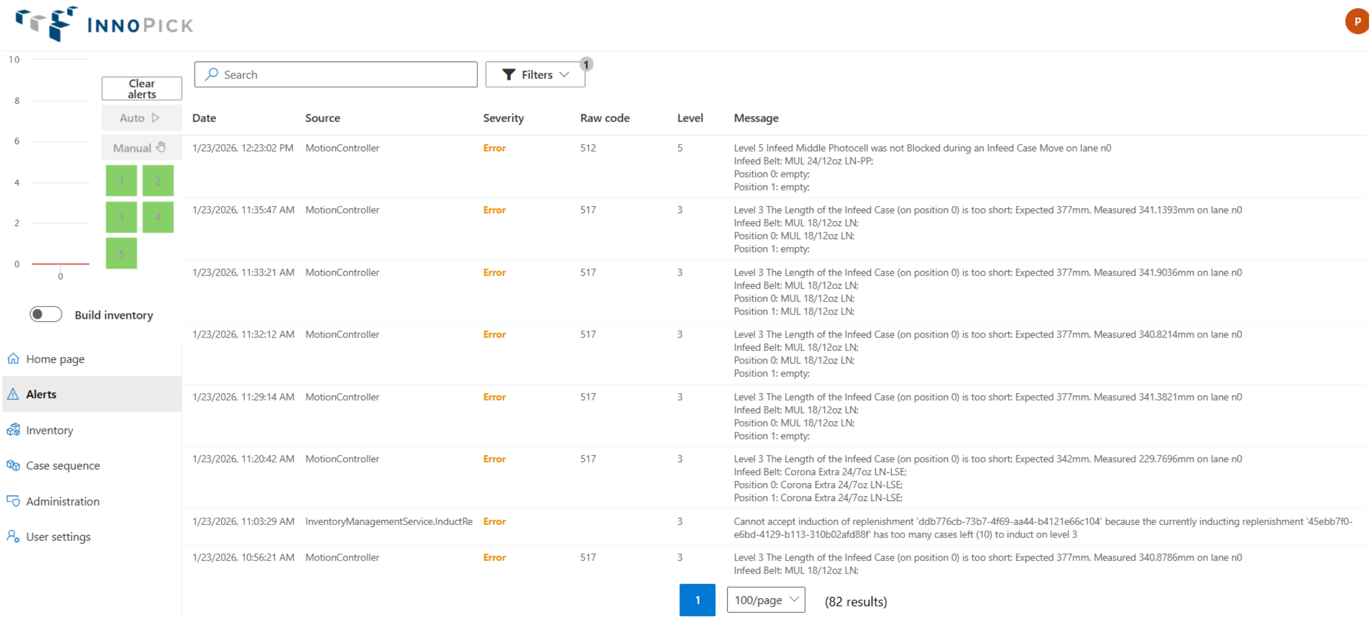
4.3 Alerts Page

Home > **Main Screens** > **Alerts Page**

Overview

The Alerts page provides a comprehensive view of current and historical alerts, with powerful search and filtering capabilities to help you identify and resolve issues quickly.

Note: If you're looking for detailed info on specific Alert Codes, see the Alert Reference Page



Accessing the Alerts Page

- Click the **Alerts** button in the left navigation menu
- Available from any screen in InnoPick Manager

Default Display

When the Alerts page first loads:

- **Only uncleared alerts** are displayed by default
- Alerts are shown in chronological order (most recent first)

Alerts Page Columns

The Alerts page displays detailed information about each alert:

Date

- The date and time when the alert first occurred
- Useful for tracking when issues began

Source

- Indicates which portion of the software generated the alert
- Examples: Motion Controller, PLC, Core Logic, Buffer Management
- Helps identify which system component detected the issue and what aspect of InnoPick is the problem

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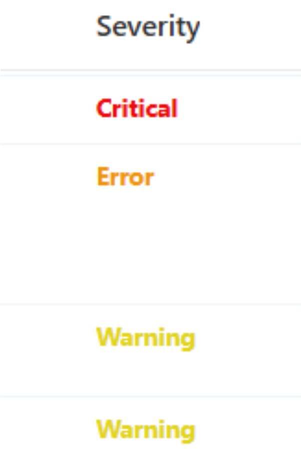
Severity

Alerts are categorized by severity level:



Warning: Minor alert that may not require immediate action

- System can often continue operating
- Should be monitored for patterns
- No audible sound

Error: A standard Alert which stops the affected level and puts it into Manual (Faulted/Red) mode.

- Other levels continue operating
- The panel Buzzer will sound to alert the operator

Critical: Major system alert requiring attention

- Production on all levels may be stopped
- Requires operator intervention
- The panel Buzzer will sound to alert the operator

Raw Code

- The numeric code associated with the alert
- Used to identify specific alert types
- Reference these codes in the Alert Reference Page
- Examples: 119, 121, 515, 555, 709

Level

- The InnoPick level where the alert occurred
- May be blank for system-wide alerts

Message

- Descriptive text explaining the nature of the alert
- Written to help operators understand what happened
- May include specific details like lane numbers or photocell locations, as well as contextual information (nearby cases, replenishment details, etc).
- Examples:

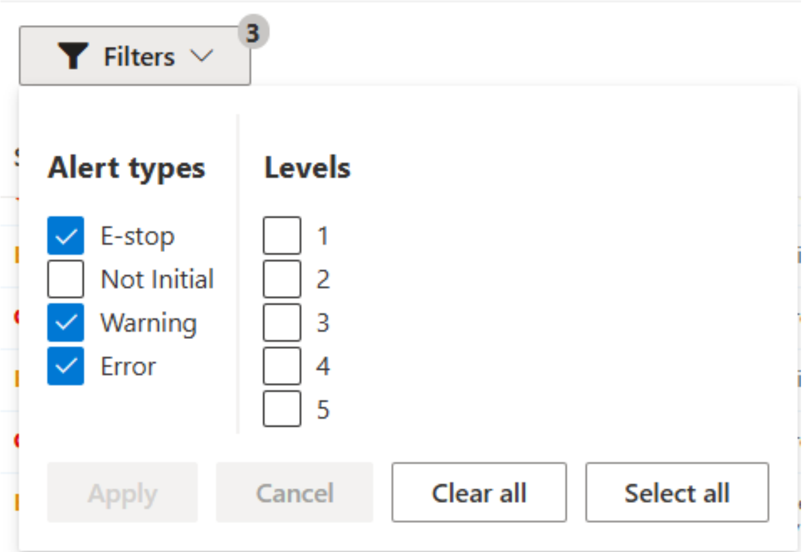
Level 3 The Length of the Infeed Case (on position 0) is too short: Expected 342mm. Measured 229.7696mm on lane n0
Infeed Belt: Corona Extra 24/7oz LN-LSE;
Position 0: Corona Extra 24/7oz LN-LSE;
Position 1: Corona Extra 24/7oz LN-LSE;

Cannot accept induction of replenishment 'ddb776cb-73b7-4f69-aa44-b4121e66c104' because the currently inducting replenishment '45ebb7f0-e6bd-4129-b113-310b02afd88f' has too many cases left (10) to induct on level 3

Warning: Clutch Prox Sensor Changed Status in the Middle of a Buffer Lane Move on lane n27 (Quantity = 24 ClientProductReference = 10224 Description = BDL 24/12oz CN) (ClientProductReference = 10224 Description = BDL 24/12oz CN in front of lane)

Warning: Clutch Prox Sensor Changed Status in the Middle of a Buffer Lane Move on lane n38 (Quantity = 9 ClientProductReference = 60424 Description = Modelo Especial 24/12oz CN) (ClientProductReference = 47646 Description = Stella Artois 24/11.2oz LN US in front of lane)

Using Search and Filters



Searching Alerts

Use the search bar to find specific alerts by:

- Alert code (e.g., "555")
- Keywords in message text (e.g., "photocell", "clutch")
- Level number
- Date/time information

Common Filtering Strategies

To find all alerts for a specific level:

- Sort by the Level column
- All alerts for each level will be grouped together

To find recent critical alerts:

- Filter by Severity = Critical
- Sort by Date (most recent first)

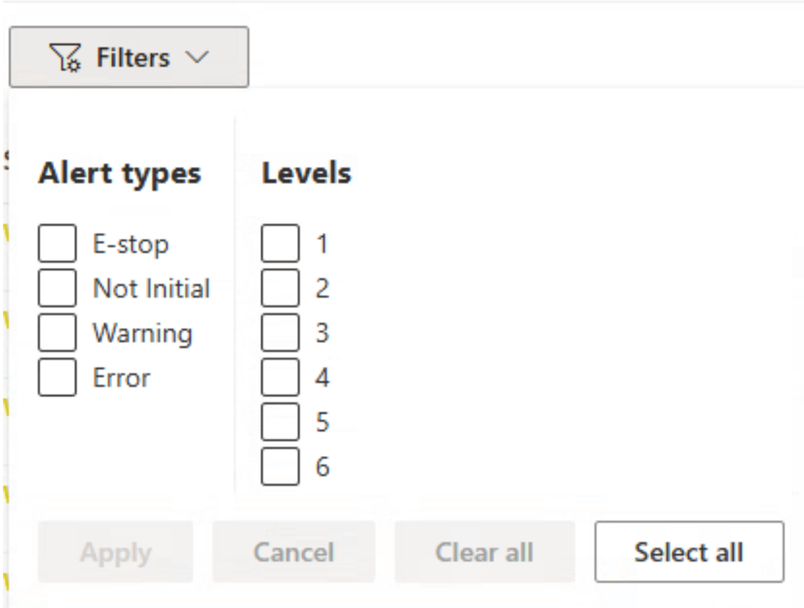
To track a specific type of fault:

- Search for the alert code or key message terms
- Review the pattern and frequency

Viewing All Alerts

To see all alerts:

- Remove all filters:



- This shows all alerts, including some Warnings and "Not Initial" alerts that are filtered out by default.
- This can be useful when a level seems to be faulted but there is no obvious reason why.

Why View Historical Alerts?

- Identify patterns of recurring issues
- Verify that alerts were properly resolved
- Troubleshoot intermittent problems
- Document issues for maintenance review

Common Alert Codes

Here are some frequently encountered alerts (see Alert Reference for complete list):

119 - Photocell still blocked after dispensing

- Photocell for a lane remained blocked after dispense move completed
- Usually indicates a case issue (broken, wrong size, poorly positioned)

121 - Photocell above lane never detected case during storing move

- InnoPick tried to store a case but photocell was never blocked
- Usually means no case was present to store

123 - Photocell never detected case during dispense move

- Case expected but not detected during dispense
- Possible causes: missing case, inventory count error

310 - Clutch Fault

- Clutch assembly failed to engage properly after multiple attempts
- Usually requires mechanical inspection

515 - Unexpected case entering infeed

- Infeed photocell blocked when no case was expected
- Check upstream conveyor system

555 - E-Stop condition active

- Safety system engaged (E-Stop pressed or door open)
- Clear E-Stop condition, then clear alert to resume

709 - Photocell blocked (PLC I/O Check)

- Photocell unexpectedly blocked during routine monitoring
- Check for debris, damaged cases, or sensor malfunction

Working with Alerts

Understanding Alert Progression

Typical Alert Lifecycle:

1. Issue occurs (sensor detects problem)
2. Alert generated and logged
3. System puts affected level in Manual or Fault mode
4. Alert appears on Home Page and Alerts Page
5. Operator resolves underlying issue
6. Operator clears alert
7. System resumes operation

When to Clear Alerts

You should clear alerts only when:

- The underlying physical issue has been resolved
- Cases are in correct positions
- Sensors are unblocked
- Mechanical issues are fixed
- Safety devices are reset

Do not clear alerts if:

- The problem still exists
- You're unsure what caused the alert
- Physical conditions don't match expectations
- Multiple related alerts are occurring

If Alerts Won't Clear

Some alerts cannot be cleared until specific conditions are met:

1. **Read the alert message carefully**
2. **Check physical conditions**
    - Is a photocell still blocked?
    - Is an E-Stop still pressed?
    - Is a case in the wrong position?
3. **Resolve the root cause**
4. **Try clearing again**
    - Use the Recovery Action: Send Clear Alarm on Level
5. The Alert in question is being filtered (not visible).
    - See how to View All Alerts
6. **Consult troubleshooting guide** if issue persists

See Alert Guidelines for best Alert resolution practices.

Alert Troubleshooting Strategy

For Single Alerts

If an alert occurs once:

1. Note the alert type and details
2. Resolve the issue
3. Clear the alert
4. Monitor to ensure it doesn't recur

For Repeated Alerts

If the same alert occurs multiple times:

1. **There's likely an underlying problem** that wasn't fully resolved
2. Check for:
    - Damaged cases causing repeated issues
    - Incorrect inventory quantities
    - Mechanical problems
    - Configuration issues
3. Resolve the root cause, not just the symptom
4. Document the issue for maintenance review

For Multiple Different Alerts

If multiple alert types occur in sequence:

1. **They may be related** - one issue causing cascading effects
2. Review the Alert Guidelines
3. Focus on resolving the earliest alert first
4. Verify physical conditions match the inventory graph
5. May indicate a larger system issue requiring maintenance

Best Practices

During Production

- Keep the Alerts page open in a second browser tab
- Check regularly for new alerts
- Address warnings before they become critical
- Document recurring issues

For Troubleshooting

- Use search to find similar past alerts
- Note the frequency and pattern
- Check if alerts correlate with specific products or lanes
- Share alert history with maintenance personnel

Related Topics

- Alert Reference Page - Complete list of all alert codes
- Alert Guidelines - Step-by-step resolution procedures

- Home Page Alerts - Quick alert overview
- Troubleshooting - General problem-solving guidance

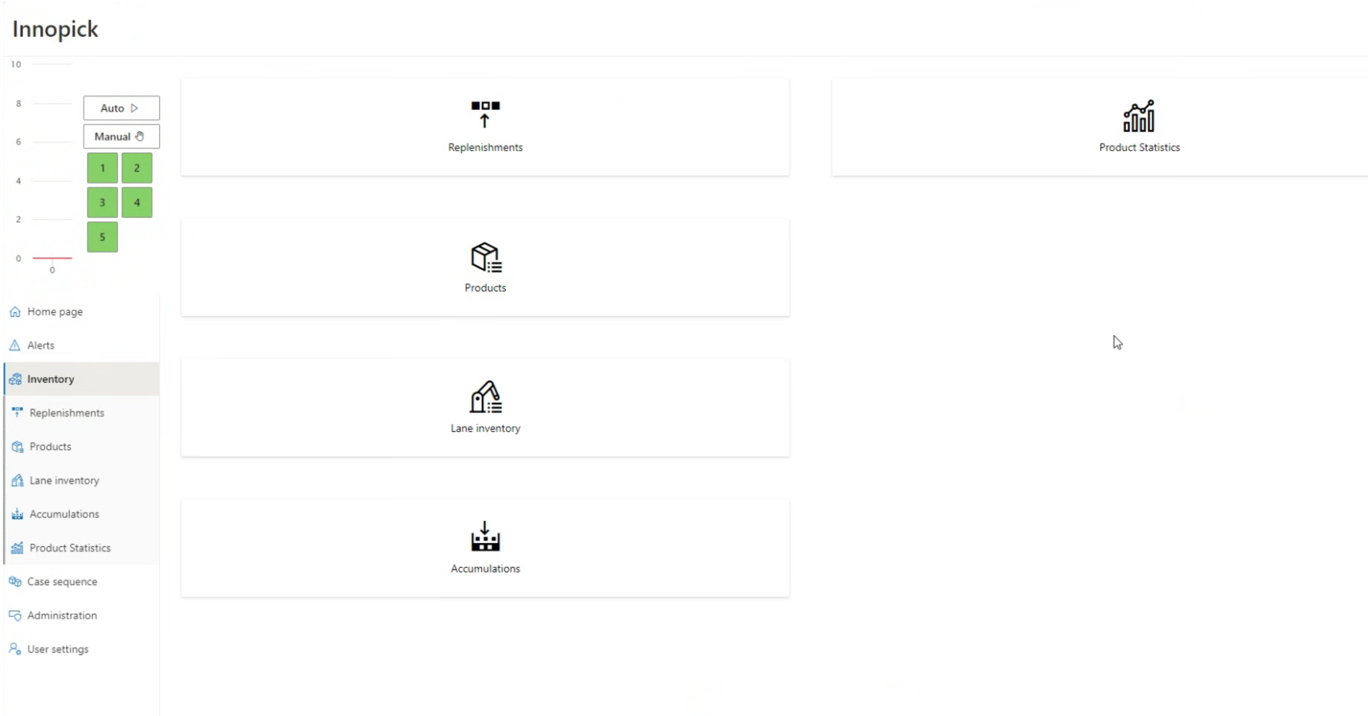
Navigation: ← Home Page | Inventory Section →

4.4 Inventory Section

Home > Main Screens > **Inventory Section**

Overview

The Inventory section provides comprehensive tools for managing product replenishments, configuring product parameters, monitoring storage lane contents, tracking accumulation, and analyzing product statistics.



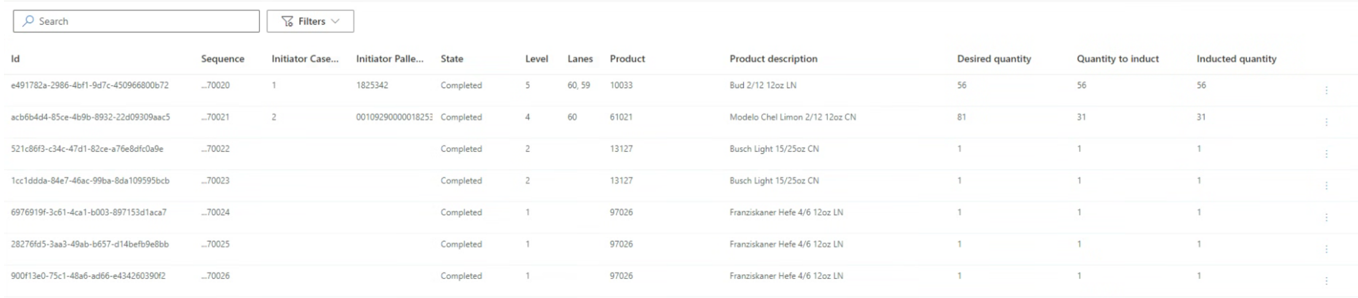
Inventory Section Pages

The Inventory section contains five pages:

1. Replenishments - Track incoming product replenishments
2. Products - Configure product settings and parameters
3. Lane Inventory - View and edit storage lane contents
4. Outbound Accumulations - Monitor outfeed accumulation space
5. Product Statistics - Analyze product usage

Replenishments Page

Back to Top - Inventory Pages Overview



Search | Filters

Id	Sequence	Initiator Case...	Initiator Palle...	State	Level	Lanes	Product	Product description	Desired quantity	Quantity to induct	Inducted quantity
e491782a-2986-4bf1-9d7c-450966800b72	...70020	1	1825342	Completed	5	60, 59	10033	Bud 2/12 12oz LN	56	56	56
acb6b4d4-85ce-4b9b-8932-22d09309aac5	...70021	2	00109290000018253	Completed	4	60	61021	Modelo Chel Limon 2/12 12oz CN	81	31	31
521c86f3-c34c-47d1-82ce-a76e8dfc0a9e	...70022			Completed	2		13127	Busch Light 15/25oz CN	1	1	1
1cc1ddda-84e7-46ac-99ba-8da109595bcb	...70023			Completed	2		13127	Busch Light 15/25oz CN	1	1	1
6976919f-3c61-4ca1-b003-897153d1aca7	...70024			Completed	1		97026	Franziskaner Hefe 4/6 12oz LN	1	1	1
28276fd5-3aa3-49ab-b657-d14befb9e8bb	...70025			Completed	1		97026	Franziskaner Hefe 4/6 12oz LN	1	1	1
900f13e0-75c1-48a6-ad66-e434260390f2	...70026			Completed	1		97026	Franziskaner Hefe 4/6 12oz LN	1	1	1

What Are Replenishments?

Replenishments occur when InnoPick doesn't have sufficient inventory for a particular order. When this happens:

1. InnoPick generates a replenishment request to MixMaster
2. A pallet of the needed product is sent to the replenishment section
3. The pallet is depalletized
4. Cases are inducted into InnoPick

Replenishment Quantities

A replenishment can range from:

- **Minimum**: A single layer
- **Maximum**: An entire pallet

Replenishment quantities are determined automatically by InnoPick's algorithms unless manually overridden. See Products Page for manual configuration options.

Replenishments Page Columns

ID

- The GUID (unique identifier) for the replenishment
- Useful for tracking in MixMaster or other systems

SEQUENCE

- Number that increases incrementally based on when the need was identified
- Indicates desired order of replenishments for a given level
- Used to prioritize replenishment requests for depalletizing

INITIATOR CASE / PALLET

- Details about what triggered the replenishment need
- Shows which customer order created the demand

STATE

Replenishments progress through these states:

- **Pending**: Replenishment is planned but not yet called for depalletizing
- **In Progress**: Anywhere in process from being requested to just before induction
- **Inducting**: Currently being inducted into InnoPick
- **Completed**: Fully inducted into storage lanes

Replenishments can also be **Canceled**

LEVEL

- Which InnoPick level the replenishment is assigned to

LANES

- Which storage lanes will receive the replenishment cases

PRODUCT

- Product SKU code

PRODUCT DESCRIPTION

- Configured name of the product

DESIRED QUANTITY

- How many cases were requested in the replenishment

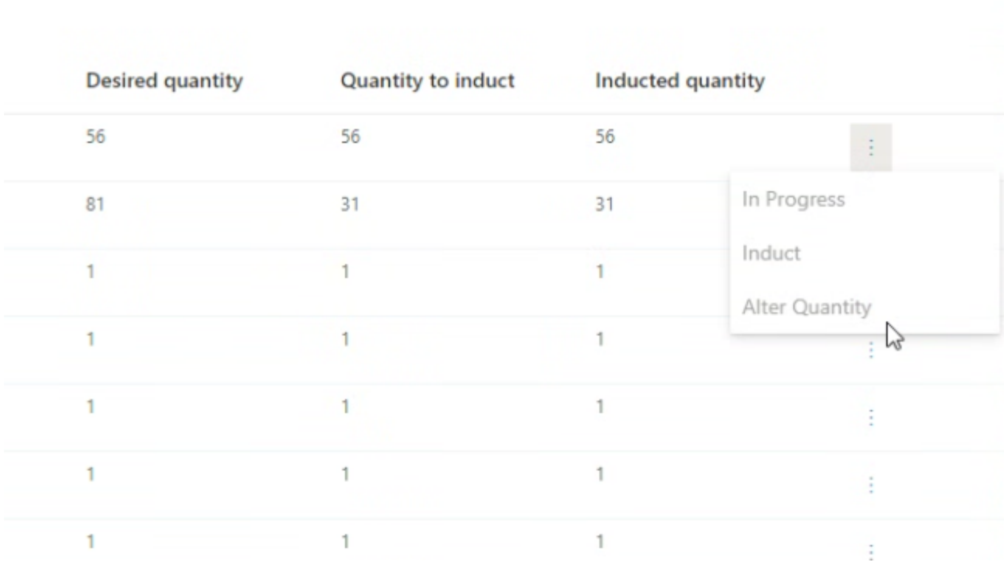
QUANTITY TO INDUCT

- Actual number of cases being sent from depalletizer to InnoPick
- May be less than Desired Quantity if:
- Source pallet has fewer layers than requested
- Replenishment was shorted by 1 or more layers during depalletization
- Cases were rejected at inbound QC station
- If quantity is insufficient to fulfill the client orders, another replenishment is automatically created

INDUCTED QUANTITY

- Number of cases successfully inducted to date
- Updates in real-time during induction

Replenishment Actions

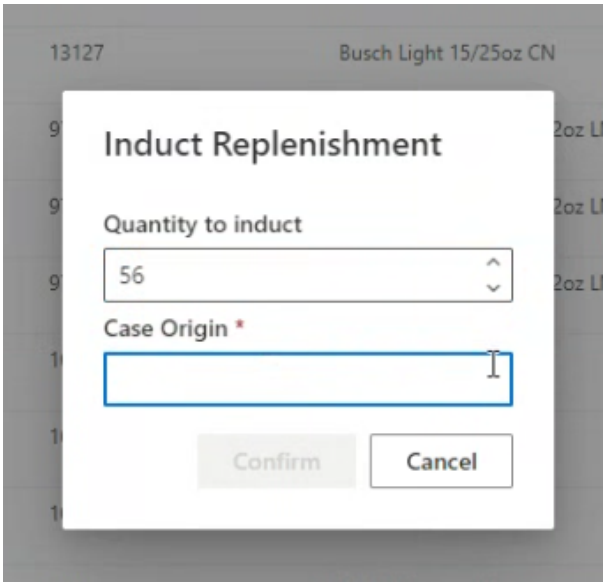


Click the three vertical dots (⋮) on the right side of any replenishment to access these options:

IN PROGRESS

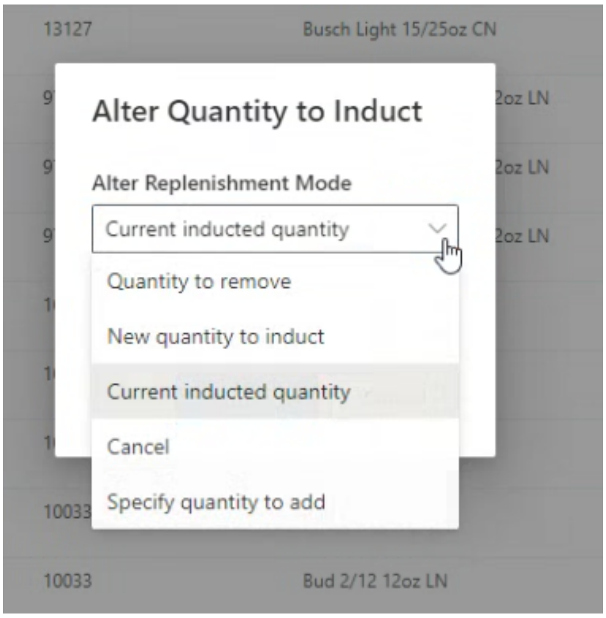
- Manually changes replenishment status from **Pending** to **In Progress**
- Only works if the replenishment is in status **Pending**
- Normally set automatically by MixMaster (WCS)
- Use only in exception scenarios when automatic update failed

INDUCT



- Changes status from **In Progress** to **Inducting**
- Only works if:
- Replenishment is already in "In Progress" state
- No other replenishment is currently "Inducting" on that level
- Use if there was a communication problem with MixMaster.

ALTER QUANTITY



- Allows manual adjustment of replenishment quantity
- Use when cases need to be manually removed before reaching InnoPick
- **Note**: Cases rejected at Inbound QC are automatically deducted

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Products Page

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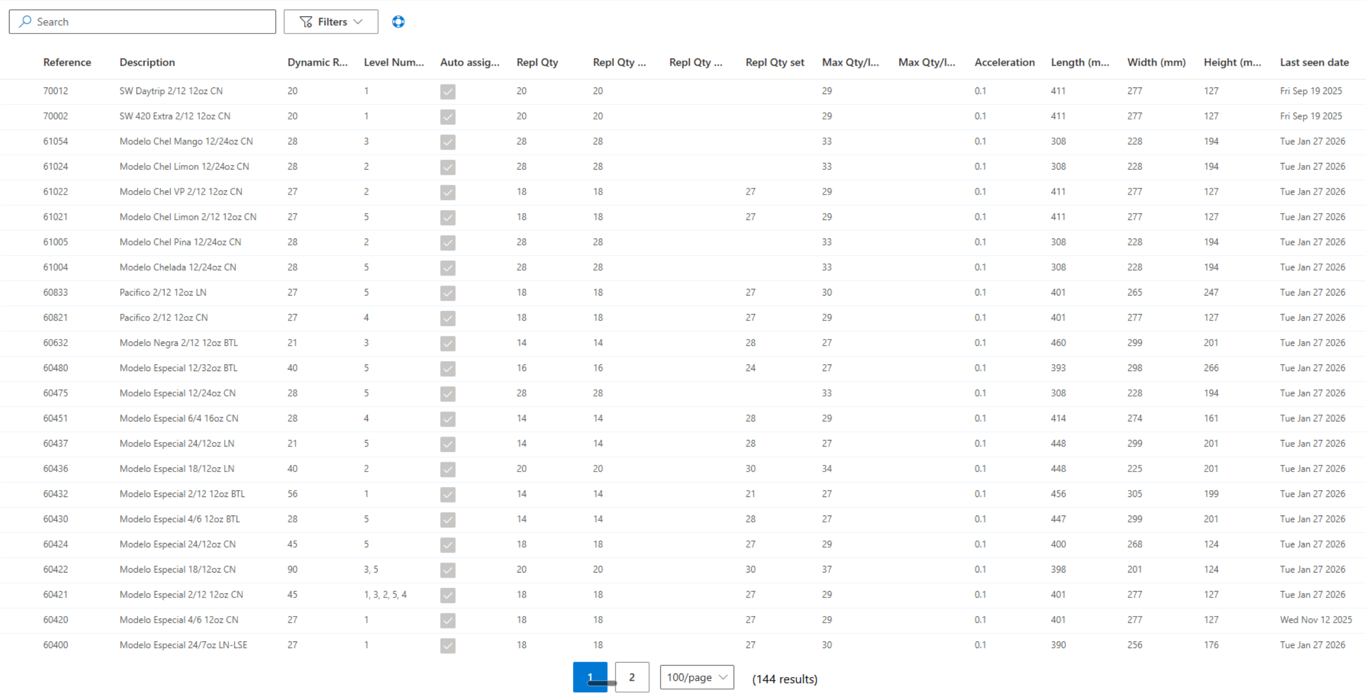


Search | Filters

Reference	Description	Dynamic R...	Level Num...	Auto assig...	Repl Qty	Repl Qty ...	Repl Qty ...	Repl Qty set	Max Qty/l...	Max Qty/l...	Acceleration	Length (m...	Width (mm)	Height (m...	Last seen date
70012	SW Daytrip 2/12 12oz CN	20	1	✓	20	20			29		0.1	411	277	127	Fri Sep 19 2025
70002	SW 420 Extra 2/12 12oz CN	20	1	✓	20	20			29		0.1	411	277	127	Fri Sep 19 2025
61054	Modelo Chel Mango 12/24oz CN	28	3	✓	28	28			33		0.1	308	228	194	Tue Jan 27 2026
61024	Modelo Chel Limon 12/24oz CN	28	2	✓	28	28			33		0.1	308	228	194	Tue Jan 27 2026
61022	Modelo Chel VP 2/12 12oz CN	27	2	✓	18	18		27	29		0.1	411	277	127	Tue Jan 27 2026
61021	Modelo Chel Limon 2/12 12oz CN	27	5	✓	18	18		27	29		0.1	411	277	127	Tue Jan 27 2026
61005	Modelo Chel Pina 12/24oz CN	28	2	✓	28	28			33		0.1	308	228	194	Tue Jan 27 2026
61004	Modelo Chelada 12/24oz CN	28	5	✓	28	28			33		0.1	308	228	194	Tue Jan 27 2026
60833	Pacifico 2/12 12oz LN	27	5	✓	18	18		27	30		0.1	401	265	247	Tue Jan 27 2026
60821	Pacifico 2/12 12oz CN	27	4	✓	18	18		27	29		0.1	401	277	127	Tue Jan 27 2026
60632	Modelo Negra 2/12 12oz BTL	21	3	✓	14	14		28	27		0.1	460	299	201	Tue Jan 27 2026
60480	Modelo Especial 12/32oz BTL	40	5	✓	16	16		24	27		0.1	393	298	266	Tue Jan 27 2026
60475	Modelo Especial 12/24oz CN	28	5	✓	28	28			33		0.1	308	228	194	Tue Jan 27 2026
60451	Modelo Especial 6/4 16oz CN	28	4	✓	14	14		28	29		0.1	414	274	161	Tue Jan 27 2026
60437	Modelo Especial 24/12oz LN	21	5	✓	14	14		28	27		0.1	448	299	201	Tue Jan 27 2026
60436	Modelo Especial 18/12oz LN	40	2	✓	20	20		30	34		0.1	448	225	201	Tue Jan 27 2026
60432	Modelo Especial 2/12 12oz BTL	56	1	✓	14	14		21	27		0.1	456	305	199	Tue Jan 27 2026
60430	Modelo Especial 4/6 12oz BTL	28	5	✓	14	14		28	27		0.1	447	299	201	Tue Jan 27 2026
60424	Modelo Especial 24/12oz CN	45	5	✓	18	18		27	29		0.1	400	268	124	Tue Jan 27 2026
60422	Modelo Especial 18/12oz CN	90	3, 5	✓	20	20		30	37		0.1	398	201	124	Tue Jan 27 2026
60421	Modelo Especial 2/12 12oz CN	45	1, 3, 2, 5, 4	✓	18	18		27	29		0.1	401	277	127	Tue Jan 27 2026
60420	Modelo Especial 4/6 12oz CN	27	1	✓	18	18		27	29		0.1	401	277	127	Wed Nov 12 2025
60400	Modelo Especial 24/7oz LN-LSE	27	1	✓	18	18		27	30		0.1	390	256	176	Tue Jan 27 2026

1 2 100/page (144 results)

The Products page displays all products in the InnoPick Manager database with their configuration settings.

Main Products Page Columns

REFERENCE

- The product's SKU code
- Primary identifier for the product

DESCRIPTION

- Configured product name
- Human-readable description

DYNAMIC REPLEN QUANTITY

- Number of cases in a standard replenishment as calculated by IPM
- Dynamically adjusted based on demand patterns

LEVEL NUMBER(S)

- Which levels InnoPick is allowed to use for this product
- Products can be restricted to specific levels

AUTO ASSIGN LEVELS

- True/False setting
- When True: InnoPick automatically determines which levels to use
- **Note:** It is possible to change the level assignment(s) when a Auto Assign is still enabled. This temporarily overrides the algorithm
- When False: Levels must be manually assigned

REPLENISHMENT QUANTITY

- Default replenishment size calculated by InnoPick's algorithm
- Based on historical usage patterns

REPLENISHMENT QUANTITY SET

- User-defined override for default replenishment quantity
- Use to manually control replenishment size

REPLENISHMENT QUANTITY MINIMUM

- Minimum cases allowed in a replenishment (calculated by InnoPick)
- Must be an even multiple of cases per layer

REPLENISHMENT QUANTITY MINIMUM SET

- User-defined minimum replenishment quantity
- Override automatic calculation if needed

MAX QUANTITY / LANE

- Maximum cases that can fit in a storage lane (calculated by InnoPick)
- Based on: lane length, configured gaps, case width

MAX QUANTITY / LANE SET

- User-defined maximum quantity per lane
- **Caution**: Do not exceed automatic calculation or cases may go past lane end

ACCELERATION

- Controls how storage lanes handle the product during store/dispense operations
- Measured in g's (gravitational force)
- **Typical range**: 0.07g to 0.15g
- **Default**: 0.1g
- **Setting guideline**: Choose highest value that doesn't cause storage faults
- Higher acceleration = faster operations but may disturb unstable cases or cause cases to slide on chains and lose the gaps between cases.

LENGTH (MM)

- Case length in millimeters
- Always the longer horizontal dimension

WIDTH (MM)

- Case width in millimeters
- Always shorter than length

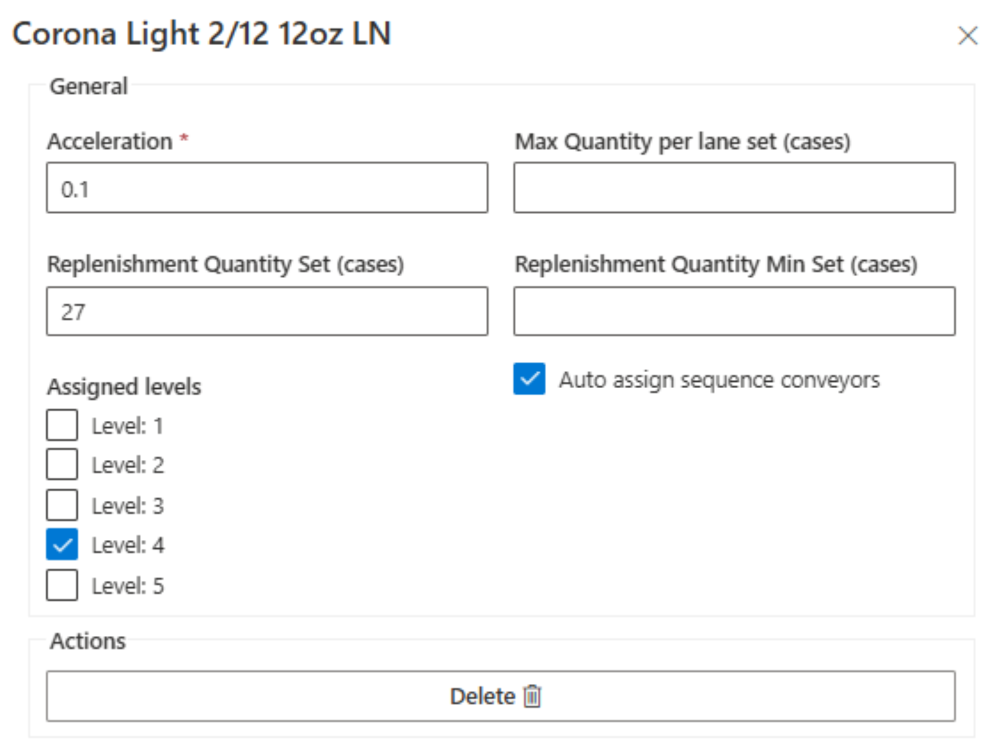
HEIGHT (MM)

- Case height in millimeters

LAST SEEN DATE

- Last time this product was present in InnoPick
- Useful for identifying inactive products

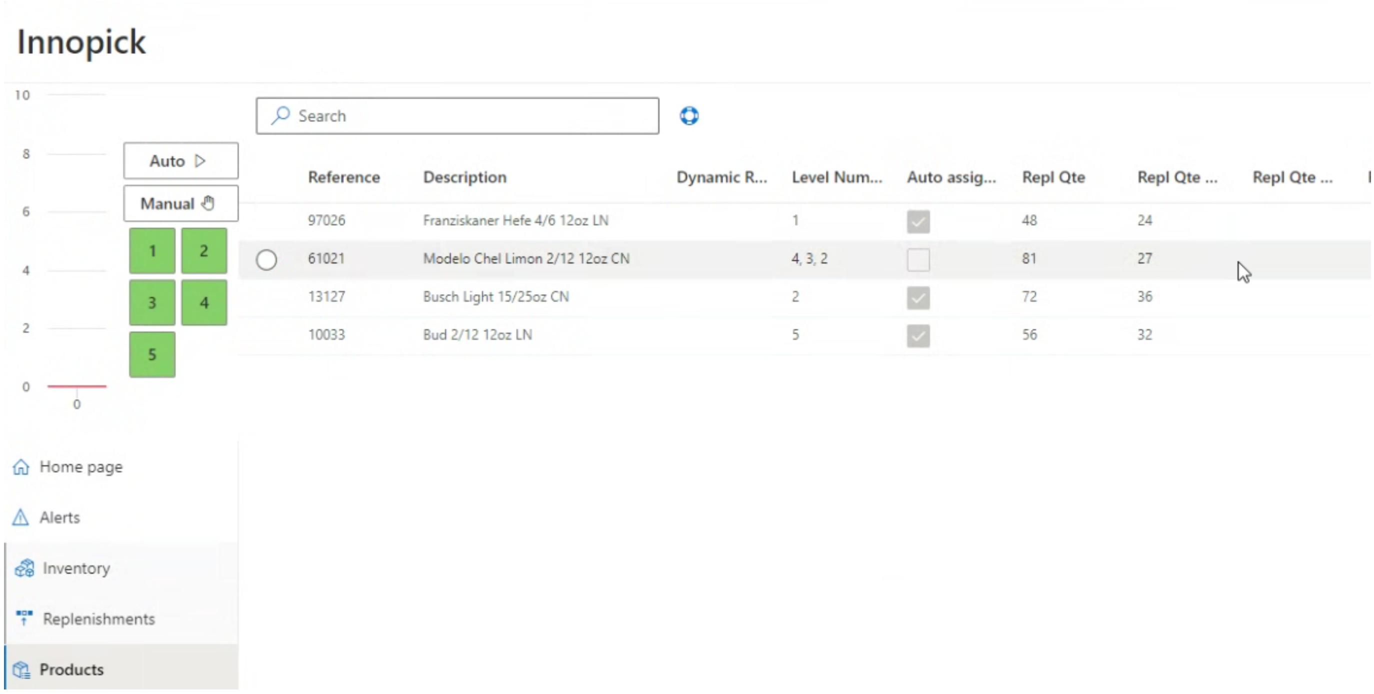
Editing Products



To edit a product:

1. Click on the product row in the table
2. Edit panel appears on the right side of the screen
3. Modify desired parameters
4. Save changes (button at the bottom of the sidebar)

EXAMPLE: MANUAL LEVEL ASSIGNMENT



When **Auto Assign** is disabled, you can manually assign the product to specific levels:

1. Uncheck "Auto Assign Levels"
2. Select which levels can store this product
3. Save the configuration

Note: It is possible to change the level assignment(s) when a Auto Assign is still enabled. This temporarily overrides the algorithm

Lane Inventory Page

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Level: 1 Level: 2 Level: 3 Level: 4 Level: 5

Lane number ↑	Quantity	Product reference	Product description	Origins	Sources	Entered inventory dates	Actions	Disable/Enabl...	Ignore sensors	Lane Types
11	0			--	--	--				
12	17	47622	Stella Artois 2/12 12oz CN	276441	Replenishment	1/27/2026, 1:01:11 PM +16				
13	9	47637	Stella Artois 18/11.2oz LN US	275767	Replenishment	1/26/2026, 11:19:58 AM +8				
14	21	26824	Select 55 24/12oz Cn	276782	Replenishment	1/27/2026, 2:29:13 PM +20				
15	8	15236	Busch Light 18/12oz LN	276409	Replenishment	1/26/2026, 10:04:10 PM +7				
16	0			--	--	--				
17	26	47646	Stella Artois 24/11.2oz LN US	MANUAL_INDUCT +	laneCaseSources.1 +1	1/27/2026, 8:13:06 AM +25				
18	2	60421	Modelo Especial 2/12 12oz CN	276809	Replenishment	1/27/2026, 3:05:46 PM +1				
19	0			--	--	--				
20	26	12431	MUL 4/6 12oz LN	276824	Replenishment	1/27/2026, 2:56:36 PM +25				
21	5	10251	BDL 6/4 16oz CN	276833	Replenishment	1/27/2026, 3:02:55 PM +4				
22	23	12433	MUL 2/12 12oz LN	276815	Replenishment	1/27/2026, 2:48:16 PM +22				
23	3	48833	Becks Pilsner 2/12 12oz LN	276798	Replenishment	1/27/2026, 2:35:55 PM +2				
24	13	10051	Bud 6/4 16oz CN	276837	Replenishment	1/27/2026, 3:02:25 PM +12				
25	10	60432	Modelo Especial 2/12 12oz BT	276823	Replenishment	1/27/2026, 2:59:32 PM +9				

The Lane Inventory page is an expanded version of the lane display available on the Home Page.

Main Lane Inventory Page Columns

LANE NUMBER

- Identifies the specific storage lane

QUANTITY

- Current number of cases in the lane

PRODUCT REFERENCE

- SKU code of product stored in lane

PRODUCT DESCRIPTION

- Configured name of the product

ORIGINS

- Pallet ID of the source replenishment
- Tracks where cases came from

SOURCES

- Indicates how cases entered the lane:
- **Replenishment**: Normal replenishment process
- **Manual Replenishment**: Manually created replenishment
- **Change Quantity Operation**: Manually added by operator

ENTERED INVENTORY DATES

- When the product was inducted into the lane
- May show multiple dates if lane has cases from different replenishments

ACTIONS

- Click the icon to edit lane contents
- Only available in Manual mode

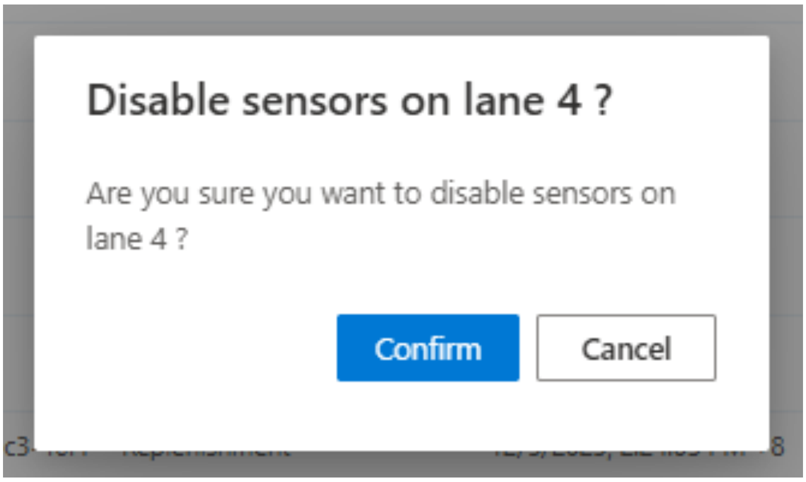
Enable/Disable Controls

DISABLE / ENABLE LANE TOGGLE

- Controls whether a specific lane can be used in production
- **Warning**: Disabling a lane during production may require manual intervention if:
- Products were already scheduled to store in that lane
- Products need to be dispensed from that lane
- See the **Disabled Lane Troubleshooting Guide** for information on how to deal with with disabling a lane that has scheduled inbound or outbound cases.

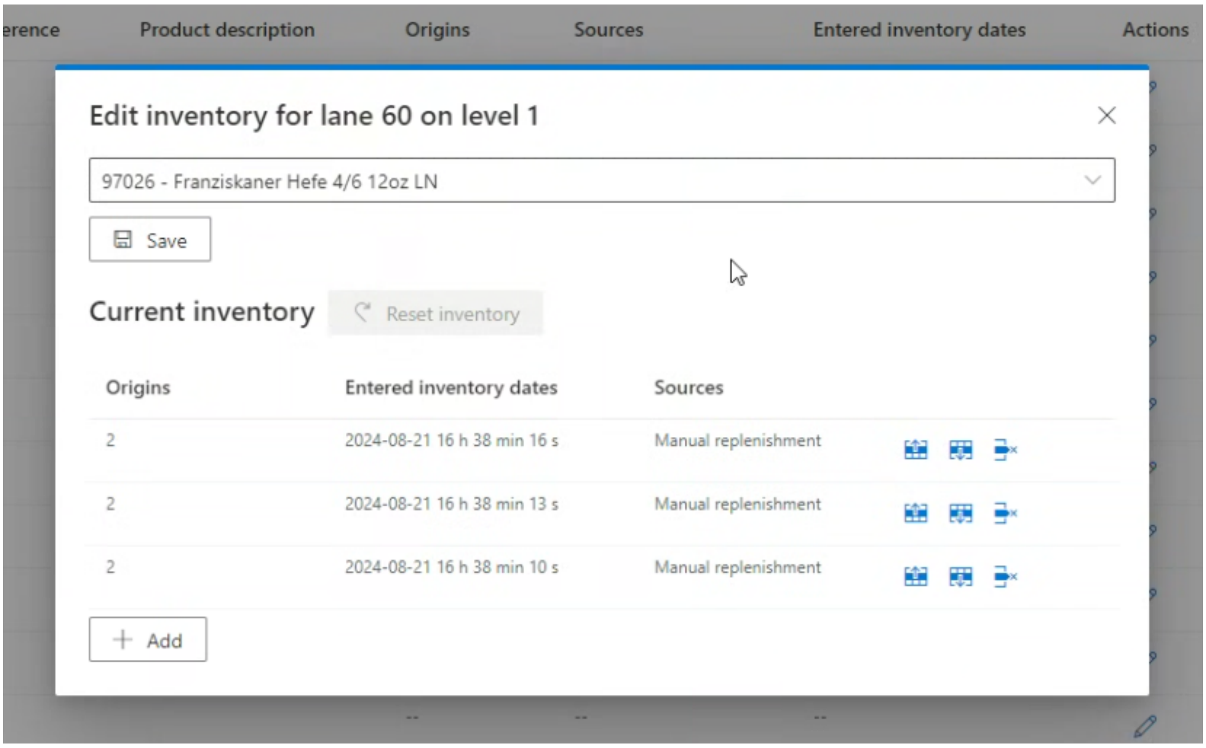
IGNORE SENSORS

- This option can be enabled only on a disabled lane.
- Normally, a sensor in an unexpected state causes an alert continually on that level of InnoPick.
- When this option is used, the system ignores the unexpected state of the sensors associated with that lane.



• This option should be used in coordination with maintenance staff responsible for mechanical interventions.

Editing Lane Inventory



PREREQUISITES FOR EDITING

You can only edit lane contents when the system is in **Manual mode** because:

• Lane contents may be changing in Automatic mode

• Editing during Automatic could cause synchronization issues

If you try to edit in Automatic mode:

• Error message appears: "No modifications are allowed during automatic mode. Please switch to manual mode to update lane inventories."

EDIT LANE DIALOG

When editing a lane, you can:

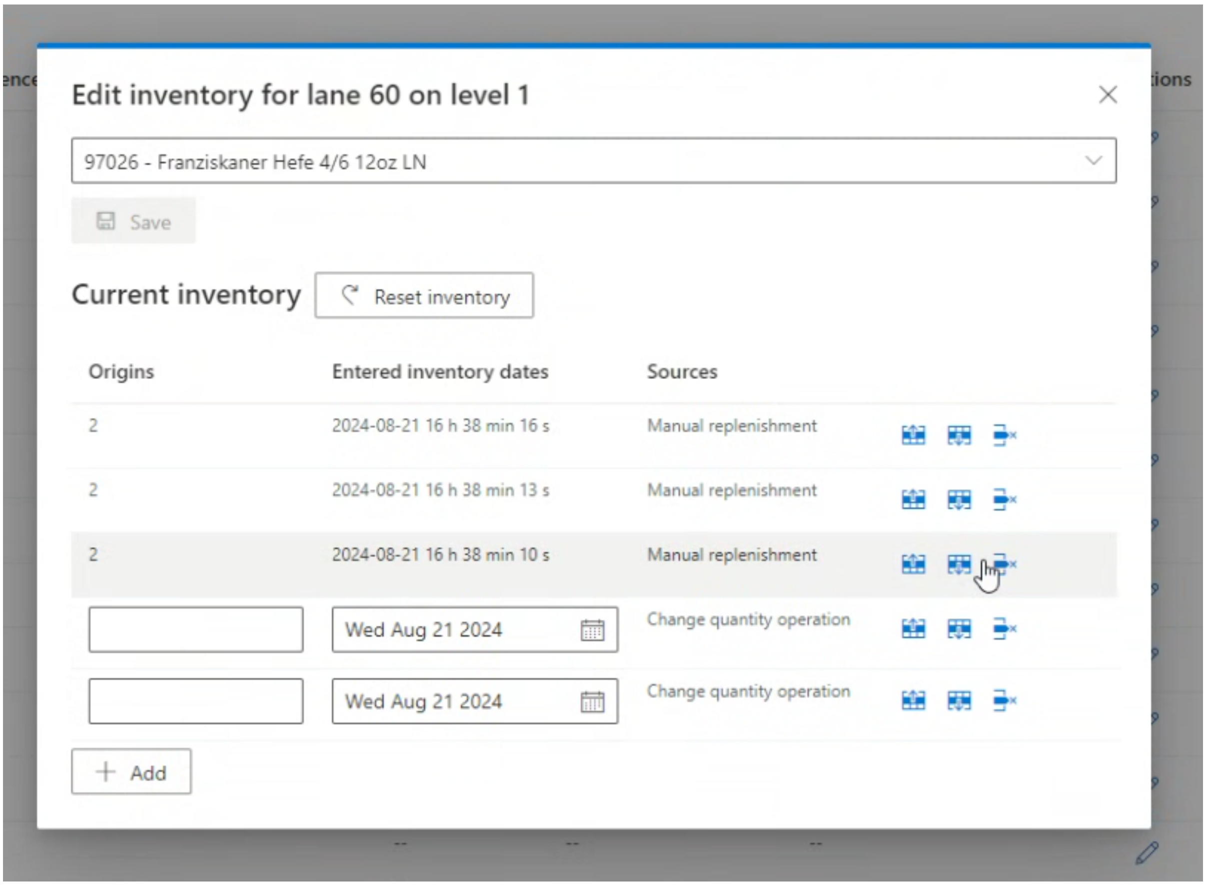
Change Product Type:

- Use the drop-down menu at top to select different product

Manage Individual Cases:

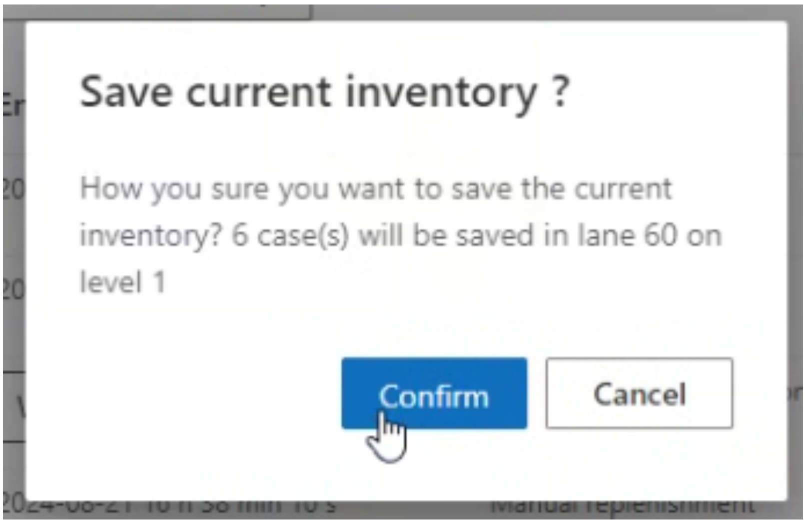
- Each case shown as a horizontal entry
- Icons on right allow you to:
- Move case up in sequence
- Move case down in sequence
- Remove case from lane

Add Cases:



- Click **+ Add** button on bottom left
- Creates new cases with "Change quantity operation" as source
- Useful for manual inventory corrections

Save Changes:



- Click **Save** button when edits are complete
- Confirm the changes in the pop-up dialog

IMPORTANT CONSIDERATIONS

Use caution when changing lane contents during production:

- Inventory may already be committed to the case sequence
- InnoPick may not always accommodate sudden changes
- Inventory changes can trigger a Deadlock -- see here for a troubleshooting guide.

Outbound Accumulations Page

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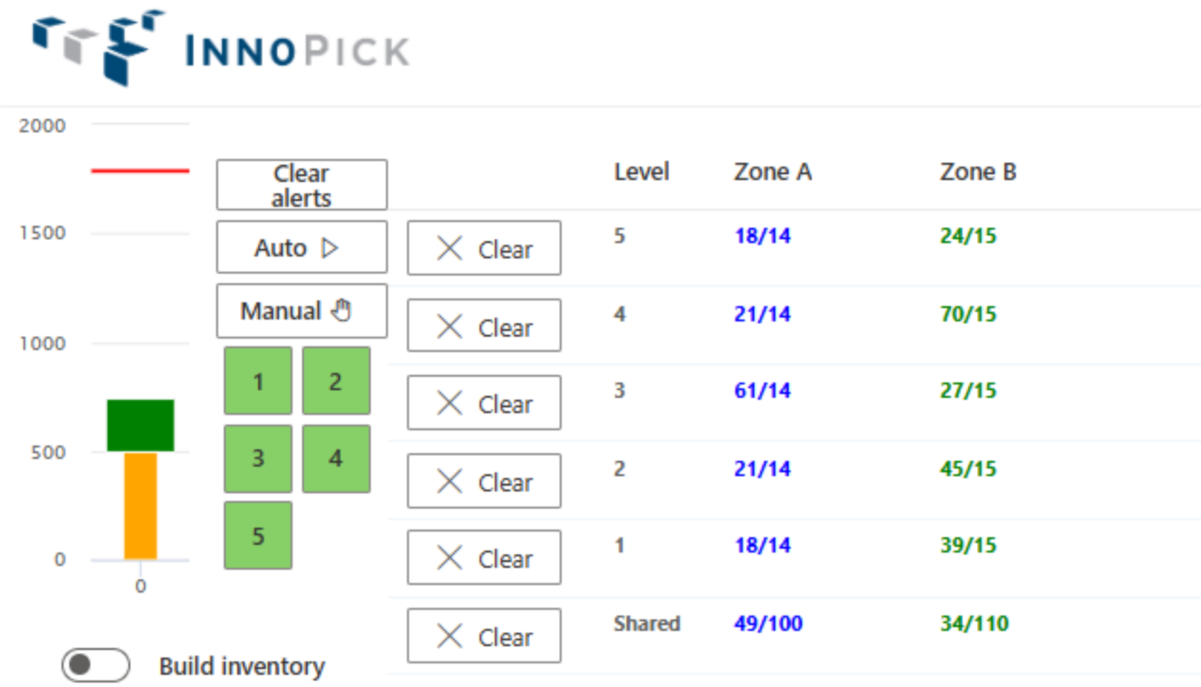

The Accumulations page displays values related to InnoPick's management of the space between InnoPick's exit and the downstream merge point (Merges or equivalent).

How Accumulation Works

InnoPick tracks:

- How many millimeters of cases it has released (if using low-pressure accumulation) or the number of cases (if using zone accumulation)
- Cases exiting the downstream merge point (Merges)
- Available accumulation space remaining

Based on these factors, InnoPick decides whether to schedule more output cases.

Accumulations Page Columns

LEVEL

- Which InnoPick level and outbound conveyor (A vs B)
- Shared: Accumulation space that is shared across all levels

IN PROGRESS MM A / B

- Millimeters of cases currently scheduled to be released
- Separate tracking for each destination (A and B)

COMPLETED MM A / B

- Millimeters of cases that have been released
- Represents space currently occupied in accumulation zone

ACCUM % A / B

- What the Completed mm represents as percentage of total configured accumulation space
- Example: 75% means accumulation zone is three-quarters full

ZONE A & ZONE B

- How many cases are currently scheduled to be released compared to the number of accumulation zones.

CLEAR

- Button to clear the accumulation tracking
- Tells InnoPick the accumulation space is now empty
- Use when:
- Downstream conveyors have been cleared
- Cases have been manually removed
- Accumulation tracking is out of sync with reality

Note: The degree to which InnoPick schedules output cases is affected by, among other things, whether the *Build Inventory Mode* is active or not. For more information on the Build Inventory Mode, see Setup InnoPick > Buffer section

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Product Statistics Page

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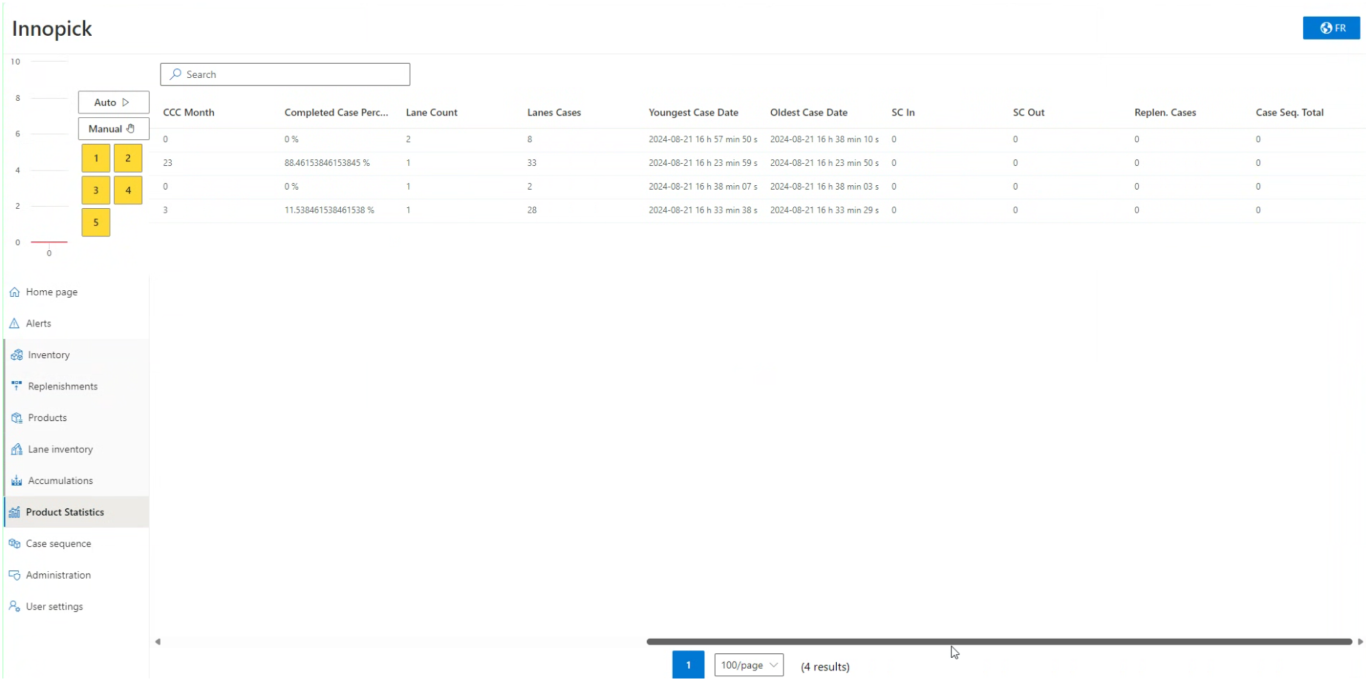
Search

Reference	Description	Completed Case Cou...	CCC Today	CCC Yesterday	CCC Week	CCC Month	Completed Case Perc...	Lane Count	Lanes Cases	Youngest Case Date
10031	Bud 4/6 12oz LN	0	0	0	0	0	0 %	0	0	
203205	NUTRL Fruit VP 3/8 12oz CN	4414	54	89	460	3053	0.5120091312548284 %	1	22	1/27/2026, 12:23:16 PM
10050	Bud 15/16oz CN	4759	55	155	622	3198	0.5520279691077771 %	1	3	1/27/2026, 11:07:48 AM
47637	Stella Artois 18/11.2oz LN US	1619	5	14	172	1340	0.1877985463302146 %	1	9	1/26/2026, 11:19:58 AM
26824	Select 55 24/12oz Cn	2102	58	82	363	1505	0.2438249193243428 %	1	21	1/27/2026, 2:29:13 PM
60233	Corona Light 2/12 12oz LN	4712	53	118	600	3355	0.5465761274292594 %	1	16	1/27/2026, 1:00:32 PM
60133	Corona Premier 2/12 12oz LN	3059	37	68	377	2181	0.3548336956294789 %	1	24	1/27/2026, 1:24:51 PM
10071	Bud 15/25oz CN	1813	24	53	266	1228	0.21030189283303213 %	1	3	1/27/2026, 11:23:07 AM
16251	Natural Ice 6/4 16oz CN	4602	75	170	728	3190	0.533816497968899 %	1	6	1/27/2026, 2:57:51 PM
10233	BDL 2/12 12oz LN	5095	70	116	622	3439	0.5910028372776054 %	1	11	1/27/2026, 1:02:42 PM
15621	Busch NA 2/12 12oz CN	2220	41	39	277	1549	0.2575125218363659 %	1	22	1/27/2026, 12:52:40 PM
60080	Corona Familiar 12/32oz BTL	6717	74	303	1024	4247	0.7791493735021935 %	1	10	1/27/2026, 2:08:59 PM
60821	Pacifico 2/12 12oz CN	1838	22	40	216	1145	0.21320180861947768 %	1	21	1/27/2026, 12:12:58 PM

The Product Statistics page provides detailed information about how InnoPick uses various products.

Product Statistics Columns



CCC Month	Completed Case Perc...	Lane Count	Lanes Cases	Youngest Case Date	Oldest Case Date	SC In	SC Out	Replen. Cases	Case Seq. Total
0	0 %	2	8	2024-08-21 16 h 57 min 50 s	2024-08-21 16 h 38 min 10 s	0	0	0	0
23	88.46153846153845 %	1	33	2024-08-21 16 h 23 min 59 s	2024-08-21 16 h 23 min 50 s	0	0	0	0
0	0 %	1	2	2024-08-21 16 h 38 min 07 s	2024-08-21 16 h 38 min 03 s	0	0	0	0
3	11.538461538461538 %	1	28	2024-08-21 16 h 33 min 38 s	2024-08-21 16 h 33 min 29 s	0	0	0	0

REFERENCE

- Product SKU code

DESCRIPTION

- Configured product name

COMPLETED CASE COUNT

- Total number of completed cases for this product (all time)

CCC TODAY / YESTERDAY / WEEK / MONTH

- Completed case count within the specified time period
- Useful for tracking demand patterns

COMPLETED CASE PERCENTAGE

- What percentage of total production this product represents
- Helps identify high-volume products

LANE COUNT

- How many storage lanes currently contain this product

LANES CASES

- Total number of cases of this product currently in InnoPick storage

YOUNGEST CASE DATE

- Date of the most recently inducted case
- Indicates how fresh the inventory is

OLDEST CASE DATE

- Date of the oldest case in inventory
- Helps identify slow-moving or stale inventory

SC IN

- Cases on sequence conveyor going to storage (inbound)
- Currently being inducted from replenishment

SC OUT

- Cases on sequence conveyor going to outfeed (outbound)
- Currently being output to palletizers

REPLEN. CASES

- Cases scheduled to be inducted as replenishments
- Pending or in-progress replenishments

CASE SEQ TOTAL

- How many cases of this product remain in the current case sequence
- Still needed to complete active orders

Using Product Statistics

For Inventory Management:

- Find slow-moving products (old Oldest Case Date)
- Track high-volume products (high Completed Case Percentage)

For Optimization:

- Products with high CCC should have adequate lane assignments
- Products with frequent replenishments may need quantity adjustments

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Related Topics

- Home Page Lane Inventory - Quick lane overview
- Monitoring Production - Using inventory data during operations

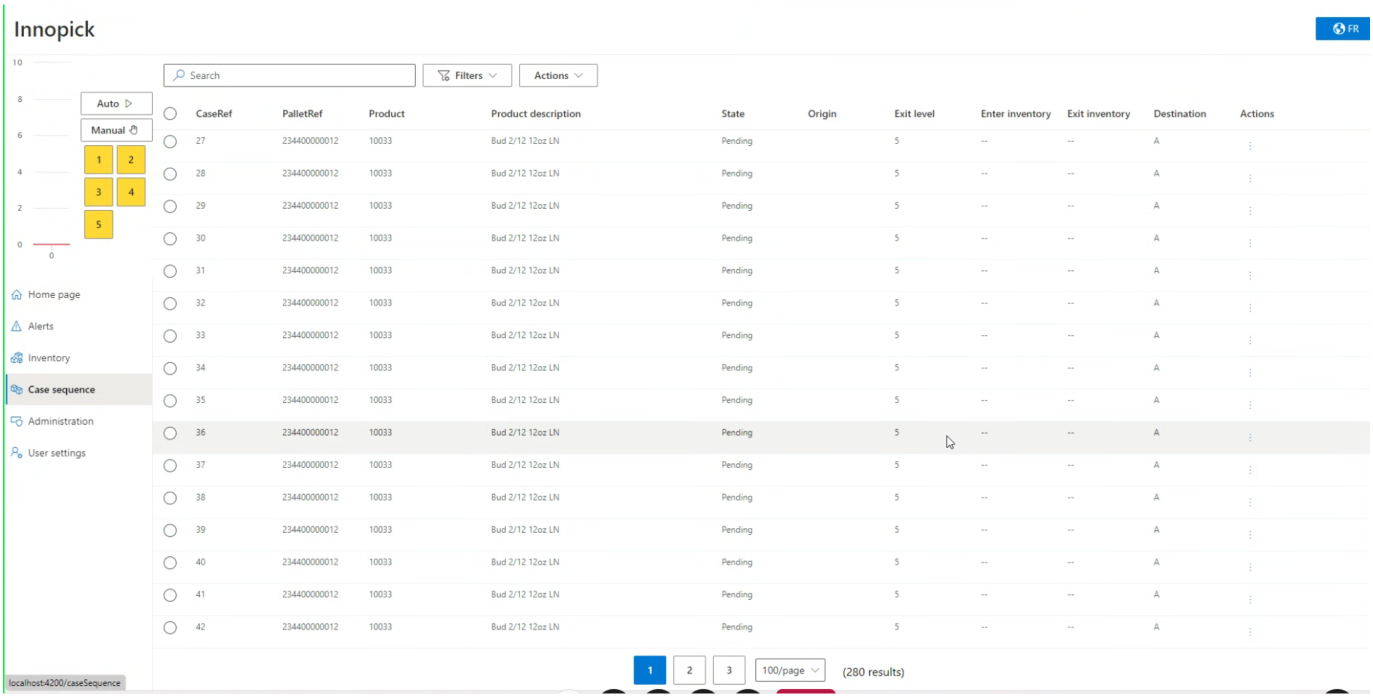
Navigation: ← Alerts Page | Case Sequence →

4.5 Case Sequence Page

Home > Main Screens > Case Sequence

Overview

The Case Sequence page displays the order in which cases will exit InnoPick and arrive at the palletizers. This powerful tool allows operators to track individual cases, verify sequencing, and troubleshoot order fulfillment issues.



Understanding the Case Sequence

The case sequence represents:

- The exact order cases will exit InnoPick (per level)
- Which cases belong to which customer pallets
- Current status of each case (pending, in progress, completed, cancelled)
- Which downstream destination each case is destined to

This information is critical for:

- Verifying correct order fulfillment
- Tracking specific cases or pallets
- Troubleshooting sequencing issues
- Monitoring production progress

Case Sequence Columns

Each case in the sequence has the following information:

Case Ref

- Internal case tracking number
- Unique identifier for each individual case
- Used for detailed case tracking and troubleshooting

PalletRef

- Pallet ID for the customer pallet this case belongs to
- Multiple cases with the same PalletRef are part of the same order
- Useful for tracking pallet completion

Product

- SKU code of the product

Product Description

- Configured product name

State

Cases progress through these states:

- **Pending**: Case is scheduled but not yet scheduled
- **In Progress**: Case is scheduled, which means currently visible on the sequence conveyor as an empty or full square (output case).
- **Completed**: Case has been successfully output from InnoPick
- **Cancelled**: Case has been removed from the sequence (see Actions below)

Origin

- Pallet ID of the replenishment that brought this case into InnoPick
- Tracks the source of the case
- Useful for tracing product batches or investigating quality issues

Exit Level

- The InnoPick level the case is assigned to
- Indicates which physical level will output this case

Enter Inventory

- Date and time the case entered InnoPick
- Useful for tracking case age and identifying stale inventory

Destination

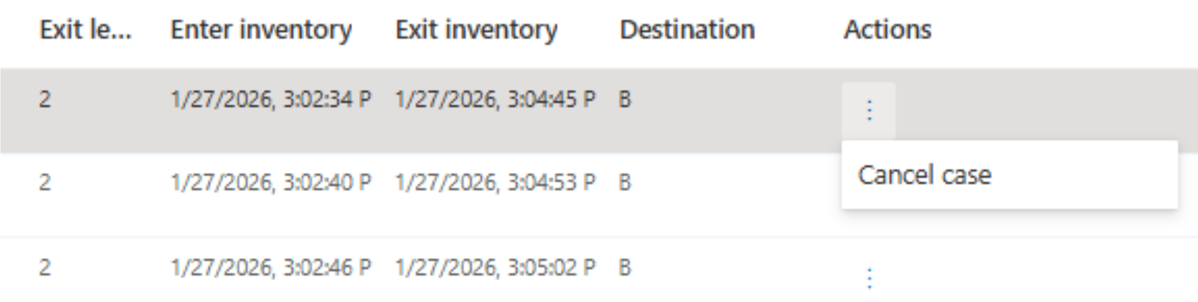
- Which downstream equipment will receive the case
- Typical values: Merge A, Merge B, or other merge point designation

• Used by InnoPick to route cases correctly

Actions

The only action available on this page is:

CANCEL CASE

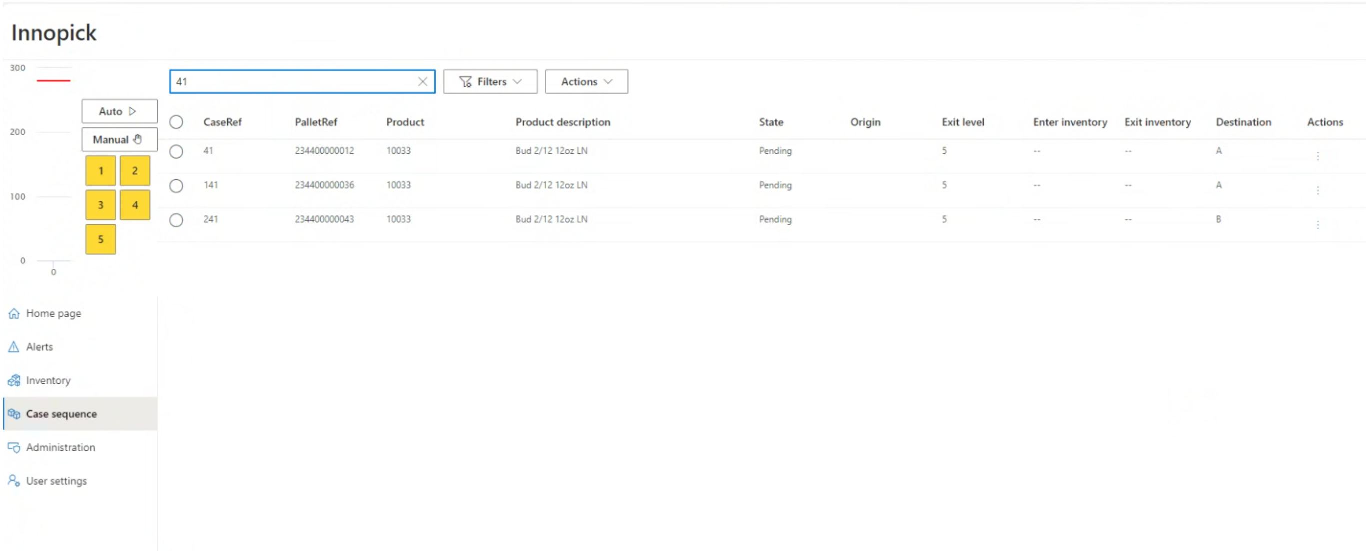


• Removes the case from the active sequence

• Changes state to "Cancelled"

• **Warning**: During normal operations, it should not be necessary to cancel cases from the case sequence. This should only be done as part of a larger recovery effort in coordination with system experts or remote support.

Using Search and Filters



The Case Sequence page becomes particularly powerful when filters are applied.

Search Bar

Use the search bar to quickly find cases by:

- Case reference number
- Pallet ID
- Product SKU
- Product description
- Any keyword in the displayed fields

Common Filtering Scenarios

FIND ALL CASES FOR A SPECIFIC PALLET

1. Enter the Pallet ID in the search bar
2. All cases for that pallet will be displayed
3. Verify the sequence is correct
4. Check that all expected cases are present

FIND CASES BY LEVEL

1. Filter by Exit Level
2. See all cases scheduled for a specific level
3. Useful for level-specific troubleshooting

FIND CANCELLED CASES

1. Filter by State = Cancelled
2. Review which cases have been removed from sequence

FIND CASES FROM A SPECIFIC REPLENISHMENT

1. Search or filter by Origin (Pallet ID of replenishment)
2. See all cases from that replenishment
3. Useful for tracking product batches or investigating quality issues

FIND CASES BY DESTINATION

1. Filter by Destination (Merge A, Merge B, etc.)
2. Verify routing is correct

Using Case Sequence for Troubleshooting

Tracking a Missing Case

If a case doesn't arrive at the palletizer when expected:

1. **Search for the case** by Pallet ID or Case Ref
2. **Check the State:**
    - Pending: Still waiting to be processed (or was somehow missed because of a system glitch)
    - Often, a pending case indicates the system is still waiting for a replenishment to provide that case
    - In Progress: On conveyor or in lane - check Home Page Inventory Graph
    - Completed: InnoPick released it - check downstream tracking
    - Cancelled: Was removed - determine why
3. **Verify the case details:**
    - Correct product?
    - Correct destination?
    - Correct level?
4. **Take appropriate action** based on findings.
    - If a case is stuck in pending mode and no replenishment for that product is on its way, contact support.

Verifying Pallet Completion

To confirm all cases for a pallet have been output:

1. **Search for the Pallet ID**
2. **Check the State of all cases:**
    - All should be "Completed" for a finished pallet
    - If any are Pending or In Progress, pallet is not complete
    - A pending case usually indicates the system is still waiting for a replenishment to provide that case
    - Note any Cancelled cases

Investigating Sequencing Problems

If cases are arriving in wrong order:

1. **Compare actual sequence** to expected order
2. **Filter by Exit Level and Destination**
3. **Check for:**
    - Cases from wrong level
    - Cases routed to wrong destination
    - Cancelled cases affecting sequence
4. **Correct the sequence** if sequence is incorrect.
5. *If InnoPick appears to be building an incorrect sequence, or the sequence in InnoPick does not match that of MixMaster, contact support.*

Related Topics

- Home Page Inventory Graph - Visual representation of In Progress cases
- Replenishments - Source of cases entering sequence

Navigation: ← Inventory Section | Administration →

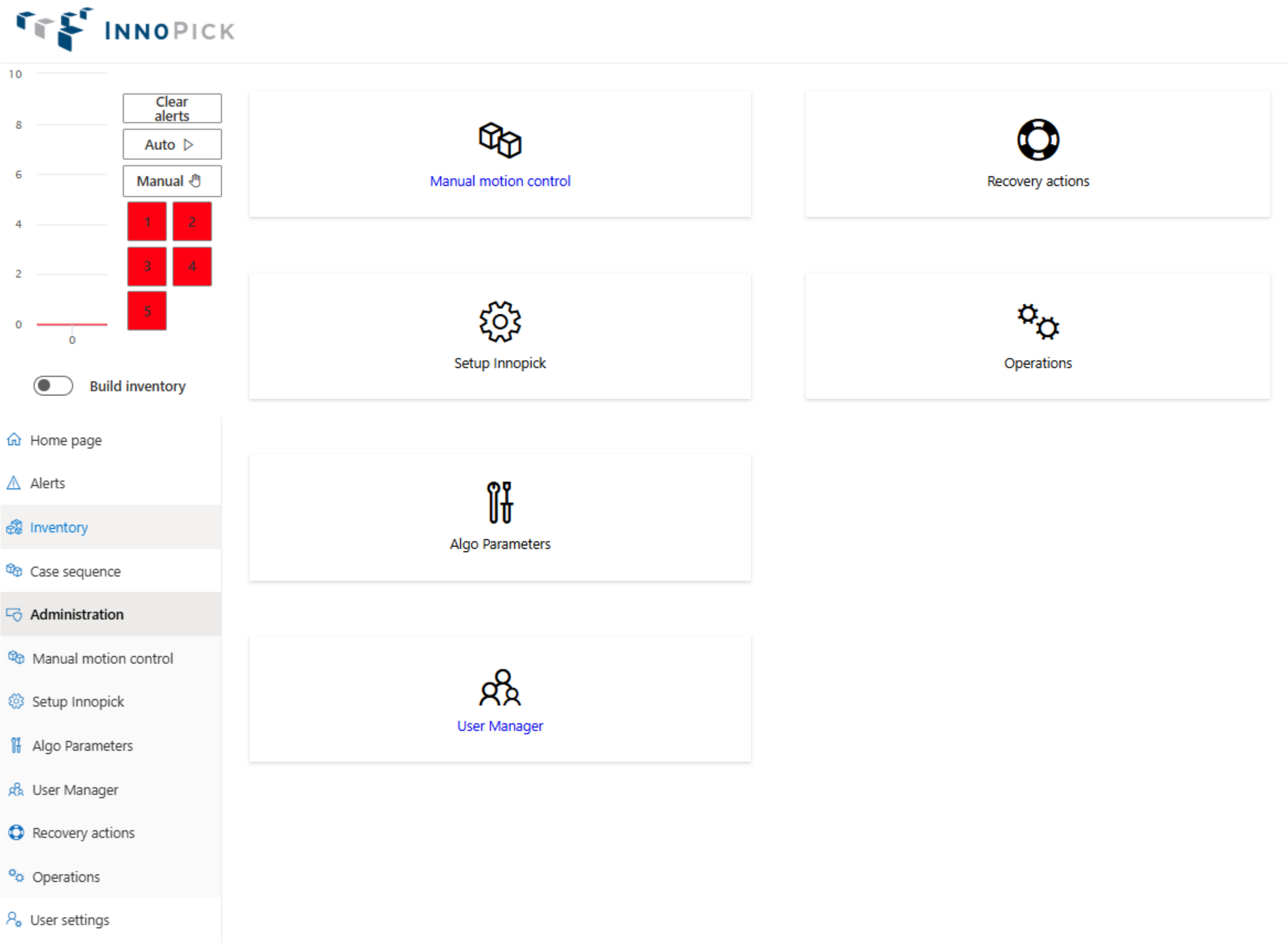
4.6 Administration

Administration

Home > Main Screens > Administration

Overview

The Administration section contains system configuration, advanced troubleshooting tools, and user management features. Access to these pages is typically restricted to Admin and Manager roles.



Administration Pages

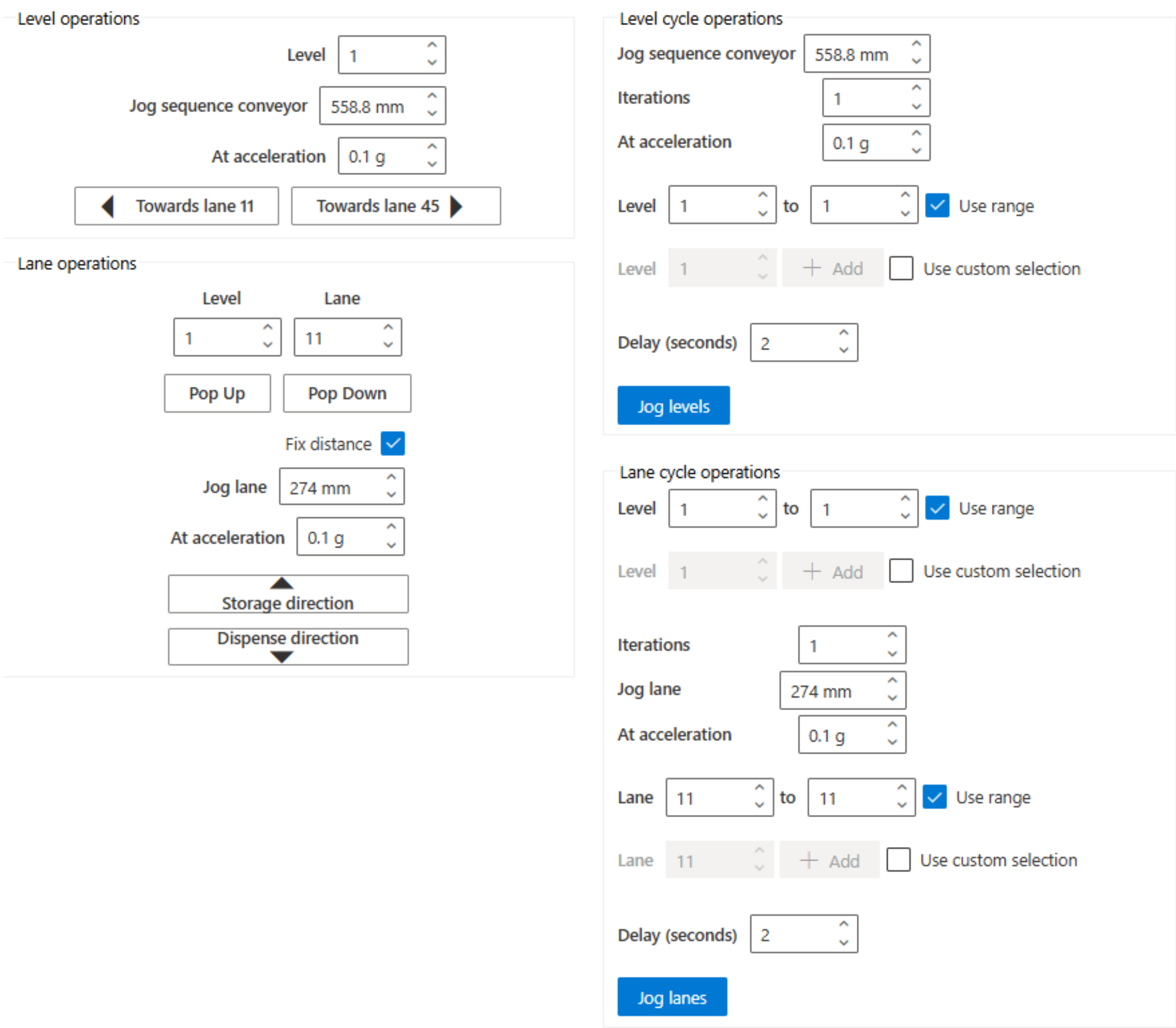
1. **Manual Motion Control** - Manual jog controls for troubleshooting
2. **Setup InnoPick** - System configuration parameters
3. **Algorithm Parameters** - Algorithm tuning

4. **User Manager** - Create and manage user accounts
5. **Recovery Actions** - Advanced troubleshooting tools
6. **Operations** - Level and Merge statuses / controls
7. **User Settings** - Personal preferences

Navigation: ← Case Sequence | Manual Motion Control →

Manual Motion Control

Home > Main Screens > Administration > Manual Motion Control



The Manual motion control page provides manual control of InnoPick conveyors for troubleshooting and maintenance purposes.

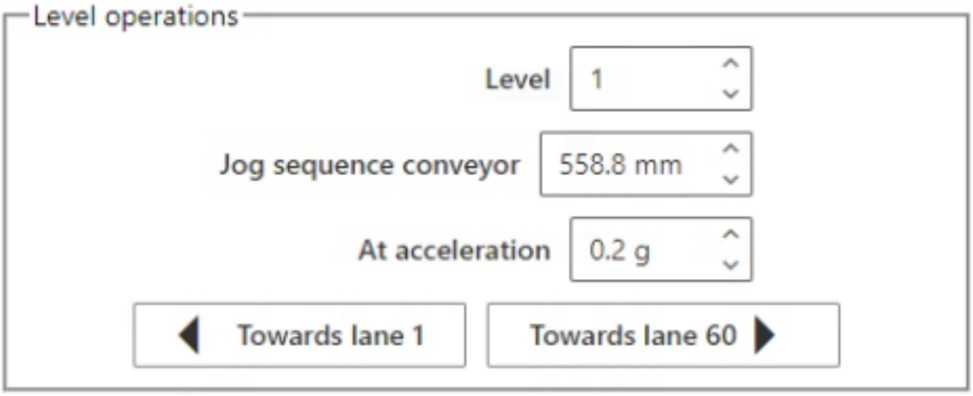
FOUR MAIN SECTIONS

1. Level Operations - Control sequence conveyors
2. Lane Operations - Control individual buffer lanes
3. Level Saga Operations - Pre-program multiple sequence conveyor moves

4. Lane Saga Operations - Pre-program multiple buffer lane moves

---

LEVEL OPERATIONS



Manually jog the sequence conveyor of a selected level.

Controls

Select Level:

- Choose which level to control

Jog Distance:

- Default: 558mm (22 inches) - standard move
- Can be customized as needed
- Measured in millimeters

Acceleration:

- Set in g's (gravitational force)
- Controls how quickly the conveyor accelerates and decelerates

Direction:

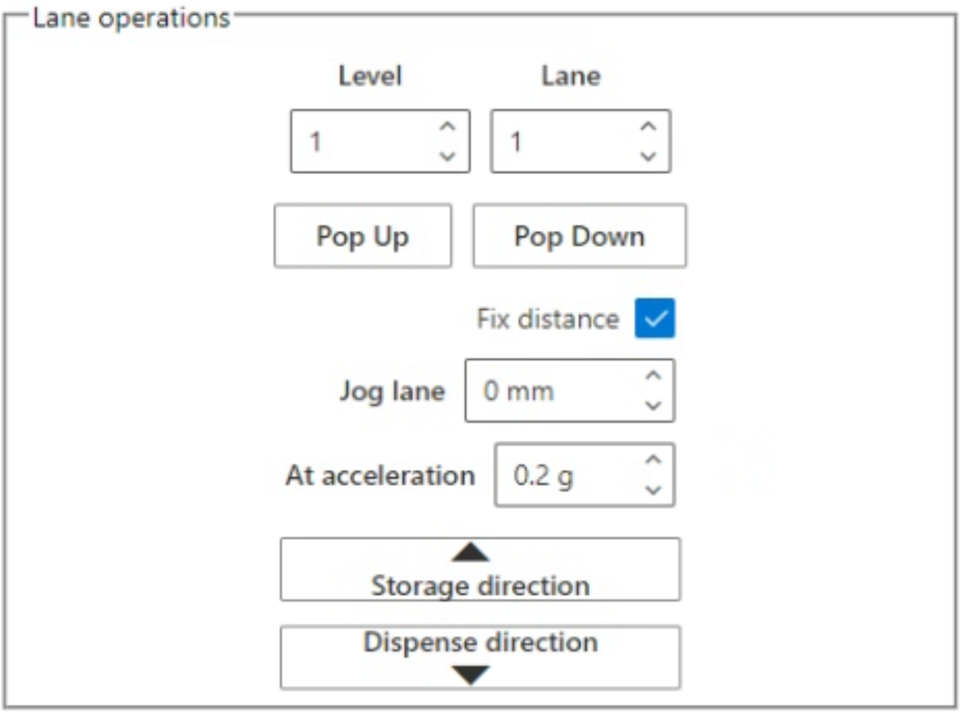
- Forward or reverse
- Typically moves toward outfeed (forward)

When to Use

- Troubleshooting positioning issues
- Manual case positioning after alerts
- Testing conveyor operation
- Maintenance and setup

Important: Only use in Manual mode when production is stopped.

LANE OPERATIONS



Manually control individual storage lanes and clutch assemblies.

Controls

Level and Lane Selection:

- Drop-down menus to select specific lane

Pop Up / Down:

- Forces the lane's pop-up assembly to move up or down
- Used for manual case handling

Distance:

- Default: Width of product in lane (if present), otherwise 0mm
- If "Fix Distance" option is unchecked, the system will automatically set the Jog lane distance to according to the product in the lane (if applicable)

Directions:

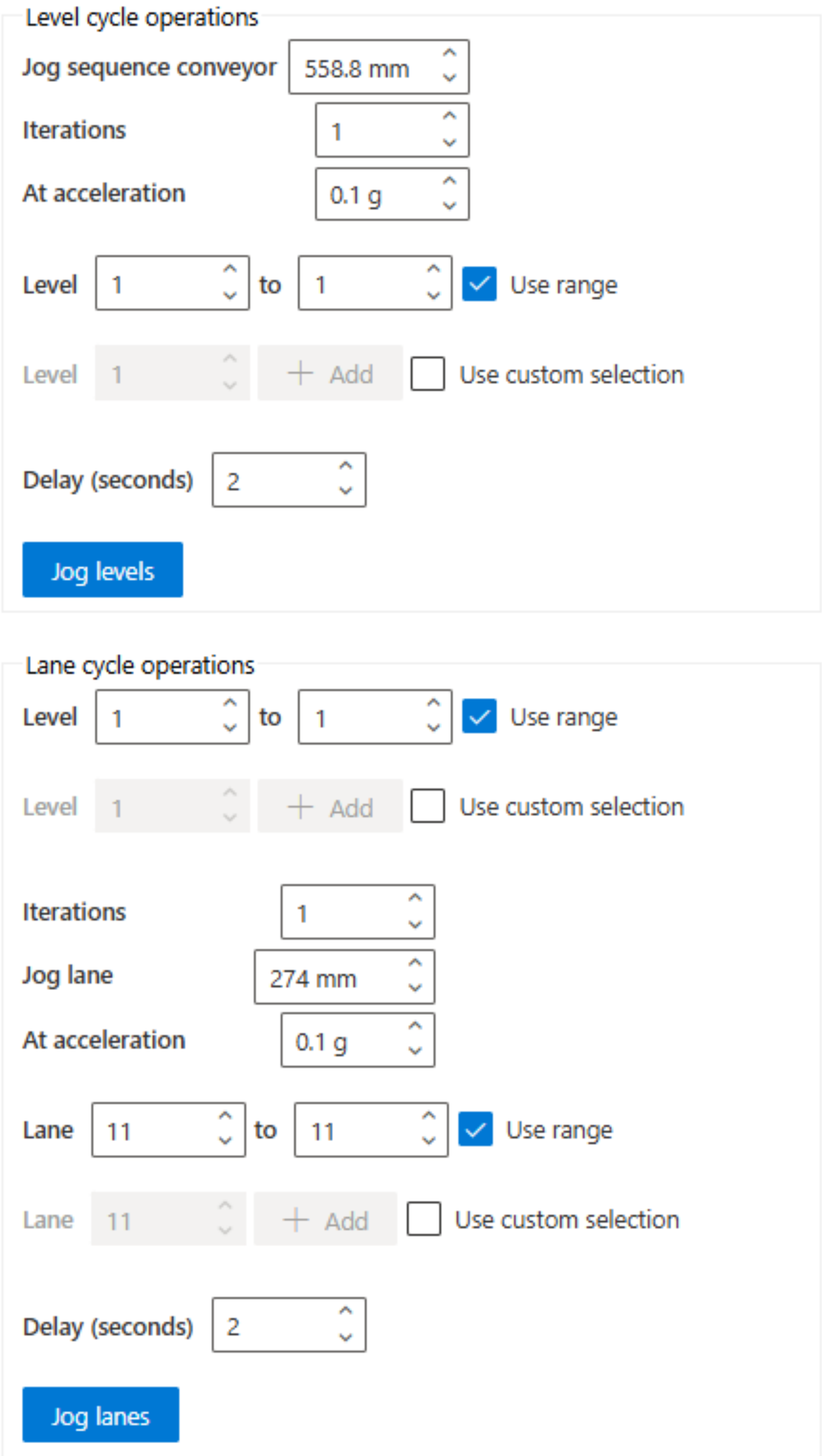
- **Storage**: Move into the lane (backward)
- **Dispense**: Move toward sequence conveyor (forward)

When to Use

- Clearing jammed cases
- Manual case positioning
- Testing lane operation
- Maintenance procedures

Caution: Be aware of case positions before jogging lanes.

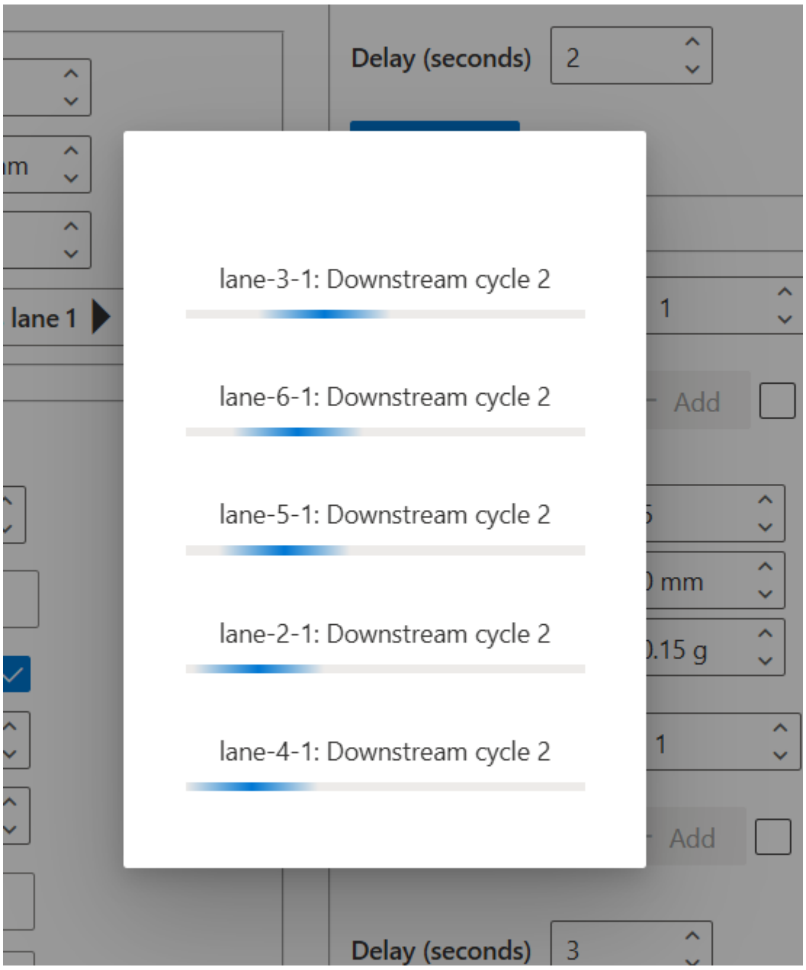
LEVEL & LANE SAGA OPERATIONS



These interfaces allow pre-programming multiple moves in sequence.

To use these controls:

- Enter in the # of iterations and specify the level(s) & lane(s), as well as a delay.
- The delay is necessary to allow the system to complete the manual move before the next move in the sequence is commanded.
- While the manual moves are being executed, the page will display something like this:



Purpose

- Execute a series of moves
- Testing and validation after mechanical work
- Break-in of equipment at installation
- Advanced troubleshooting scenarios

Usage

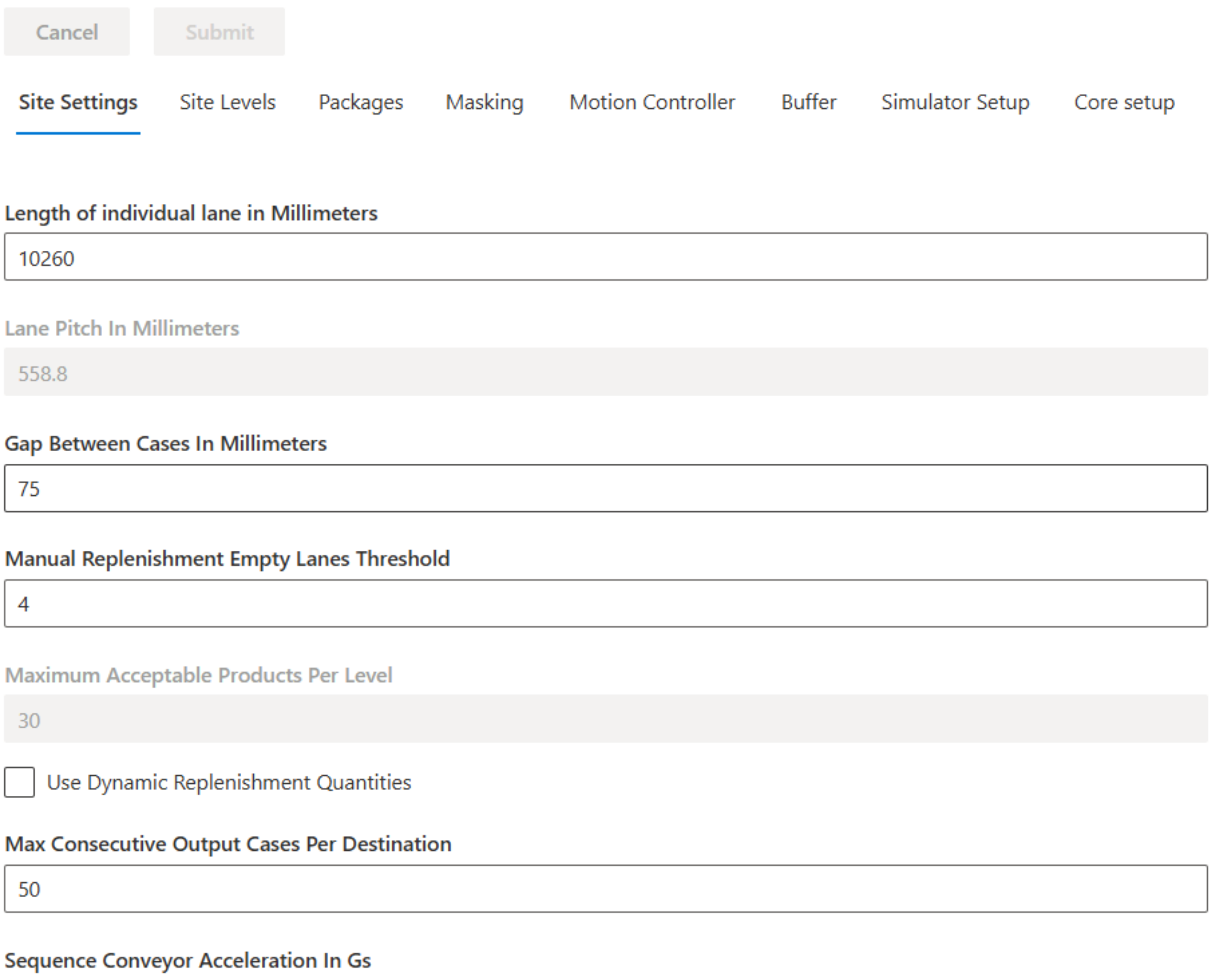
Primarily used by:

- System integrators
- Advanced maintenance personnel
- During commissioning and testing

Navigation: ← Administration | Setup InnoPick →

Setup InnoPick

Home > Main Screens > Administration > Setup InnoPick



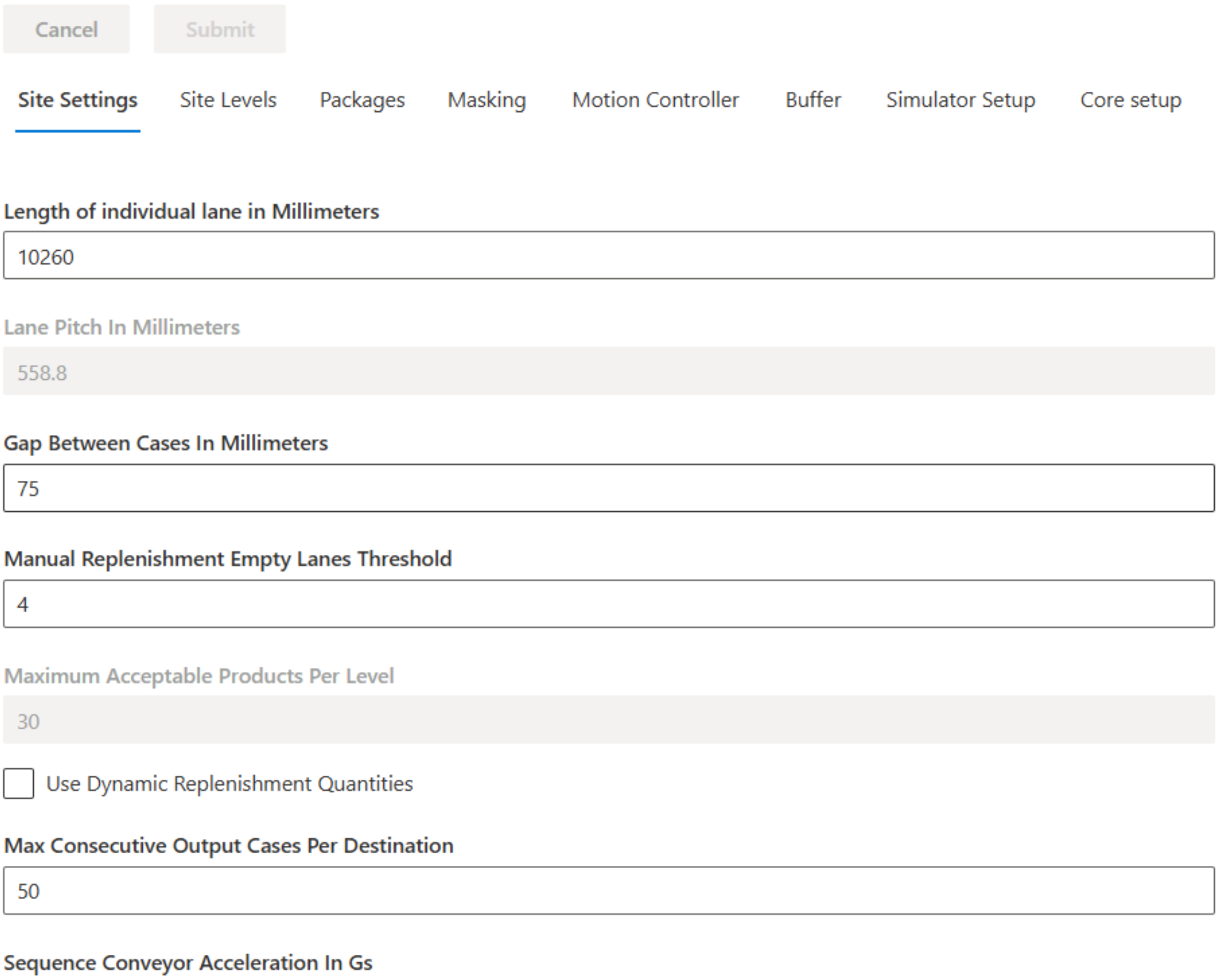
Setup InnoPick Sub-Sections:

- Site Settings
- Site Levels
- Packages
- Masking
- Motion Controller
- Buffer
- Simulator Setup
- Core Setup

The Setup InnoPick page contains configuration settings for InnoPick's internal operation. Access is typically restricted to Admin users and system integrators.

Site Settings

[Back to Setup InnoPick Sub-Sections]



Critical system-wide configuration parameters.

GAP BETWEEN CASES IN MILLIMETERS

The gap InnoPick maintains between cases stored in buffer lanes.

Purpose:

- Prevents cases from pressing together
- Allows smooth dispensing
- Accommodates case size variations
- **Note:** Do not change this parameter unless directed to system expert / remote support.

MANUAL REPLENISHMENT EMPTY LANES THRESHOLD

Minimum number of empty lanes required before IPM allows manual replenishment creation.

Purpose:

- Ensures sufficient space for manual replenishments
- Prevents overfilling storage capacity

USE DYNAMIC REPLENISHMENT QUANTITIES

Enable/disable IPM's ability to dynamically adjust replenishment quantities based on demand patterns.

When enabled:

- IPM analyzes usage patterns
- Adjusts quantities automatically
- Optimizes inventory levels

When disabled:

- Uses static configured quantities
- More predictable but less adaptive

Note: Should be managed by system integrators.

MAX CONSECUTIVE OUTPUT CASES PER DESTINATION

Limits consecutive cases sent to a single destination (Merge A, B, etc.).

Purpose:

- Balances output across destinations
- Prevents overwhelming single destination
- Contingent on other function being enabled

Note: Should be managed by system integrators.

SEQUENCE CONVEYOR ACCELERATION / DECELERATION IN GS

Acceleration rate for sequence conveyor indexing moves.

Measured in: g's (gravitational force)

Considerations:

- Higher = faster moves but more case disturbance
- Lower = gentler but slower throughput
- Balance between speed and stability

ADDITIONAL CONFIGURATION OPTIONS

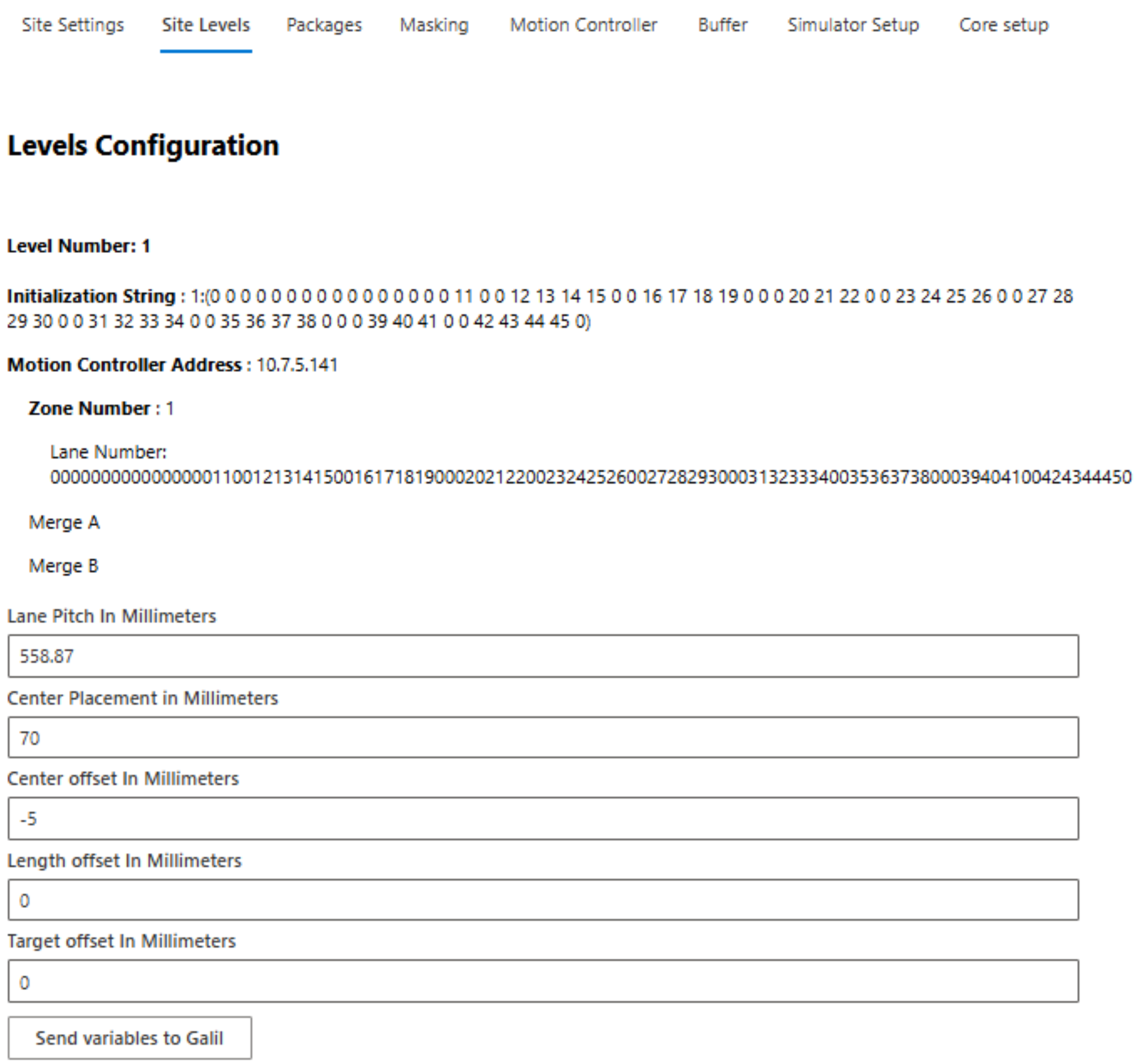
Other settings visible on this page are for:

- Initial system setup
- System integration
- Advanced configuration

Warning: Do not change these without consulting system integrators.

Site Levels

Back to Setup InnoPick Sub-Sections



This page contains level-specific configuration settings:

Lane Pitch in Millimeters
- Value (in mm) representing the standard distance between storage lanes
- **Note**: Values may differ slightly per level due to:

```
- Chain stretch over time
- Tensioning adjustments
- Mechanical variations
```

- **See the Lane Pitch Adjustment Guide for guidance on how to adjust these parameters to ensure good centering on all lanes**

Center Placement in millimeters
- Controls the initial positioning of a case as it first enters InnoPick infeed.
- **See the Lane Pitch Adjustment Guide for guidance on how to adjust these parameters to ensure good centering on all lanes**

Center offset in millimeters
- Adjusts the expected centerpoint of infeed cases.
- If there are midpoint alarms being generated despite the cases looking nicely centered, then this parameter needs to be adjusted.
- Contact system integrators if adjustment is needed.

Length offset in millimeters
- Adjusts the expected length by this amount

Target offset in millimeters
- Adjusts the final case placement when dispensing a case

Send variables to Galil
- Press this after making a change to one of the variables in this section, followed by the 'Submit' button at the top of the page.

Packages

Back to Setup InnoPick Sub-Sections

Product configuration page for system setup.

Access: System integrators during initial setup

Masking

Back to Setup InnoPick Sub-Sections

Configuration for masking certain system moves.

Access: System integrators during initial setup

Motion Controller

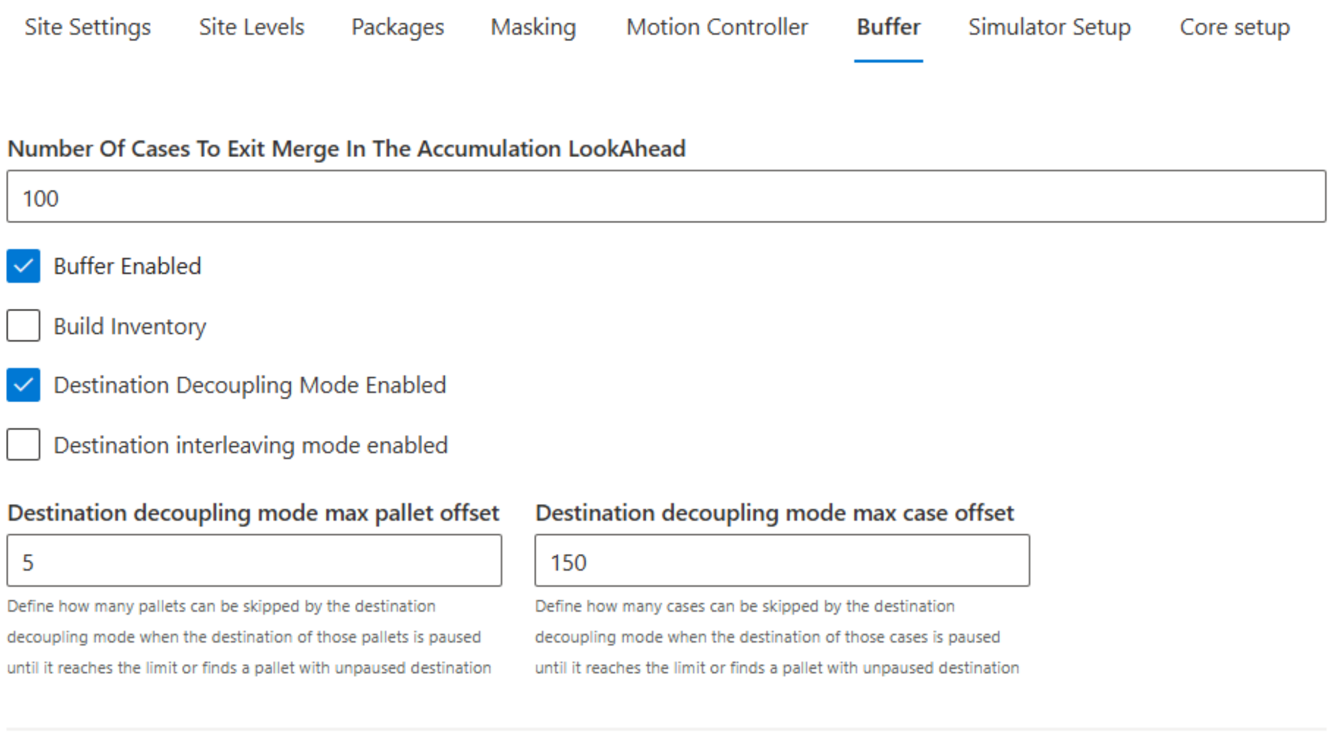
Back to Setup InnoPick Sub-Sections

Connection and configuration for the motion controller.

Access: System integrators during initial setup

Buffer

Back to Setup InnoPick Sub-Sections



Configuration for accumulation management and output buffering.

NUMBER OF CASES TO EXIT MERGE IN THE ACCUMULATION LOOKAHEAD

How many cases IPM considers when forecasting and scheduling output. This is only applicable if Destination Decoupling is Disabled. This # is meant to reprensent how many cases can exit the Merge during the time it takes a case to travel the entire length of InnoPick.

Impact:
- **Higher number**: More aggressive scheduling, uses more accumulation
- **Lower number**: More conservative, leaves accumulation space unused
- **Too high**: Risk of misjudging available space, may block output more often
- **Too low**: Underutilizes accumulation, reduces throughput

Recommendation: Determine empirically based on system performance

---

BUFFER ENABLED

(Default = On)

When enabled, InnoPick keeps track of the cases in the outbound accumulation to optimize outbound scheduling when space is available and pause outbound scheduling to allow inbound to keep entering when the outbound is full.

When enabled:
- IPM actively manages accumulation
- Schedules output based on available space
- Prevents overloading downstream equipment

When disabled:
- IPM schedules output independently of accumulation status
- Simpler but may cause downstream congestion issues

BUILD INVENTORY

InnoPick can operate in two modes regarding accumulation:

Normal Operation Mode

- IPM schedules as many output cases as possible, using Buffer logic if that option (see above) is enabled
- Assumes downstream conveyors advance at nominal rate
- Maximizes throughput
- May occasionally have to stop if outbound flow stops unexpectedly

Build Inventory Mode

- IPM prioritizes building lane inventory over outputting cases
- Only schedules output cases that will definitely fit in current accumulation space
- More conservative approach
- Level should not have to stop due to full accumulation
- Allows continuous replenishment induction

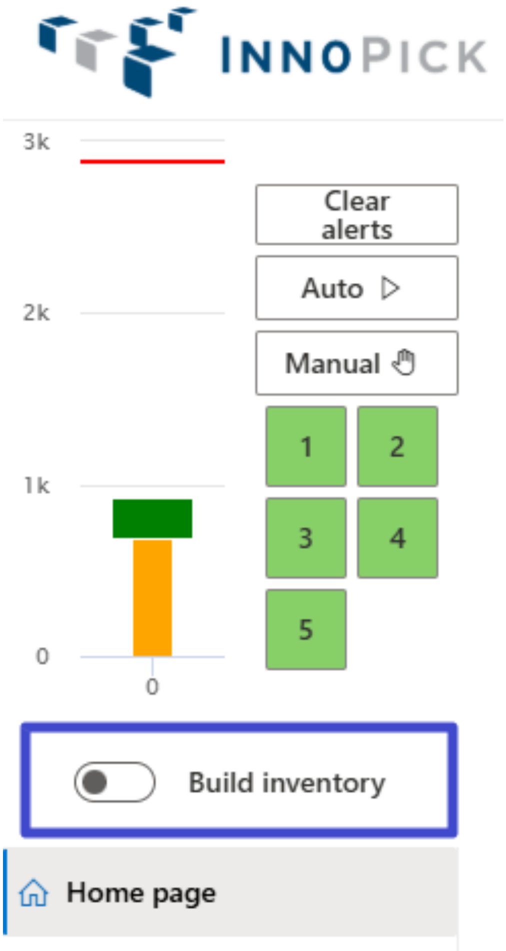
When to enable:

- Starting production with low inventory
- Recovering from downstream stoppage
- Building to nominal inventory levels

When to disable:

- Normal production operations
- When inventory is at nominal or above
- To maximize throughput

Note: Build Inventory Mode can be turned on and off directly from the set of controls that are always visible in the top left corner of the InnoPick Manager screen:



DESTINATION DECOUPLING MODES

Enabled

- Allows InnoPick to schedule cases for multiple destinations in parallel based on downstream accumulation availability and inventory.
- If InnoPick cannot schedule the next outbound case for the next priority pallet, it is allowed to look for pallets further down the sequence for other destinations and attempt scheduling them if accumulation and inventory allows it.
- This mode is affected by the two parameters below:

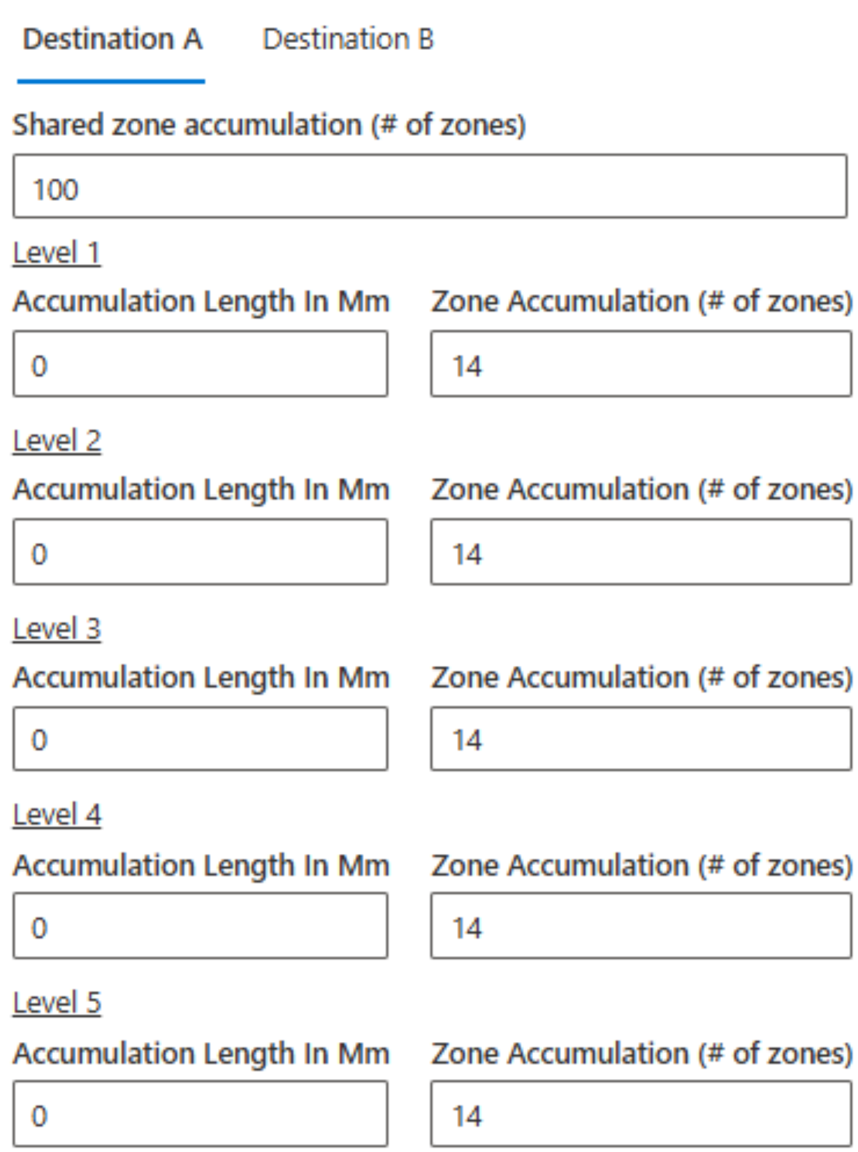
Destination decoupling mode max pallet offset

- This is configuration parameter for the "Destination Decoupling Mode Enabled" above.
- See the description beneath the setting.
- Note: The decoupling level is limited by whichever limit is reached first: # of pallets or # of cases.

Destination decoupling mode max case offset

- This is configuration parameter for the "Destination Decoupling Mode Enabled" above.
- See the description beneath the setting.
- Note: The decoupling level is limited by whichever limit is reached first: # of pallets or # of cases.

DESTINATION A / B ACCUMULATION



These settings teach InnoPick about various accumulation zones for its algorithmic calculations.

Shared zone accumulation (# of zones)

- This setting teaches InnoPick how many cases can fit between the exit of the merge and the palletizer.

Level 1 to 5: Accumulation Length in Mm

- This setting teaches InnoPick the length of low pressure accumulation where cases are accumulated in a way that they bunch up together with no gaps.

Level 1 to 5: Zone Accumulation (# of zones)

- This setting teaches InnoPick how many individual cases can be accumulated between the exit of InnoPick and the merge.

Simulator Setup

Back to Setup InnoPick Sub-Sections

Configuration page for developers and system testing.

Access: System integrators and developers

Core Setup

Back to Setup InnoPick Sub-Sections

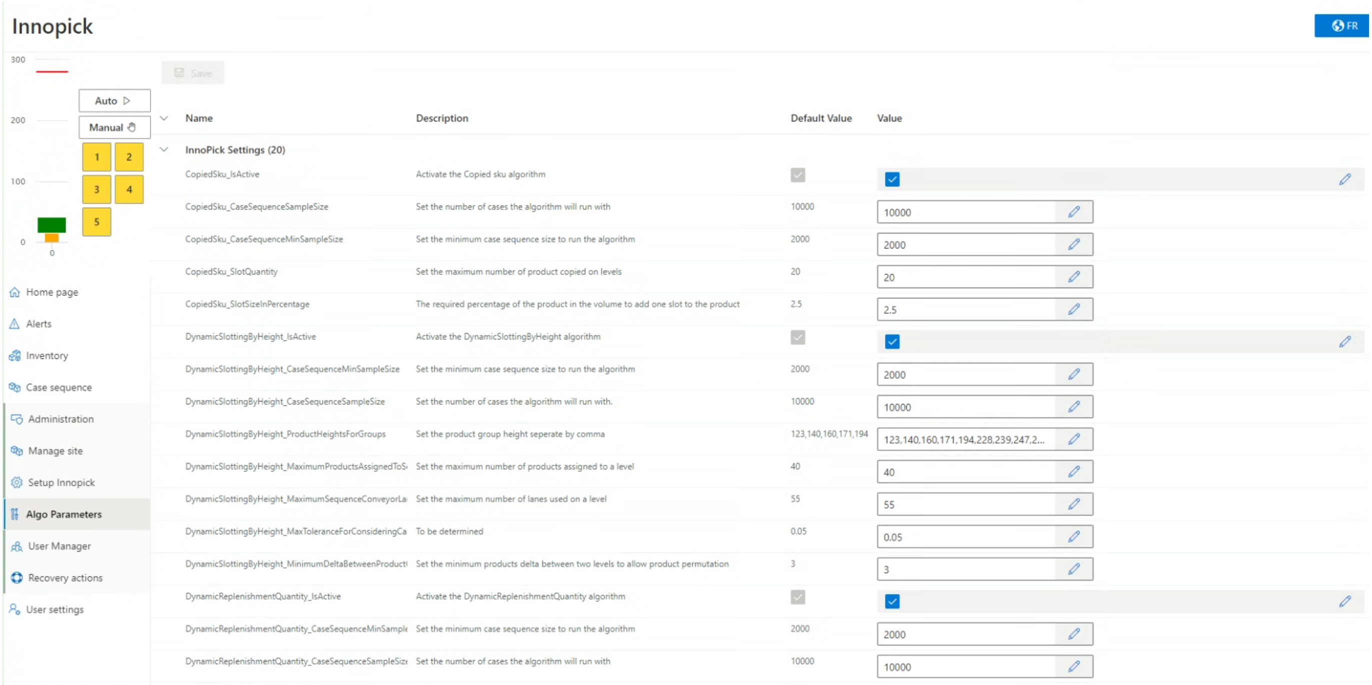
Core system parameters for developers.

Access: System integrators and developers

Navigation: ← Manual Motion Control | Algorithm Parameters →

Algorithm Parameters

Home > Main Screens > Administration > Algorithm Parameters



The Algo Parameters page lists parameters that control various elements of InnoPick's embedded algorithms.

Purpose

- Fine-tune system performance
- Adjust algorithm behavior
- Optimize for specific products or operations

Access Level

Parameters are meant to be altered by:

- System architects
- System administrators
- Personnel with deep understanding of InnoPick operation

Documentation

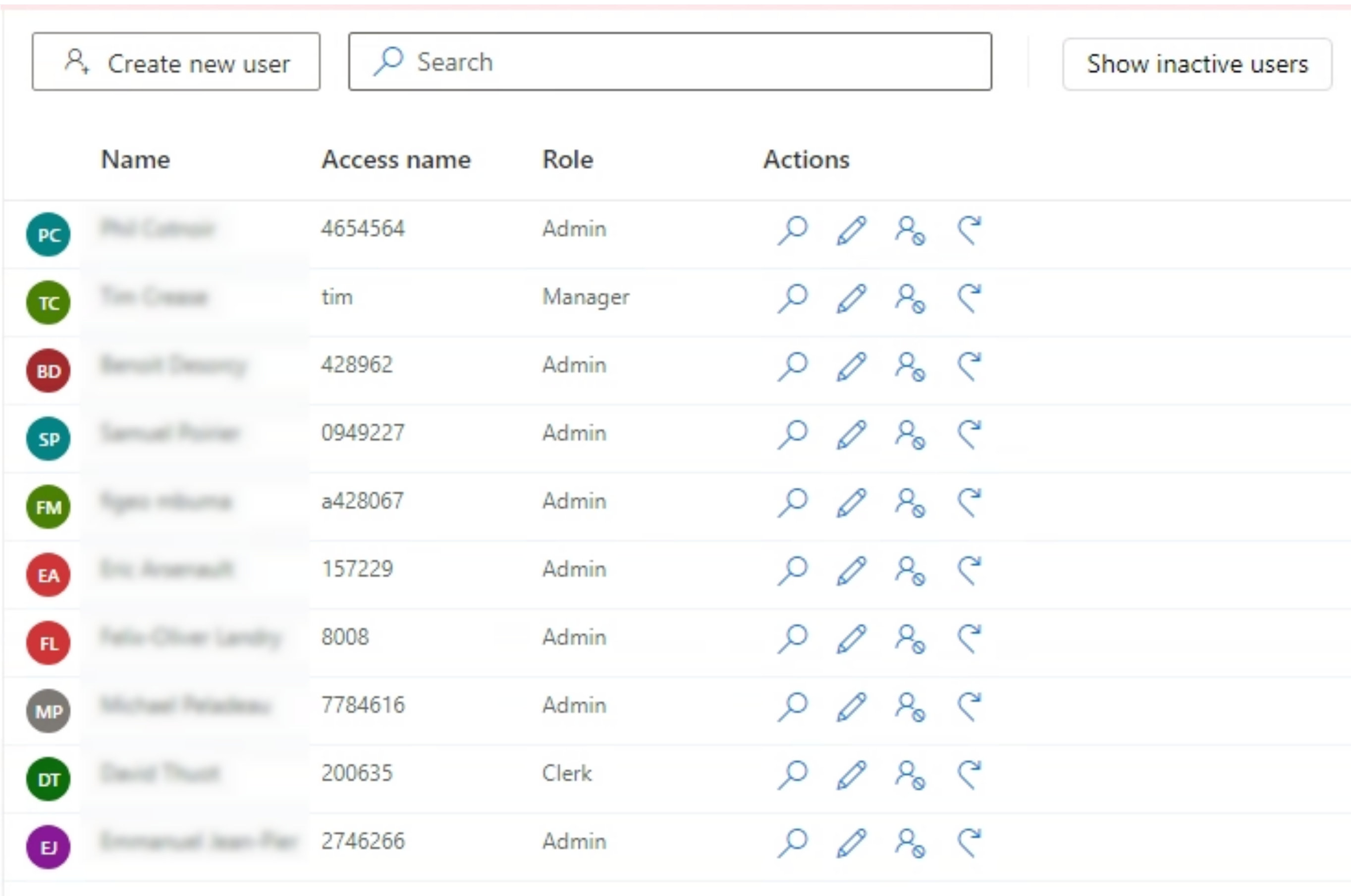
Each parameter includes a description on the page explaining its purpose and effect.

Note: Changes to algorithm parameters can significantly affect system behavior.

Navigation: ← Setup InnoPick | User Manager →

User Manager

[Home](#) > [Main Screens](#) > [Administration](#) > **User Manager**



The User Manager page allows Admin-level users to create and manage user accounts for InnoPick Manager.

User Roles

Each user account is assigned one of these roles:

ADMIN

- Full access to all controls and options
- Can create and manage other user accounts
- Can modify system configuration
- Can access all pages and features

MANAGER

- Access to all operational settings
- Cannot modify site settings or core configuration

- Can manage production operations
- Can resolve alerts and manage inventory

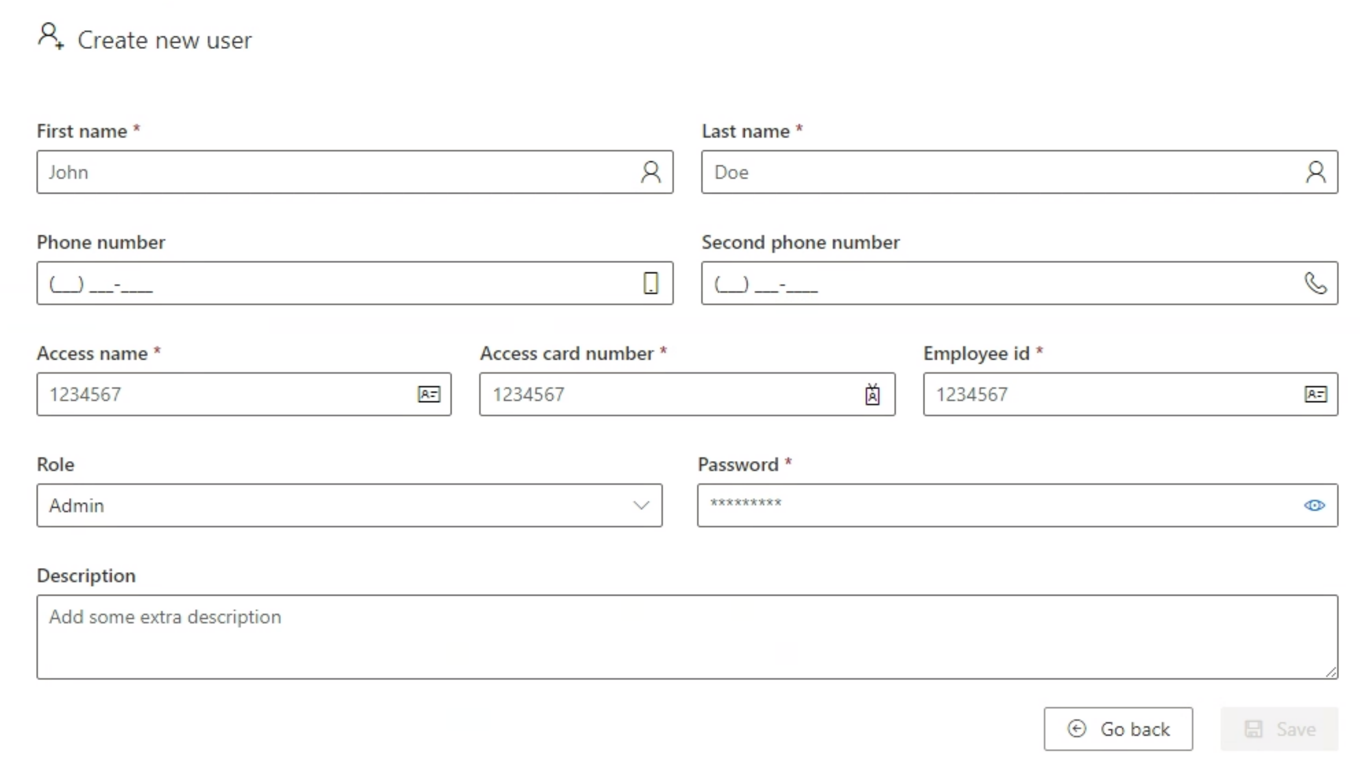
CLERK

- Access necessary to operate system for production
- Cannot change configuration settings
- Can monitor status and clear alerts
- Limited administrative access

TERMINAL USER

- Access only to MixMaster Terminals
- There are no Terminals within InnoPick Manager, but since User Accounts are shared between the systems, the option appears here.

Creating a New User



To create a new user account:

1. Click "Create New User" or similar button
2. Enter required information:
    - **Username**: Login identifier
    - **Password**: Initial password
    - **Full Name**: User's actual name
    - **Email**: Contact email address
    - **Role**: Select from Admin, Manager, Clerk, Terminal User
3. Save the new user

Managing Existing Users

From the User Manager page, you can:

- View all user accounts
- Edit user information
- Change user roles
- Disable or delete accounts
- Reset passwords

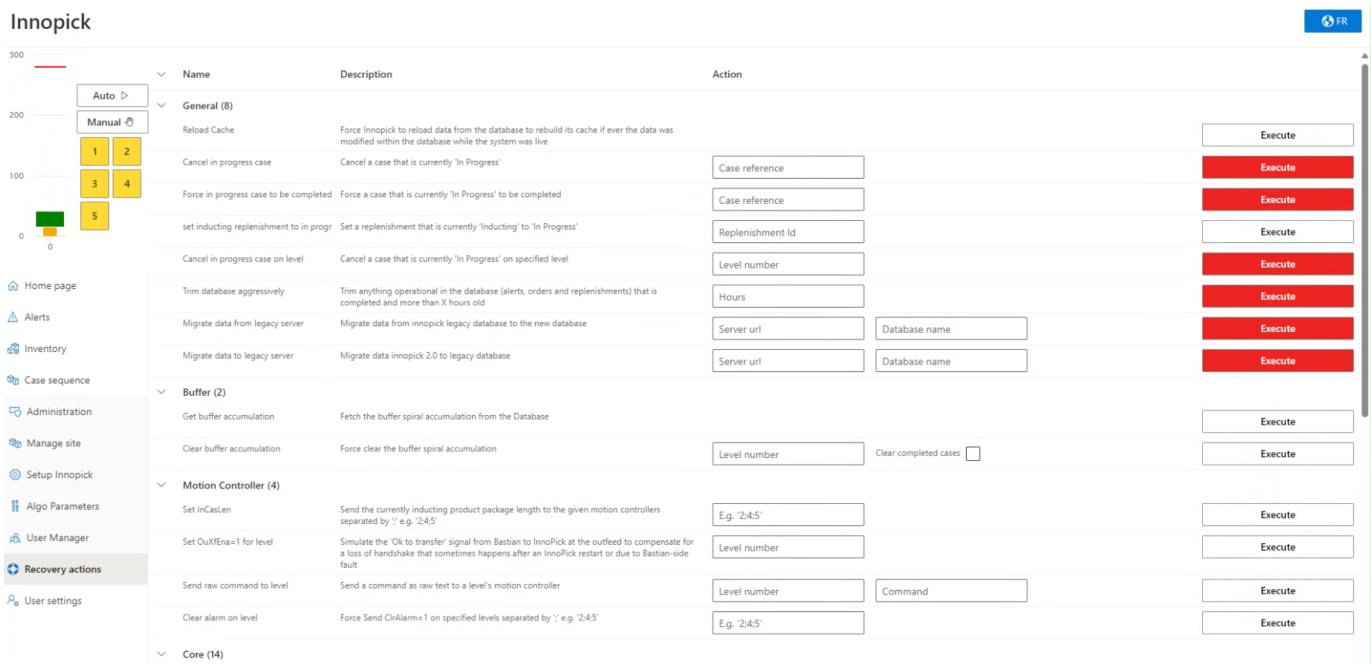
Security Best Practices

- Assign the minimum necessary role for each user
- Review user accounts periodically
- Disable accounts for personnel no longer needing access
- Use strong passwords
- Change default passwords immediately

Navigation: ← Algorithm Parameters | Recovery Actions →

Recovery Actions

Home > **Main Screens** > **Administration** > **Recovery Actions**



Recovery Actions are built-in features for dealing with specific issues that arise during operation.

Purpose

Provide operators and support personnel with:

- Tools for recovering from unusual situations
- Options for resolving issues that can't be handled through normal procedures
- Advanced troubleshooting capabilities

Recovery Action Groups

GENERAL

- System-wide recovery actions
- General troubleshooting tools

BUFFER

- Actions related to accumulation management
- Output buffering issues

MOTION CONTROLLER

- Actions related to conveyor motion
- Drive and motor issues

CORE

- Core system recovery actions
- Advanced system state management

Using Recovery Actions

Recovery actions include a description of what it does and some of them require the operator to put in some required information (in the Action column).

- If a recovery action has an input field for the operator to put in information, it typically will not work unless it has that information.

Important:

- Understand the action before executing
- Document which recovery actions were used
- Consult with support if unsure

Note: The list of available Recovery Actions may change depending on:

- System version
- Integrator customizations
- Specific installation needs

Most Commonly Used Recovery Actions

SET CASE PENDING COMPLETION

This recovery action resends the outfeed case data to the downstream conveyor system.

- This is used when recovering from issues at the outbound transition (handshake), when the downstream conveyor system does not have the tracking data for the case currently on the last InnoPick position.

SET INCASLEN

This recovery action re-sends the case length to the motion controller for the currently inducting product on a given level.

- This is used when after a system restart or some other abnormal stoppage, cases are not entering InnoPick infeed even though everything else is good, and the replenishment is in the *Inducting* state.

SET OUXFENA=1 FOR LEVEL

This recovery action simulates the "OK to Receive" signal from the downstream conveyor system to InnoPick, which is the last step in the outfeed handshake process.

- This can be used when the downstream conveyor system has tracking data for the case, but InnoPick is not advancing the physical case from the last position.

CLEAR ALARM ON LEVEL

This recovery action sends a 'clear alarm' signal to the motion controller for a given level.

- This is used when a fault or alert on the InnoPick system refuses to clear normally, but the underlying conditions that caused the alarm have been resolved (meaning it should be able to clear).

CLEAR LEVEL PROCESSOR ERROR

This recovery action clears internal processor errors and restarts the level processor for a given level.

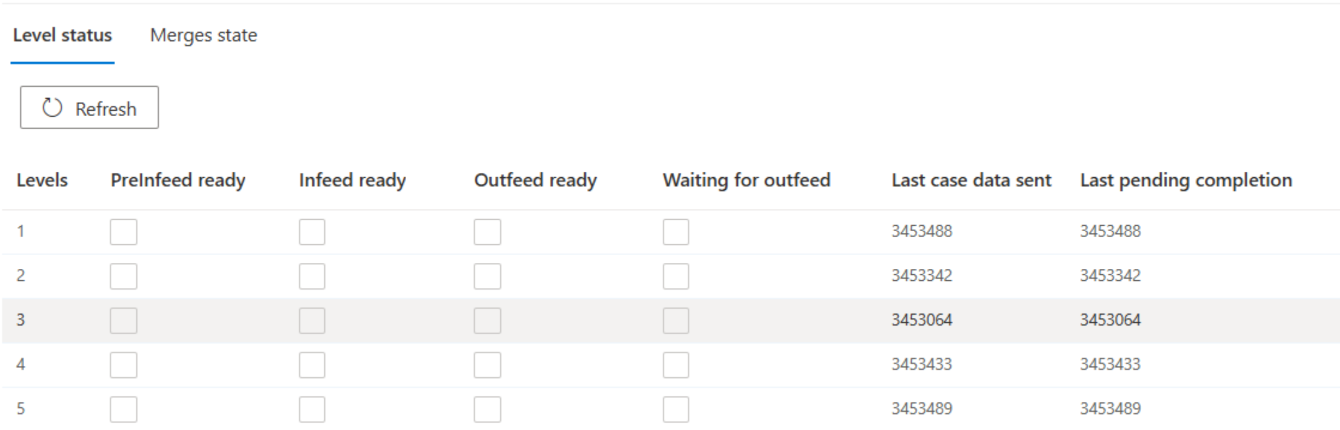
- This is used after certain critical faults whose descriptions indicate to the operator that the level processor has been stopped.

Navigation: ← User Manager | Operations →

Operations

Home > Main Screens > Administration > Operations

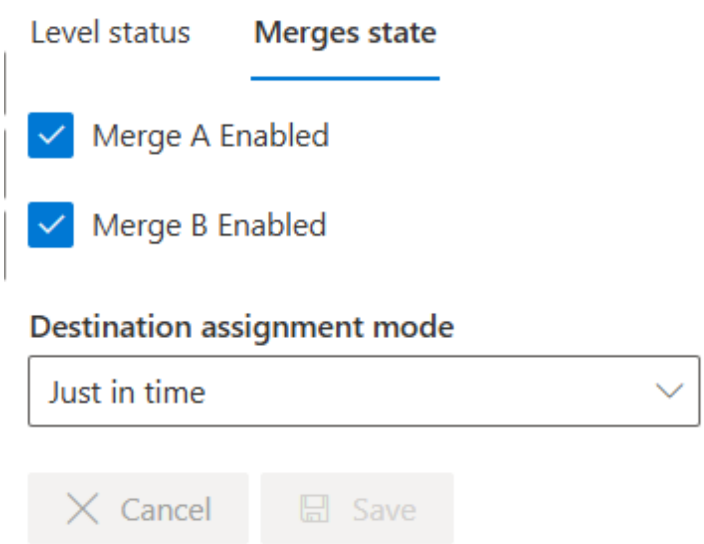
Level status



Levels	PreInfeed ready	Infeed ready	Outfeed ready	Waiting for outfeed	Last case data sent	Last pending completion
1	☐	☐	☐	☐	3453488	3453488
2	☐	☐	☐	☐	3453342	3453342
3	☐	☐	☐	☐	3453064	3453064
4	☐	☐	☐	☐	3453433	3453433
5	☐	☐	☐	☐	3453489	3453489

- This page provides an overview of each level's various statuses as well as:
- Last case data sent (CaseRef #)
- Last pending completion (CaseRef #)

Merges state



Configuration for downstream merge points (where the different levels combine to a single conveyor that feeds one of the palletizers).

MERGE A / B ENABLED

Enable or disable individual downstream destinations.

Purpose:

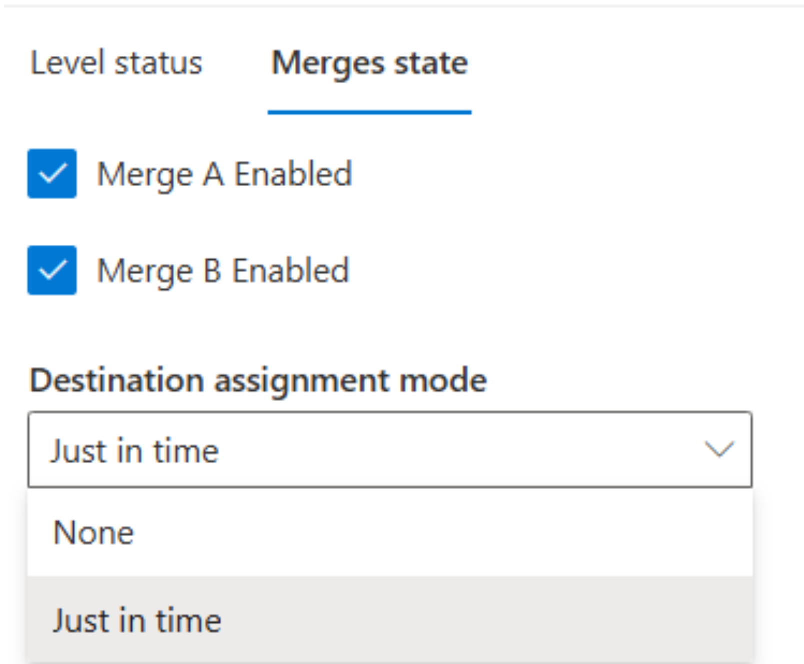
- Route all cases to operational Merge if one fails
- Handle downstream equipment failures
- Temporary operational adjustments

Usage:

- Last resort when downstream fault cannot be resolved
- Will not affect cases already on sequence conveyors -- those cases that are already scheduled for a certain destination will retain their destination.
- High likelihood that remote support interventions will be required if this is changed during production.

Caution: *Only use when absolutely necessary and coordinate with support. Can have unintended effects and complications.*

DESTINATION ASSIGNMENT MODE



There are two possible settings for the destination assignment mode:

1. None
2. Just-in-time

None: This is the regular destination mode:

- InnoPick will initially load balance pallet assignment when receiving a new case sequence
- Will never change a destination unless a destination is disabled

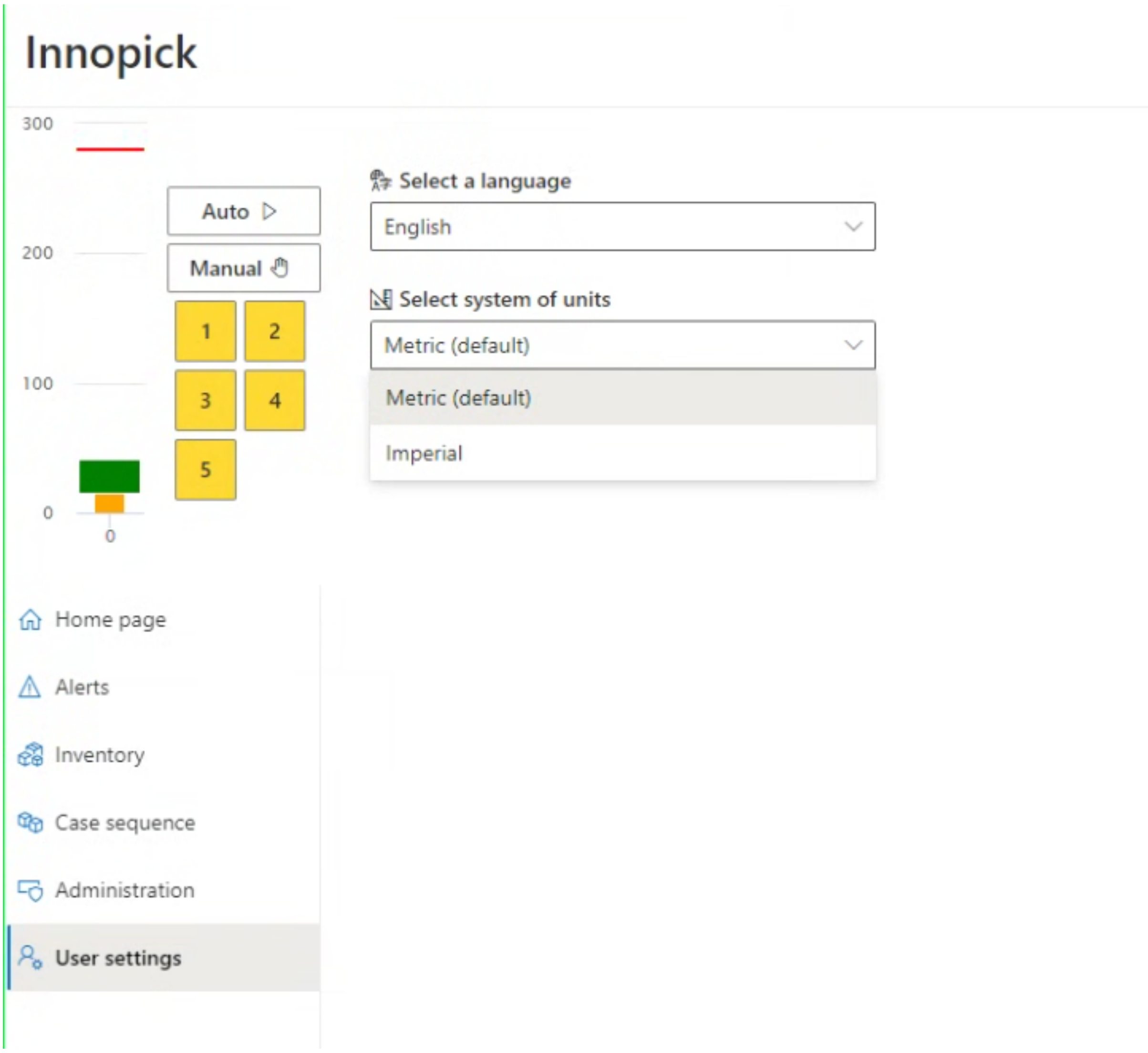
Just in time:

- InnoPick will initially load balance pallet assignment when receiving a new case sequence (like normal)
- InnoPick may change the destination when the first case of a pallet gets scheduled, based on the remaining accumulation available to load balance assignments "Just-in-Time"
- In this mode, InnoPick responds dynamically to current accumulation levels and assigns pallets to the destination with the most capacity.

Navigation: ← Recovery Actions | User Settings →

User Settings

Home > Main Screens > Administration > User Settings



The User Settings page contains personal preferences that individual users can configure.

Available Settings

Settings include:

- Language preferences (if applicable)
- System of units (metric vs imperial)

Accessing User Settings

- Click **User Settings** in the left navigation menu
- Settings are specific to your user account

Related Topics

- Monitoring Production - Using admin tools during operation
- Troubleshooting - When to use recovery actions
- Products Page - Product configuration

Navigation: ← Operations | Troubleshooting →

5. Troubleshooting

5.1 Troubleshooting

Home > Troubleshooting

Overview

This section provides guidance for resolving alerts, troubleshooting common issues, and recovering from problems during InnoPick operation.

Troubleshooting Resources

Guidelines for Resolving Alerts

General principles and best practices for resolving alerts successfully, including:

- Understanding level status colors
- General guidelines for operating InnoPick
- When alerts indicate bigger issues
- Important system settings that affect behavior

Alerts Reference

Complete listing of all InnoPick alerts with:

- Alert code and description organized by ranges
- Explanation and common causes
- Step-by-step resolution procedures
- Quick navigation by alert code ranges

Common Issues

Frequently encountered problems and their solutions:

- Outfeed faults
- Infeed issues
- Sequencing problems
- Performance issues

Case Replacement

Procedures for replacing broken, missing, or incorrect cases during operation.

Case Centering Adjustments

Procedures for adjusting the centering of cases on a given level.

Deadlocks

Explanations of and guidance for resolving deadlock situations.

Disabling an Active Lane

Guidance on how to handle disabling a lane that has incoming replenishment cases or already-scheduled output cases.

Quick Reference by Symptom

System won't start:

→ Check for E-Stop condition
→ Verify no unresolved or hidden alerts

Alert won't clear:

→ See Alert Guidelines
→ Verify physical issue is resolved
→ Use Recovery Action to send a clear alarm to the motion controller.

Case missing from sequence:

→ See Case Replacement
→ Check Case Sequence Page

Repeated same alert:

→ See Alert Guidelines
→ Check for underlying issues

Photocell fault:

→ See Alert Reference codes 109, 119, 121, 123, 709

Clutch fault:

→ See Alert Reference codes 310, 333, 334, 340

Infeed problem:

→ See Alert Reference codes 512, 515, 517

Troubleshooting Approach

Step 1: Identify the Problem

1. **Check the Alerts Page** for current alerts
2. **Note the alert code and message**
3. **Identify which level** is affected
4. **Determine if this is** a new issue or recurring problem

Step 2: Understand the Situation

1. **Read the alert description** in Alert Reference
2. **Check the Home Page Inventory Graph** - does it match physical reality?
3. **Consider recent events** - what was happening when the alert occurred?
4. **Look for patterns** - have similar alerts occurred recently?

Step 3: Resolve the Issue

1. **Follow the resolution steps** in the Alert Reference
2. **Verify physical conditions** match expected state
3. **Make corrections** as needed
4. **Clear the alert** only when underlying issue is resolved

Step 4: Verify Resolution

1. **Resume operation** (put level back in Auto)
2. **Monitor closely** for recurrence
3. **Report persistent problems** to maintenance

Safety Reminders

- **Always use E-Stop** if there's immediate danger to personnel or equipment
- **Never bypass safety interlocks** or disable safety devices
- **Lock out equipment** before entering guarded areas
- **Follow lockout/tagout procedures** for maintenance work
- **Report safety concerns** immediately to supervision

Prevention Tips

Reduce Alerts Through Good Practices

- Perform visual inspections during startup
- Monitor production actively as described in Monitoring
- Address minor warnings before they become critical

- Keep inventory graphs matched to physical reality
- Maintain clear communication during shifts
- Follow proper shutdown procedures

Maintain System Health

- Report unusual sounds or movements immediately
- Note patterns in alerts and report to maintenance
- Keep documentation updated
- Follow scheduled maintenance procedures
- Address root causes, not just symptoms

Related Topics

- Daily Operations - Preventing issues through proper procedures
- Alerts Page - Viewing and filtering alerts
- Home Page - Monitoring system status
- Administration - Recovery actions for advanced issues

Navigation: ← User Settings | Alert Guidelines →

5.2 Common Issues

Home > Troubleshooting > Common Issues

Overview

This page addresses frequently encountered issues that may not have specific alert codes but require operator attention and intervention.

Outfeed Issues

Cases Not Exiting InnoPick

Symptoms:

- Cases stuck at outfeed position
- Output has stopped
- Accumulation showing as full
- Levels remain in Auto but no output

Common Causes:

- Downstream conveyor system stopped or faulted
- Accumulation space completely full
- HOLD active on outbound conveyor system
- Communication error with downstream equipment
- Photocell at exit blocked

Solution:

1. **Check downstream equipment status:**
    - Check for faults on outbound conveyor HMI
    - Look for HOLDs or stopped conditions
2. **Check Inventory Graph Outfeed statuses:**
    - On the Home Page, look at the output statuses
3. **Coordinate with downstream operators:**
    - Communicate the issue
    - Verify when downstream will be ready
    - May need to wait for accumulation to clear
4. **Check InnoPick settings:**
    - Check if Pause Outfeed is active
    - Verify Buffer settings
5. **If issue persists:**
    - If no explanation can be found for the stoppage, contact support

---

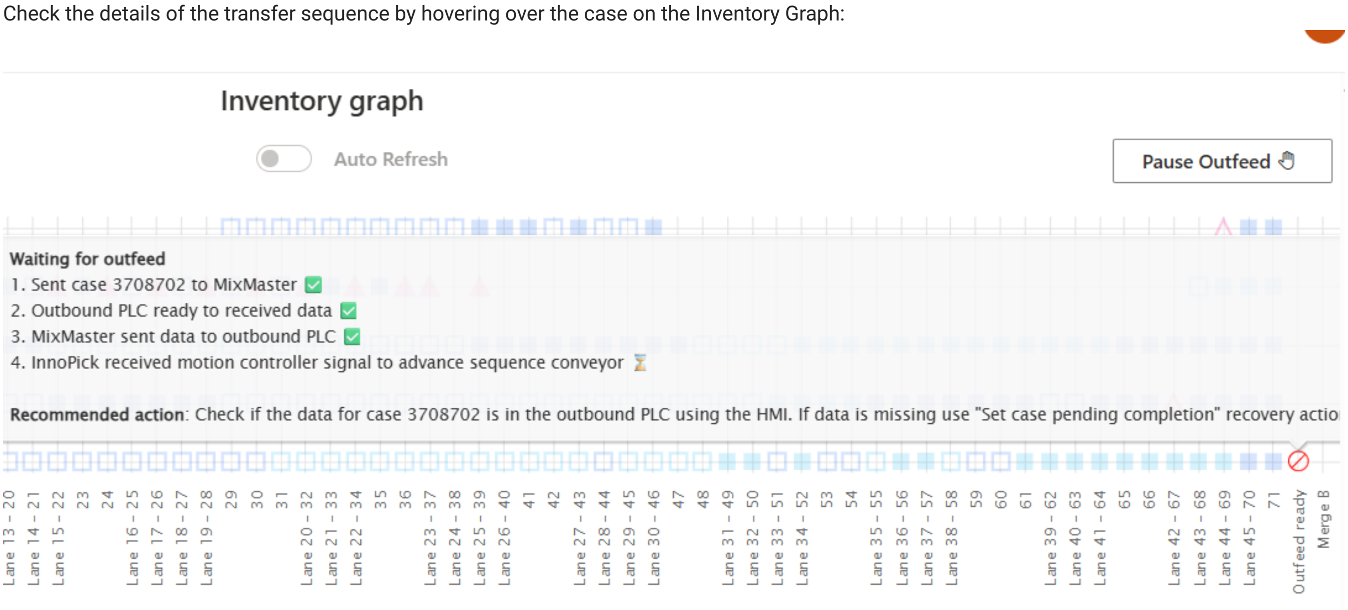
Next Outfeed Case Data is not in Outbound Conveyor System

Symptoms:

- The case at the outfeed position of InnoPick is waiting for the "Ok to Receive" signal from the downstream conveyor system
- It is determined that the case's data has not been received by that system

Solution:

- Check the details of the transfer sequence by hovering over the case on the Inventory Graph:



The steps for each case transfer are as follows:

1. Send case to MixMaster
    - This is done automatically when the case arrives at the last InnoPick position
2. Outbound PLC ready to receive data
    - InnoPick checks to see if the outbound PLC is ready for new case data
3. MixMaster sends data to outbound PLC
    - MixMaster sends case data to the outbound PLC
4. Outbound PLC sends the "Ok to Receive" signal to InnoPick
5. InnoPick advances the case and transfers it to the outbound conveyor system.

Additional Info:

- The pop-up which appears when the cursor hovers over the case on the outfeed position shows what steps have been completed to date for this case.
- If step 3 has a checkmark next to it (as in the screenshot above), then MixMaster has already sent the case data to the outbound PLC, and it will not re-send it unless a recovery action is used.
- Therefore, if the outbound PLC does not have the case data, the operator can use the recovery action **"Set case pending completion"**

Note: This recovery action re-sends the case data to the outbound PLC. If the data was already there, the downstream system will have a duplicate data which can cause serious tracking issues. Use with caution and monitor downstream system for any issues after using.

Next Outfeed Case not advancing despite all conditions met

Symptoms:

- If the outbound PLC does have case data for the next outfeed case
- And the Inventory Graph shows that all steps have been completed for the case transfer
- And all other likely causes have been ruled out (system is not in manual, not faulted, downstream conveyor is not full, faulted, or held, etc)
- But the case still does not advance

Solution:

- It may be necessary to force the case forward using the recovery action called **"Set OuXfEna=1 for level"**

Note: This recovery action will force the sequence conveyor to move forward. If it is used incorrectly or in the wrong scenario, it will make the situation worse. Use with caution.

Infeed Issues

Cases Not Being Inducted

Symptoms:

- Replenishments show "Inducting" but cases not entering InnoPick infeed
- Upstream conveyor has cases but won't release them

Common Causes:

- HOLD active on inbound conveyor system
- No Replenishment in the **Inducting** status
- InnoPick at capacity (all lanes full) on that level
- Communication / handshake error between systems
- **Pause Infeed** enabled in InnoPick
- If following a system reboot, a recovery action is needed

Solution:

1. **Check upstream conveyor system:**
    - Verify no HOLD is active
    - Check HMI for faults or stopped conditions
    - Verify that cases are not physically jammed
    - Confirm cases have correct tracking data and are ready to release
2. **Check InnoPick capacity:**
    - Review Lane Inventory
    - Verify lanes available for induction
3. **Check InnoPick settings:**
    - Verify Pause Infeed is not active on the Home page
4. **Check Replenishments Status on that level**
    - Go to the Replenishments Page
    - Filter for the level in question
    - Make sure the replenishment associated with the cases at the infeed is in the state **Inducting**
    - If not, investigate why, and then manually set the replenishment to **Inducting**
    - See here for more info on the Replenishments Page
5. **If a system or motion controller reboot occured recently:**
    - Use the recovery action Set InCasLen for the level in question.

Wrong Product Being Inducted

Symptoms:

- Alert 517 (incorrect case length)
- Cases don't match expected replenishment
- Lane contains mixed products

Common Causes:

- Operator error in selecting pallet
- Tracking data mismatch in inbound conveyor system
- Upstream case routing mixup
- Cases were removed without updating upstream conveyor system or the InnoPick replenishment

Solution:

1. **Stop induction immediately:**
    - Put level in Manual mode
    - Prevent more wrong cases from entering
2. **Verify what was expected:**
    - Check Replenishments Page
    - Note expected product SKU
    - Check inducted quantity
    - Check recently filled lanes for incorrect inventory
3. **Coordinate with upstream operators:**
    - Inform depalletizer operator or appropriate personnel
4. **Resolve the issue:**
    - Remove wrong cases from InnoPick
    - Adjust replenishment quantities as needed
    - Adjust buffer lane inventory as needed
    - Correct upstream tracking if needed
5. **Resume InnoPick production**
    - Pay close attention to the first transitions between replenishments

---

Inventory Tracking Not Matching Physical Reality

Symptoms:

- Alert 121 or Alert 123 (cases not found where expected)
- Outfeed alerts (expecting case but none present, or vice versa)
- Lane inventory counts don't match physical count

Common Causes:

- Improper alert recovery
- Skipped verification after clearing alerts
- Cases manually moved without updating system
- Motion controller thread crash (Alert 791)

Solution:

1. **Put level in Manual mode**

1. **Conduct physical inventory check:**
    - Compare each affected lane to Lane Inventory Page
    - Compare sequence conveyor to Inventory Graph
    - Fix all discrepancies
2. **Correct lane inventory:**
    - Use Edit Lane Inventory feature
    - Adjust quantities to match physical reality
    - Update product types if wrong
3. **Correct sequence conveyor tracking:**
    - May need to jog conveyor to align cases
    - Remove or add cases as needed to match Inventory Graph
    - Follow Case Replacement procedures
4. **Clear all alerts** after corrections
5. **Resume in Auto and monitor closely:**
    - Watch first several cycles carefully
    - Verify tracking stays accurate
6. **Prevent recurrence:**
    - Always verify Inventory Graph after clearing alerts
    - Never assume - always check
    - Train operators on proper procedures

Frequent Alerts on Specific Products

Symptoms:

- Certain SKUs causing repeated alerts
- Same product repeatedly has storage/dispense issues
- Product-specific fault patterns

Common Causes:

- Case dimensions incorrectly configured
- Case quality issues (weak, damaged, irregular)
- Acceleration settings too aggressive for product

Solution:

1. **Review product configuration:**
    - Check acceleration settings in the Products Page
    - Verify dimensions are correct in MixMaster product configurations (length, width, height)
2. **Inspect physical cases:**
    - Look for damage, weak boxes, irregular shapes
3. **Adjust settings:**
    - Lower acceleration if cases are unstable
    - This is modified in the **Products Page**
4. **Test and monitor:**
    - Monitor results over several cycles
5. **Consider removing product from InnoPick**
    - If the product is very problematic such that it is slowing down the entire production and the problems cannot be resolved by parameter changes, it may be better to remove the product from automation.

Related Topics

- Alert Reference - Specific alert code resolutions
- Alert Guidelines - General troubleshooting principles
- Monitoring Production - Catching issues early
- Case Replacement - Replacing problem cases

Navigation: ← Alert Reference | Deadlocks →

5.3 Alert Guidelines

Home > Troubleshooting > Alert Guidelines

Overview

This page provides general principles and best practices for successfully resolving InnoPick alerts. Following these guidelines will help you avoid cascading faults and minimize downtime.

Understanding Level Status Colors

When alerts occur, InnoPick changes the level status indicators to show the severity and type of issue.

Level Color Meanings

- **Green**: Automatic mode - system is operating normally
- **Yellow**: Manual mode - system is paused
- **Red**: Fault mode - system has an active alert that stopped operation
- **Dark Red**: E-Stop active - safety system engaged (door open or E-Stop pressed)
- **Blue**: Communication Loss

Alert Progression

When an alarm occurs:

1. InnoPick stops the affected level
2. Level enters Fault mode (Red) or E-Stop status (Dark Red)
3. Alert appears on Home Page and Alerts Page

After clearing the alarm:

1. Level goes to Manual mode (Yellow)
2. Operator can verify conditions are correct
3. Level can be put back into Automatic mode (Green)

Six General Guidelines for Operating InnoPick Successfully

Following these guidelines will prevent most cascading faults and reduce troubleshooting time.

Guideline 1: Always Compare Physical Reality to the Inventory Graph

When responding to an Alert, always verify physical cases match their position on the Inventory Graph.

The Inventory Graph is the gold standard for what should be where. Physical cases should ALWAYS match the graph.

WHY THIS MATTERS

- A good operator always double-checks visuals after clearing faults
- This is the absolute best way to avoid knock-on faults from incorrect recoveries
- Most cascading faults occur because this step was skipped

Guideline 2: After Store/Dispense Alerts, Check Physical Case Position

Always verify where the case actually ended up.

AFTER A STORE FAULT:

- Case should be in the storage lane
- Verify it's not blocking the photocell
- Confirm it's properly positioned

AFTER A DISPENSE FAULT:

- Case should be on the sequence conveyor
- Verify it cleared the photocell
- Confirm proper spacing to adjacent cases

IF IN DOUBT, CHECK THE LANE CASE COUNT (INVENTORY).

This is a general guideline, but remember Guideline #1 - the Inventory Graph is your ultimate reference.

Guideline 3: Check Replenishment Status After Any Infeed Alarm

Infeed alarms require extra vigilance about several details.

Infeed alarms are confusing because they occur where two different systems connect. Problems can originate on either side, and if not properly identified and corrected, more issues will follow.

AFTER EVERY INFEED ALARM, CHECK:

1. **Product (SKU) is correct**
    - Case matches what was expected
    - Replenishment is for the right product
2. **Number of cases inducted (or yet-to-be inducted) is correct**
    - Go to Replenishments Page
    - Verify inducted quantity
    - Check that 'quantity to induct' is accurate
3. **Upstream conveyor system has correct tracking data**
    - Cases are properly tracked upstream
    - No missing or extra cases in upstream system
4. **If cases had to be moved, follow Guideline 1**
    - Resume in Automatic mode for a few moves
    - Verify case placement on sequence conveyor
    - Ensure inventory graph matches physical reality

Guideline 4: Determine if Alert is Stand-Alone or a Symptom

Consider whether the alert is an isolated incident or indicates a bigger issue.

SYMPTOM ALERTS

Certain alerts are **rarely stand-alone problems** and usually indicate previously mishandled faults or bigger issues. Correctly identifying this makes **the difference between:**

- One 4-minute stoppage (properly handled)
- Five stoppages costing 25 minutes (symptom not recognized)

COMMON SYMPTOM ALERTS:

These typically indicate inventory graph doesn't match physical reality:

- **Fault Code 121**: *Photocell above lane never detected case during storing move*
- **Fault Code 123**: *Photocell never detected case during dispense move*
- ***Expecting a case but there is no case present at the outfeed***
- ***Not expecting a case but there is a case at the outfeed***

These faults typically show up when someone has not followed Guideline #1.

WHEN YOU SEE SYMPTOM ALERTS:

1. Stop and assess the situation
2. Review recent alert history

3. Check if earlier fault was improperly resolved
4. Verify that the inventory graph matches physical reality
5. Correct any discrepancies before clearing alert and resuming production

---

Guideline 5: Be Aware of InnoPick Settings

Always understand current system settings and their implications.

A good operator is aware of settings that affect InnoPick's behavior. Failing to understand current settings leads to serious operator mistakes.

KEY SETTINGS TO MONITOR:

PAUSE INFEED / PAUSE OUTFEED

- Controls whether cases enter or exit InnoPick
- Check: Home Page

BUILD INVENTORY MODE

- When ON: Prioritizes storing replenishments over outputting cases
- When OFF: Balances input and output for maximum throughput
- Check: Build Inventory

DESTINATION (MERGE) A OR B ENABLED / DISABLED

- Controls which downstream destinations are active
- If a Merge is disabled, all cases route to other destination(s)
- Check: Administration > Setup InnoPick > Merges

Guideline 6: Be Aware of Other System Settings

InnoPick is connected to other systems that directly affect its behavior.

SYSTEMS THAT AFFECT INNOPICK:

MIXMASTER (OR EQUIVALENT WMS/WCS)

Impact Examples:

- **HOLD on a pallet**: InnoPick won't schedule output cases for it or subsequent pallets
- **Order changes**: May affect case sequence and replenishment needs

INBOUND CONVEYOR SYSTEM (PRE-INNOPICK)

Impact Examples:

- **HOLD enabled**: Cases won't be released to InnoPick
- **Tracking errors**: Wrong cases or quantities arrive
- **Photocell issues**: Cases arrive unexpectedly

OUTBOUND CONVEYOR SYSTEM (POST-INNOPICK)

Impact Examples:

- **HOLD or fault active**: InnoPick cannot release cases at outfeed
- **Accumulation full**: Output stops even if InnoPick is ready
- **Downstream equipment down**: Cases back up into InnoPick

WHEN TROUBLESHOOTING:

1. **Check downstream first** if output issues occur
2. **Check upstream first** if infeed issues occur
3. **Coordinate with other system operators** to understand full picture
4. **Verify communication** between systems is working

Clearing Alerts

When to Clear Alerts

Clear alerts only when:

- ✓ Underlying physical issue is resolved
- ✓ Cases are in correct positions
- ✓ Inventory graph matches reality
- ✓ Sensors are unblocked
- ✓ Mechanical issues are fixed
- ✓ Safety devices are reset

When Alerts Won't Clear

If alerts cannot be cleared, it usually means the underlying issue has not been resolved.

Common reasons alerts won't clear:

- Photocell sensor still blocked
- E-Stop button still pressed
- Door still open
- Case still in wrong position
- Mechanical issue not resolved
- The Alert in question is being filtered (not visible).
- See how to View All Alerts

What to do:

1. Read the alert message carefully
2. Inspect physical conditions
3. Resolve the actual problem
4. Try clearing again
    - Use the Recovery Action: Send Clear Alarm on Level
5. See Alert Reference for specific alert resolution steps

Recognizing Symptoms of Bigger Issues

Single Occurrence vs. Pattern

If an alert occurs once:

- Likely an isolated incident
- Resolve and monitor
- Document but may not indicate systemic issue

If same alert repeats:

- Indicates underlying problem not fully resolved
- Stop and reassess
- Don't just keep clearing and restarting
- Look for root cause

Multiple Related Alerts

If different but related alerts occur in sequence:

- May be cascading effects from one root cause
- Example: Infeed issue → Storage fault → Dispense fault
- Address the earliest/root issue first
- Verify inventory graph accuracy
- May need to stop and reset entire level

Best Practices Summary

1. **Inventory Graph is Truth** - Always verify physical matches visual
2. **Verify Case Position** - After store/dispense, check where case ended up
3. **Check Replenishments** - After any infeed alarm, verify quantities and tracking
4. **Look for Root Causes** - Some alerts are symptoms, not standalone issues
5. **Know Your Settings** - Understand how system is configured
6. **Check Connected Systems** - Upstream and downstream affect InnoPick

7. **When in Doubt, Ask** - Better to escalate than cause more problems

---

Related Topics

- Alert Reference - Specific resolution procedures for each alert code
- Home Page Inventory Graph - Understanding the visual display
- Monitoring Production - Catching issues early

Navigation: ← Troubleshooting | Alert Reference →

5.4 Alert Reference Page

Home > Troubleshooting > **Alert Reference**

Overview

This page provides detailed information about every InnoPick alert code, including explanations, common causes, and step-by-step resolution procedures.

How to Use This Reference

1. **Find your alert code** in the list below
2. **Read the explanation** to understand what happened
3. **Review the possible causes**
4. **Follow the solution steps** to resolve the issue
5. **Verify resolution** before clearing the alert

Alert Code Format

Each alert entry follows this format:

- **Alert Code - Description**
- **Explanation & Causes**
- **Solution**

---

98 - ServoDrive Faulted while Moving (Sys_Flt)

Explanation:

The servo drive faulted during a motion operation.

Common Causes:

- Drive overload or overheating
- Electrical fault in drive
- Motor connection issue
- Drive parameter error
- Excessive load on axis

Solution:

1. Check drive status indicators/LEDs
2. Verify motor connections
3. Check for mechanical binding or excessive load
4. Review drive fault code in servo drive diagnostics
5. Contact maintenance if drive hardware fault suspected
6. Clear alert after resolving issue

---

99 - ServoDrive Time-out Fault occurred during Start-up (Sys_Flt)

Explanation:

The servo drive did not respond in time during system start-up.

Common Causes:

- Connection issue in plug or cabling
- Communication timeout
- Rarely occurs when clearing and resuming from faults

Solution:

1. Attempt to clear the fault
2. If it clears successfully, safe to put level back in automatic
3. If fault persists or repeats, contact maintenance
4. Notify supervisors for maintenance follow-up even if it clears

---

109 - Photocell above lane was still blocked after the end of the storing move

Explanation:

After storing a case into a buffer lane, the photocell remained blocked when it should be clear.

Common Causes:

- Turned or skewed case
- Damaged case (torn cardboard, collapsed)
- Wrong product (different case dimensions)
- Case dimensions incorrectly referenced in system
- Case too far forward in lane

Solution:

1. Visually inspect the lane and case
2. Identify why photocell is still blocked
3. Correct the issue:
    - Adjust case position if needed
    - Replace damaged case
    - Retrieve correct product if wrong case
    - Remove case entirely if necessary
4. Ensure photocell is clear
5. Verify other cases in lane are properly positioned
6. Clear the alert
7. Resume operation

---

113 / 117 - Buffer lane tried to dispense a case but the case was too far back

Explanation:

InnoPick tried to dispense a case from buffer lane, but no case blocked the photocell during the dispense move, or case was detected later than expected.

Common Causes:

- No case present in lane (inventory count wrong)
- Case too far back from front of lane
- Case fell or was removed
- Gaps between cases too large
- Lane position tracking error

Solution:

If no case is present:

1. Find replacement case with correct product
2. Place it on the **sequence conveyor** in front of the lane (NOT in the buffer lane)
3. Verify position matches Inventory Graph
4. Clear alert

If case is too far back:

1. InnoPick reached maximum programmed advance distance
2. Either:
    - Manually adjust cases in lane, OR
    - Jog the lane forward to fully dispense the case
3. Ensure case is now on sequence conveyor
4. Verify position on Inventory Graph
5. Clear alert

---

115 - Buffer lane tried to dispense a case but the case was too close to the front

Explanation:

Dispense case was detected earlier than expected (case seen too early).

Common Causes:

- Case positioned too far forward in lane
- Case moved forward unexpectedly
- Lane position tracking error

Solution:

1. Verify actual case position in lane
2. Check other cases in lane for proper spacing
3. Adjust case position if needed
4. Verify case dimensions match product configuration
5. Clear alert once position is correct

---

119 - Photocell still blocked after dispensing

Explanation:

The photocell for that lane remained blocked at the end of the dispense move.

Common Causes:

- Broken case (collapsed, torn cardboard)
- Ripped cardboard hanging down
- Crooked or skewed case
- Wrong case dimensions (didn't fault on storing)
- Cases poorly referenced with irregular gaps

Solution:

1. Inspect the lane photocell area
2. Identify what is blocking the photocell
3. Correct the issue:
    - Remove damaged cardboard or case parts
    - Straighten crooked case
    - Verify dispensed case is properly positioned on sequence conveyor
4. Check other cases in buffer lane:
    - Ensure proper positioning
    - Verify adequate gaps
    - Confirm ready to dispense next case
5. Clear photocell completely
6. Verify on Inventory Graph
7. Clear alert
8. Monitor next few dispenses from this lane

---

121 - Photocell above lane never detected the case during a storing move

Explanation:

InnoPick attempted to store a case, but the photocell was never blocked during the move, indicating there was no case present.

Most Common Causes:

- No case present at all
- Inventory count was wrong
- Bad recovery from previous infeed issue
- Case removed and not replaced

Less Common Causes:

- Case snagged or displaced
- Case in wrong position to be picked up by pop-up
- Sequence conveyor offset by one or more positions

Solution:

If case is simply missing:

1. Determine correct product from alert details
2. Find replacement case
3. Place on sequence conveyor in expected position
4. Verify Inventory Graph matches
5. Clear alert

If case is displaced:

1. Locate the case
2. Position correctly for pop-up to engage
3. Retry the store move
4. Clear alert if successful

Important: See Alert Guidelines - this is often a symptom of improper previous recovery.

---

123 / 124 - Photocell above lane never detected the case during a dispense move

Explanation:

No case detected during dispense move when one was expected.

Common Causes:

- Case is missing (inventory count was wrong)
- Case fell off conveyor
- Case caught on something
- Wrong lane actuated
- Previous storage fault not properly resolved

Solution:

1. Check inventory count for this lane
2. Locate the missing case or find replacement
3. Place case in the storage lane at the front
4. Jog lane forward to dispense case onto sequence conveyor
5. Verify case position matches Inventory Graph
6. Update lane inventory count if necessary
7. Clear alert

Important: See Alert Guidelines - often indicates inventory tracking issue.

---

130 - Could not Execute a Sequence Conveyor Move. Timed Out Waiting for a previous Dispense Move to Finish

Explanation:

The sequence conveyor was ready to move but had to wait too long for a buffer lane dispense operation to complete, causing a timeout.

Common Causes:

- Lane dispense move taking excessive time
- Mechanical issue slowing lane operation
- Pop-up or clutch not disengaging properly
- Communication delay between systems
- Lane conveyor running slowly

Solution:

1. Check which lane was dispensing when timeout occurred
2. Inspect that lane for:
    - Mechanical binding or slowness
    - Pop-up not retracting fully
    - Clutch not disengaging cleanly
3. Test lane operation manually if needed
4. Verify lane motor and drive functioning properly
5. Clear alert
6. Monitor lane performance
7. Contact maintenance if issue persists

---

152 - PopUp stayed down during store (300)

Explanation:

Pop-up assembly was detected in incorrect position (down) when it should have been up during a store operation.

Common Causes:

- Pneumatic system issue
- Pop-up mechanism mechanical failure
- Position sensor malfunction
- Obstruction preventing pop-up from rising

Solution:

1. Verify pneumatic components:
    - Air pressure adequate
    - Solenoid valves functioning
    - Air lines not blocked or leaking
2. Check mechanical components:
    - Pop-up shaft moves freely
    - No obstructions
    - Linkage intact
3. Verify position sensors:
    - Clean sensor faces
    - Check sensor alignment
    - Verify sensor wiring
4. Remove any obstructing objects
5. Test pop-up operation manually if safe
6. Clear alert and test
7. Contact maintenance if issue persists

---

153 - PopUp stayed down during dispense (301)

Explanation:

Pop-up assembly was detected in incorrect position during dispense operation.

Common Causes & Solution:

See alert 152 above - same troubleshooting applies.

---

154 - PopUp stayed up after dispense (301)

Explanation:

Pop-up assembly did not retract after completing dispense operation.

Common Causes & Solution:

See alert 152 above - same troubleshooting applies, but focus on retraction mechanism and down position sensor.

---

155 - PopUp never went down before store (300)

Explanation:

Pop-up assembly failed to fully retract before a store operation was attempted.

Common Causes & Solution:

See alert 152 above - same troubleshooting applies, with emphasis on:

- Air cylinder air release adjustment
- Solenoid valve operation
- Down position sensor verification

---

156 - PopUp never went down before dispense (301)

Explanation:

Pop-up assembly failed to retract before dispense operation.

Common Causes & Solution:

See alert 152 above - same troubleshooting applies.

---

310 - Clutch Fault: Mechanical engagement was not successfully completed after 15 attempts

Explanation:

The clutch assembly failed to engage properly after multiple attempts.

Common Causes:

- Mechanical failure in clutch assembly
- Intermittent mechanical issue
- Misalignment
- Worn components
- Obstruction

Solution:

1. Visually inspect clutch assembly if accessible
2. Retry the move in Automatic mode while observing clutch
3. Look for:
    - Mechanical binding
    - Worn parts
    - Obstructions
    - Misalignment
4. If engagement seems to work sometimes, note conditions
5. Document behavior for maintenance
6. Mechanical intervention may be necessary
7. Consider deactivating lane if repairs cannot be immediate
8. Resume production on other lanes
9. Contact maintenance

---

320 - Clutch Fault: Calculation Error in Mechanical Engagement Position

Explanation:

The system calculated an incorrect position for clutch mechanical engagement, indicating a software or calibration issue.

Common Causes:

- Clutch calibration drift
- Position calculation error in software
- Encoder feedback issue
- Mechanical wear affecting expected positions
- System parameters need updating

Solution:

1. Document current clutch position and behavior
2. Note if this is first occurrence or recurring
3. Check clutch engagement manually if safe
4. Verify encoder feedback is working
5. May require clutch recalibration
6. Contact system expert or maintenance
7. System parameters may need adjustment
8. Clear alert after issue resolved

---

333 - Warning: Actual sensor transition was not seen, but system is assuming clutch was engaged properly

Explanation:

Proximity sensor did not detect expected transition, but system continued based on assumptions.

Common Causes:

- Sensor misalignment
- Intermittent sensor issue
- Mechanical timing variation
- Sensor malfunction

Monitor and Inspect Clutch Assembly if it occurs repeatedly on the same lane

---

334 - Clutch Fault: Prox sensor off after mechanical transition. Probably clutch mechanism bounce

Explanation:

Proximity sensor turned off after mechanical transition, likely due to clutch mechanism bouncing.

Common Causes:

- Mechanical bounce in clutch engagement
- Worn clutch components
- Improper clutch timing
- Sensor positioning

Solution:

1. Verify clutch assembly condition
2. Check for worn components
3. May require mechanical adjustment
4. Contact maintenance if repeats
5. Clear alert

---

340 - Clutch Fault: Mechanical disengagement was not successfully completed after 15 attempts

Explanation:

Clutch assembly failed to disengage properly after multiple attempts.

Common Causes & Solution:

Similar to alert 310, but focus on disengagement mechanism:
1. Inspect clutch release mechanism
2. Verify return spring operation
3. Check for binding during retraction
4. Look for obstructions
5. Contact maintenance
6. May need to deactivate lane

---

345 - Warning: Clutch Disengaged the Mechanism but the Prox Sensor is still Off

Explanation:

Clutch successfully disengaged mechanically, but the proximity sensor still indicates "off" state when it should be "on".

Common Causes:

- Sensor misalignment
- Sensor malfunction
- Wiring issue
- Sensor dirty or obstructed
- Mechanical position slightly off from sensor detection range

Monitor and Inspect Clutch Assembly if it occurs repeatedly on the same lane

---

350 - Warning: Clutch Prox Sensor Changed Status in the Middle of a Buffer Lane Move

Explanation:

The clutch proximity sensor unexpectedly changed state during a buffer lane operation when it should have remained stable.

Common Causes:

- Vibration affecting sensor
- Loose sensor mounting
- Intermittent wiring connection
- Mechanical movement or flex during operation
- Sensor malfunction

Monitor and Inspect Clutch Assembly if it occurs repeatedly on the same lane

---

371 - Auxiliary Encoders (at Outfeed & Infeed) are not Aligned to each other on the Buffer Shaft

Explanation:

The two auxiliary encoders that monitor the buffer shaft position are showing misalignment, indicating potential mechanical slippage or encoder calibration drift.

Common Causes:

- Coupling slip on one or both encoders
- Mechanical slippage in shaft system
- Encoder calibration drift
- Belt or chain stretch
- Mounting looseness

Solution:

1. Stop production on affected level
2. Visually inspect auxiliary encoder couplings
3. Check for loose mounting
4. Check shaft for mechanical issues
5. Contact maintenance - this indicates potential mechanical problem
6. Clear alert after mechanical verification

---

372 - Auxiliary Encoder C (Infeed side) is not Aligned to Motor Encoder on Buffer Shaft

Explanation:

The auxiliary encoder on the infeed side of the buffer shaft is out of alignment with the main motor encoder, indicating slippage.

Common Causes:

- Coupling slip on infeed encoder
- Mechanical slippage between motor and encoder
- Encoder mounting looseness
- Belt or coupling issue

Solution:

1. Contact maintenance
2. Inspect infeed-side auxiliary encoder coupling
3. Check encoder mounting
4. Verify mechanical connection to shaft
5. Check for any slip indicators on coupling
6. Clear alert after inspection and correction

---

373 - Auxiliary Encoder D (Outfeed side) is not Aligned to Motor Encoder on Buffer Shaft

Explanation:

The auxiliary encoder on the outfeed side of the buffer shaft is out of alignment with the main motor encoder, indicating slippage.

Common Causes & Solution:

Same as alert 372, but for outfeed-side encoder.

---

374 - Auxiliary Encoders Offsets have reached the Cumulative threshold

Explanation:

The cumulative offset errors between encoders have exceeded acceptable threshold, indicating ongoing slippage or calibration drift.

Common Causes:

- Repeated small slippages accumulating over time
- Gradual mechanical wear
- Encoder drift
- Inadequate coupling tension
- System running near limits

Alert Maintenance Team when this fault occurs
- It is safe to clear and resume operation
- If the fault returns soon after, a mechanical inspection of shaft couplings and encoder attachment hardware will be necessary.

---

375 - Auxiliary Encoder C (Infeed side) Coupling Slip was Detected on Buffer Shaft

Explanation:

Actual slippage detected in the coupling connecting the infeed-side auxiliary encoder to the buffer shaft.

Common Causes:

- Loose coupling
- Worn coupling
- Inadequate coupling clamping force
- Shaft wear
- Excessive torque or shock loads

Solution:

1. Contact maintenance
2. Inspect couplings for:
    - Tightness
    - Wear
    - Damage
    - Proper seating
3. Tighten or replace coupling as needed
4. Clear alert after mechanical repair
5. Monitor closely after resuming

---

376 - Auxiliary Encoder D (Outfeed side) Coupling Slip was Detected on Buffer Shaft

Explanation:

Actual slippage detected in the coupling connecting the outfeed-side auxiliary encoder to the buffer shaft.

Common Causes & Solution:

Same as alert 375, but for outfeed-side encoder coupling.

---

377 - Buffer Motor Gearbox / Shaft Slip Detected. Critical Fault. Inspect Mechanics.

Explanation:

Slippage detected between the buffer motor gearbox and the shaft - a critical mechanical fault indicating major component failure or imminent failure.

Common Causes:

- Gearbox internal failure
- Coupling failure
- Mounting looseness
- Major mechanical breakdown

Solution:

1. **Stop production**
2. **DO NOT attempt to resume** without maintenance inspection
3. Contact maintenance urgently
4. Inspect for:
    - Gearbox condition
    - Shaft coupling integrity
    - Any signs of mechanical failure
5. Clear alert only after repairs completed and verified

---

378 - Buffer Motor Gearbox / Shaft and Coupling Slip Detected. Critical Fault. Inspect Mechanics.

Explanation:

Multiple slip points detected simultaneously - gearbox, shaft, and coupling - indicating severe mechanical problem or system-wide failure.

Common Causes:

- Multiple component failures
- Catastrophic mechanical event
- System overload
- Major mechanical breakdown
- Multiple loose connections

Solution:

1. **Stop production**
2. **DO NOT attempt to resume** without maintenance inspection
3. Contact maintenance and supervision urgently
4. Full mechanical inspection required for:
    - Gearbox
    - All couplings
    - Shaft integrity
    - Mounting systems
    - Drive components
5. Clear alert only after complete repairs and testing

---

400 & 401 - Communication loopback timeout (Sys_Flt)

Explanation:

Communication timeout in system loopback check.

Common Causes:

- Network issue
- Communication module fault
- Intermittent connection problem

Solution:

1. If this is the first occurrence, safe to clear and resume
2. If it happens repeatedly, contact support
3. Most likely indicates network infrastructure issue

---

402 - Timeout Clutch Fault - Waiting for Clutch to be Disengaged to Resume (Sys_Flt)

Explanation:

System timed out while waiting for a clutch to fully disengage before resuming operation, indicating clutch is stuck or slow to respond.

Common Causes:

- Clutch mechanically stuck in engaged position
- Pneumatic system slow or weak
- Clutch return mechanism failure

Solution:

1. Check clutch physical position
2. Check for mechanical binding
3. Contact maintenance if issue persists
4. Clear alert after clutch verified disengaged

512 - Infeed middle photocell was not blocked during an infeed case move

Explanation:

The infeed cycle started and initial photocell was blocked by incoming case, but the second photocell (infeed mid photocell) was never blocked.

Common Causes:

- Case jammed between upstream conveyor and InnoPick infeed
- Case turned 90 degrees
- Case fell or was displaced
- Case too short to reach second photocell

Solution:

1. Locate the case
2. **Important**: System expects a case on infeed belts since cycle started
3. Clear any jamming condition
4. Place the proper expected case correctly on infeed belts
5. Position case carefully to avoid photocell fault during storage
6. Verify case orientation
7. Check case dimensions match expected product
8. Clear alert
9. Resume operation

See Alert Guidelines for additional infeed troubleshooting steps.

515 - Unexpected case entering onto the infeed belts

Explanation:

The first infeed photocell was blocked when InnoPick was not expecting to receive a case.

Common Causes:

- Back-to-back cases without gap
- Upstream conveyor motor advanced case too far
- Debris blocking photocell
- Extra case sent by upstream system
- Tracking error in upstream system

Solution:

1. **Key Point**: InnoPick is NOT expecting this case
2. Remove whatever is blocking the photocell:
    - If extra case: Remove from infeed belt
    - If debris: Clear obstruction
    - If case sent too far: Pull back or remove
    - **If removing an extra case:**
    - Adjust count in upstream system
    - Document the discrepancy
    - Verify replenishment quantities
    - Once photocell is clear, operator can resume production
    - **Do not** leave unexpected case on infeed belt
    - Clear alert
    - Coordinate with upstream system operator

See Alert Guidelines for complete infeed procedures.

---

516 - Infeed Case's Length Too Long

Explanation:

The measured case length was greater than expected length plus tolerance during infeed transition.

How Measurement Works:

- Case length measured as it transitions from infeed belts to first sequence conveyor position (position 0)
- Compared to configured product length
- Fault if outside tolerance (too long)

Common Causes:

- Wrong product/case
- Case dimensions incorrectly configured in system
- Turned case (length and width swapped)
- Damaged case extended beyond normal dimensions
- Flaps or damage extending case length

Solution:

1. Inspect the case that triggered the fault
2. Determine if:
    - Wrong product → Replace with correct case
    - Turned case → Orient properly (length should be longer dimension)
    - Case damaged/extended → Replace with proper case
    - Dimensions wrong in system → Check Products Page
    - Verify case matches expected product for this replenishment
    - Place correct case on infeed
    - Adjust replenishment quantity if case was rejected
    - Clear alert

See Alert Guidelines.

---

517 - Infeed Case's Length Too Short

Explanation:

The measured case length was less than expected length minus tolerance during infeed transition.

How Measurement Works:

- Case length measured as it transitions from infeed belts to first sequence conveyor position (position 0)
- Compared to configured product length
- Fault if outside tolerance (too short)

Common Causes:

- Damaged or collapsed case
- Wrong product/case
- Case dimensions incorrectly configured
- Turned case

Solution:

1. Inspect the case that triggered the fault
2. Determine if:
    - Case is damaged → Replace with proper case
    - Wrong product → Replace with correct case
    - Turned case → Orient properly
    - Verify case matches expected product for this replenishment
    - Check product dimensions in Products Page
    - Place correct case on infeed
    - Adjust replenishment quantity if case was rejected
    - Clear alert

See Alert Guidelines.

---

518 - Infeed Case (on position 0) is too far downstream

Explanation:

The infeed case on position 0 (first sequence conveyor position) is positioned too far downstream from where it should be.

Common Causes:

- Case advanced too far during infeed transition
- Timing error in infeed cycle
- Case slipped or was pushed downstream
- Conveyor speed mismatch
- Case measurement started late

Solution:

1. Visually verify case position on position 0
2. Check if case can be adjusted to correct position
3. If case is only slightly off, may be safe to continue
4. If significantly misplaced:
    - Remove case
    - Replace with properly positioned case
    - See Case Centering Guide
5. Clear alert
6. Monitor subsequent cases for repeat
7. Contact maintenance if issue recurs

---

519 - Infeed Case (on position 0) is too far upstream

See Fault 518 above.

---

520 - Infeed Fault: Previous Length Check was not Completed

Explanation:

A new infeed case arrived before the previous case's length measurement was completed, indicating timing or sequencing issue.

Common Causes:

- Cases too close together at infeed
- Previous length check taking too long
- Infeed cycle timing error

Solution:

1. Inspect sensors for proper alignment
2. Clear alert
3. Monitor subsequent infeed moves
4. Contact maintenance if alert re-occurs

---

525 - Pop-up shaft reached final upper limit - probably did not engage the pop-up mechanism

Explanation:

The steel finger measuring pop-up height tripped the uppermost limit photocell, which is higher than the pop-up should normally reach.

Common Causes:

- Air cylinder air release adjustment too slow
- Solenoid valve malfunctioning
- Pop-up mechanism not engaging properly
- Mechanical binding or misalignment

Solution:

1. Check air cylinder settings:
    - Verify air release adjustment
    - Ensure proper timing
2. Test solenoid valve operation
3. Inspect pop-up mechanism:
    - Verify engagement
    - Check for mechanical issues
4. Adjust timing if needed
5. Contact maintenance for pneumatic system tuning
6. Clear alert after resolving

---

555 - E-Stop condition active - safety is disarmed (door opened or E-Stop pressed)

Explanation:

A safety device is not in the proper 'safe' condition. This puts all levels into E-Stop status (Dark Red).

Common Causes:

- E-Stop button pressed
- Safety door opened
- Safety interlock open
- Safety device malfunction
- Wiring issue to safety system

Solution:

1. **Identify the E-Stop condition:**
    • Check all E-Stop buttons → ensure all are pulled out (released)
    • Check all safety doors → ensure all are fully closed
    • Check safety interlocks → verify all are engaged
    • Check guard position switches → ensure guards are in place
2. **Clear the E-Stop condition:**
    • Release/rotate E-Stop buttons to reset
    • Close all doors completely
    • Verify interlocks are engaged
    • Check safety system indicator lights
3. **After E-Stop is cleared:**
    • Press Clear Alerts button
    • All levels should return to Manual mode (Yellow)
    • Perform visual inspection before resuming
    • Put levels back into Automatic mode (Green)
4. **If E-Stop won't clear:**
    • One or more safety devices still not satisfied
    • Systematically check each safety device
    • Look for damaged switches or wiring
    • Contact maintenance if cause not obvious
5. **Resume operation:**
    • Only after all safety conditions are satisfied
    • Verify safe to restart
    • Monitor closely on restart

Safety Note: Never bypass E-Stop or safety interlocks. If safety device seems to be malfunctioning, resolve the issue properly or contact maintenance.

---

556 - Open Door Preventing return to RUN Status (Door# 1-4)

Explanation:

A specific safety door (numbered 1-4) is open and preventing the system from returning to RUN status.

Common Causes:

• Door not fully closed
• Door latch not engaging
• Door switch malfunction
• Door misaligned
• Safety interlock issue

Solution:

1. Locate the problem door
2. Ensure door is fully closed
3. Check door latch engages properly
4. Verify door is properly aligned
5. Check door safety switch operation
6. Clear alert after door is secured
7. If door repeatedly shows open when closed, contact maintenance

---

570 - UPS power supply on battery (power loss)

Explanation:

Loss of main power. InnoPick PC is now running on UPS battery backup.

Common Causes:

- Building power failure
- Electrical panel issue
- Power supply fault

Solution:

1. Investigate cause of power loss
2. Restore main power as soon as possible
3. UPS battery has limited runtime
4. If power not restored quickly, perform orderly shutdown
5. Contact facility maintenance

---

571 - Mico Circuit Tripped (24V Short-Circuit)

Explanation:

The Mico circuit breaker has tripped, indicating a short circuit in the 24V system.

Common Causes:

- Short circuit in 24V wiring
- Damaged cable or connector
- Water or moisture in electrical components
- Failed 24V device
- Pinched or crushed wire

Solution:

1. **DO NOT immediately reset** - find cause first
2. Visually inspect 24V wiring for:
    - Damaged insulation
    - Pinched wires
    - Water or moisture
    - Burn marks
3. Check recently moved or adjusted equipment
4. Disconnect suspect circuits if safe to do so
5. After finding and correcting short:
    - Reset Mico circuit breaker
    - Power up system
    - Clear alert
6. Contact maintenance if cause not obvious
7. **Never bypass** Mico protection

---

572 - Air pressure switch off - no air in system

Explanation:

Loss of air pressure in the pneumatic system.

Common Causes:

- E-Stop engaged (normal when doors open)
- Compressor shut down
- Air line leak or blockage
- Pressure switch malfunction

Solution:

1. Check if E-Stop condition is active (doors open, E-Stop pressed)
2. If E-Stop is cause:
    - Close all security doors
    - Clear E-Stop
    - Power up system
3. If fault persists after clearing E-Stop:
    - Check compressor status
    - Inspect for air leaks
    - Verify air pressure gauge readings
    - Contact maintenance if issue not obvious
4. Clear alert once air pressure is restored

---

573 - Input/Output Murr module fault (read variable Murr_Flt)

Explanation:

Loss of communication with Murr Modules (electrical modules mounted on front of InnoPick for every 8-lane section).

Common Causes:

- Communication disruption
- Module hardware fault
- Wiring issue
- E-Stop condition

Solution:

1. Ensure all security doors are closed
2. Verify E-Stop is cleared
3. Power up the system
4. If fault persists:
    - Perform visual inspection of all Murr Modules for affected level
    - Check BUS Node Modules
    - Compare with modules from other levels
    - Look for any irregularities (LEDs, physical damage)
5. Contact system expert if issue continues
6. Document specific module or level affected

---

575 - 12-Volt Power Supply for Auxiliary Encoders Not OK

Explanation:

The 12V power supply that powers the auxiliary encoders is not functioning properly or voltage is out of range.

Common Causes:

- 12V power supply failure
- Voltage out of specification
- Wiring issue to encoders
- Circuit breaker or fuse blown
- Power supply overload

Solution:

1. Check 12V power supply status indicators
2. Measure voltage if possible (should be ~12VDC)
3. Check circuit breakers/fuses for 12V circuit
4. Inspect wiring to auxiliary encoders
5. Look for damaged cables or connectors
6. Contact maintenance - encoder operation critical
7. May need power supply replacement
8. Clear alert after power restored and verified

---

576 - Warning: UPS Battery Faulted

Explanation:

The UPS (Uninterruptible Power Supply) battery has a fault condition, indicating the battery is failing or has failed.

Common Causes:

- Battery end of life
- Battery cell failure
- Internal battery fault
- Charging system failure
- Battery disconnected

Solution:

1. Check UPS status indicators
2. System can continue running on main power
3. **However**: No backup power protection available
4. Schedule UPS battery replacement urgently
5. Avoid intentional power interruptions until battery replaced
6. Contact maintenance or UPS service provider
7. Document battery age and fault details
8. Clear alert after battery replaced and tested

---

577 - Warning: UPS Battery Low

Explanation:

The UPS battery charge level is low, reducing available backup runtime.

Common Causes:

- Recent power outage depleted battery
- Battery aging and reduced capacity
- Charging system not fully recharging battery
- Battery nearing end of life
- UPS continuously on battery power

Solution:
1. Check if main power is currently available
2. If on main power, battery should recharge automatically
3. Allow several hours for full recharge
4. If battery remains low after recharge time:
- Battery may be failing
- Charging system may have issue
- Contact maintenance
5. Monitor battery status
6. Consider battery replacement if issue persists
7. Alert may clear automatically after battery recharges

---

660 - Motion Control Following Error - Critical Fault

Explanation:

A motion control following error has occurred, meaning the actual position of a motor-driven axis has deviated too far from the commanded position. This is a critical fault indicating loss of position control.

Common Causes:

- Mechanical binding or obstruction
- Motor or drive fault
- Excessive load on axis
- Broken belt or coupling
- Drive tuning issue
- Encoder failure

Solution:

1. Identify which axis faulted (see specific axis alerts 661-663, 667)
2. Check for mechanical obstructions or binding
3. Verify motor and drive operation
4. Check belts, chains, couplings for breaks
5. Verify encoder feedback working
6. Contact maintenance - inspection suggested before restarting
7. Clear alert

---

661 - Motion Control Following Error - Axis A (Infeed Conveyor) - Critical Fault

Explanation:

Following error on Axis A (Infeed Conveyor), indicating loss of position control on the infeed system.

Common Causes & Solution:

See alert 660 above, with specific focus on:

- Infeed conveyor belt condition
- Infeed motor and drive
- Mechanical binding at infeed
- Encoder on infeed axis

---

662 - Motion Control Following Error - Axis B (Sequence Conveyor) - Critical Fault

Explanation:

Following error on Axis B (Sequence Conveyor), indicating loss of position control on the main sequence conveyor.

Common Causes & Solution:

See alert 660 above, with specific focus on:

- Sequence conveyor chain condition
- Sequence conveyor motor and drive
- Mechanical binding on sequence conveyor
- Encoder on sequence conveyor axis

---

663 - Motion Control Following Error - Axis C (Buffer Conveyor) - Critical Fault

Explanation:

Following error on Axis C (Buffer/Lane Conveyor), indicating loss of position control on the buffer lane system.

Common Causes & Solution:

See alert 660 above, with specific focus on:

- Buffer lane belt/chain condition
- Buffer motor and drive
- Mechanical binding in buffer lanes
- Encoder on buffer axis

---

667 - Motion Control Following Error - Axis A B & C (All) - Critical Fault

Explanation:

Following error on all axes simultaneously (Infeed, Sequence, and Buffer conveyors), indicating a system-wide motion control failure.

Common Causes:

- Motion controller fault or crash
- Power loss to all drives
- Emergency stop activation
- Communication loss to all drives
- System-wide mechanical seizure (unlikely)
- Motion controller program error

Solution:

1. Check motion controller status
2. This is likely a controller or system-level fault, not mechanical
3. Contact system expert
4. May require controller restart or program reload
5. Verify all mechanical systems after restoration
6. Clear alert only after full system verification

---

671 - Galil Program Thread Command Error

Explanation:

An error occurred in the Galil motion controller program thread, indicating a command execution problem or program fault.

Common Causes:

- Invalid command in Galil program
- Program logic error
- Variable out of range
- Array bounds exceeded
- Syntax error in downloaded program

Solution:

1. Note any recent program changes
2. Check Galil controller for error code using:
    - TC1 (Tell Code)
    - MG _TC (Message Tell Code)
    - MG _ED0 (The line number of the last error)
    - MG _ED1 (The thread where the error occurred)
3. Document the specific error code
4. Contact system expert or programmer
5. May require program correction and reload
6. Verify program version matches expected
7. Clear alert after program issue corrected
8. Test operation carefully after program changes

---

701 - Clutch Prox / Sensor Not On Before Starting Routine I/O Check

Explanation:

Before starting the routine I/O check, the clutch proximity sensor was not in the expected "on" state, indicating the clutch may not be in the proper starting position.

Common Causes:

- Clutch not in home/rest position
- Proximity sensor misaligned
- Sensor malfunction
- Clutch mechanism stuck in wrong position
- Previous move not completed properly

Solution:

1. Check physical clutch position
2. Verify clutch is in expected rest position
3. Check proximity sensor alignment
4. Test sensor operation
5. May need to manually cycle clutch to home position
6. Verify clutch returns properly after moves
7. If the issue cannot be resolved quickly, it is possible to disable the lane and ignore its sensors. This allows the fault to be cleared and production to resume.
8. Clear alert after clutch position verified
9. Contact maintenance if sensor or clutch issue persists

---

702 - Clutch Prox Not Off after Clutch Activation - Routine I/O Check

Explanation:

During routine I/O check, after the clutch was activated (engaged), the proximity sensor did not turn off as expected.

Common Causes:

- Sensor stuck or malfunctioning
- Clutch not moving fully when activated
- Mechanical binding preventing full travel
- Sensor misaligned
- Wiring issue

Solution:

1. Observe clutch operation during activation
2. Verify clutch moves fully when activated
3. Check proximity sensor for proper operation
4. Verify sensor alignment and mounting
5. Check wiring and connections
6. May need sensor adjustment or replacement
7. Contact maintenance
8. Clear alert after verification

---

703 - Clutch Prox Not On after clutch De-activation - Routine I/O Check

Explanation:

During routine I/O check, after the clutch was deactivated (disengaged), the proximity sensor did not turn on as expected.

Common Causes:

- Sensor malfunction
- Clutch not returning to rest position
- Return spring weak or broken
- Mechanical binding
- Sensor misaligned

Solution:

1. Observe clutch operation during deactivation
2. Verify clutch returns fully to rest position
3. Check return spring operation
4. Check proximity sensor operation
5. Verify sensor alignment
6. Look for mechanical binding
7. Contact maintenance
8. Clear alert after clutch verified returning properly

---

706 - Missing Y Pop-Up Flag (H) after Lift - Routine I/O Check

Explanation:

During routine I/O check, after the pop-up lifted, the "Y" flag sensor (H - intermediate height) did not detect the expected flag position.

Common Causes:

- Pop-up not lifting to expected height
- Sensor malfunction
- Flag missing or damaged
- Sensor misaligned
- Pneumatic pressure insufficient

Solution:

1. Check pop-up operation and lift height
2. Verify air pressure adequate
3. Check "H" height sensor alignment
4. Verify flag is present and undamaged
5. Test sensor operation
6. May need pneumatic or sensor adjustment
7. Contact maintenance
8. Clear alert after pop-up verified reaching proper height

---

707 - Z Pop-Up Flag (HH) Present after Lift - Routine I/O Check

Explanation:

During routine I/O check, after pop-up lift, the "Z" flag sensor (HH - upper limit) is detecting the flag when it shouldn't, indicating pop-up may be going too high.

Common Causes:

- Pop-up lifting too high
- Air cylinder over-extending
- Mechanical stop not functioning
- Sensor misaligned or malfunction
- Pneumatic pressure too high

Solution:

1. Check pop-up maximum lift height
2. Verify mechanical stops functioning
3. Check air pressure - may be too high
4. Verify "HH" sensor alignment
5. Check for over-travel conditions
6. May need pneumatic adjustment
7. Contact maintenance
8. Clear alert after pop-up height verified correct

---

708 - Missing W Pop-Up Flag (LL) After Pop-Up De-activation - I/O Check

Explanation:

During routine I/O check, after the pop-up was deactivated (lowered), the "W" flag sensor (LL - lower limit) did not detect the flag, indicating pop-up may not be fully retracted.

Common Causes:

- Pop-up not fully retracting
- Air cylinder not fully extending downward
- Mechanical obstruction
- Sensor misaligned
- Flag damaged or missing

Solution:

1. Verify pop-up fully retracts when deactivated
2. Check for mechanical obstructions
3. Verify air exhaust functioning properly
4. Check "LL" sensor alignment and operation
5. Verify flag present and undamaged
6. May need pneumatic or mechanical adjustment
7. Contact maintenance
8. Clear alert after pop-up verified retracting fully

---

709 - Buffer Lane Photocell Blocked during Routine I/O Check

Explanation:

InnoPick continuously monitors every lane's photocell and clutch proximity sensor inputs. This fault indicates one sensor was in an unexpected state.

Common Causes:

For Photocells:

- Broken or damaged case
- Case inducted or dispensed too close to photocell
- Random debris (plastic, cardboard)
- Malfunctioning photocell sensor
- Case positioned incorrectly

For Clutch Proximity Sensors:

- Proximity sensor in wrong state
- Clutch assembly not in expected position
- Sensor misalignment
- Wiring issue

Solution:

1. Note which lane the alert specifies
2. Inspect that lane:
    - Check photocell area for obstructions
    - Look for broken case parts
    - Verify clutch position (if clutch prox fault)
3. Clear the obstruction or correct the issue
4. Verify sensor is now in correct state
5. Clear alert
6. Monitor to ensure issue doesn't recur
7. If the issue cannot be resolved quickly, it is possible to disable the lane and ignore its sensors. This allows the fault to be cleared and production to resume.

---

715 - Case Detected at Far Back of Buffer Lane

Explanation:

A case has been detected at the far back position of a buffer lane, which may indicate the lane is full or a case has moved too far back.

Common Causes:

- Lane is full to capacity
- Case pushed too far during storage
- Case slipped backward in lane
- Photocell at back of lane detecting case

Solution:

1. Visually inspect the affected lane
2. Remove extra cases or jammed cases
3. Update lane inventory count if necessary
4. Clear alert
5. Monitor lane operation

---

719 - Clutch Prox Sensor Off during Routine I/O Check

Explanation:

During routine I/O monitoring, the clutch proximity sensor is in the "off" state when it should be "on", or vice versa.

Common Causes & Solution:

See alert 709 above - same troubleshooting applies for clutch proximity sensors.

---

725 - Photocell above Lane was Blocked on Activated Lane

Explanation:

When a lane was activated (clutch engaged, ready to operate), the photocell above the lane was already blocked, which is unexpected and may indicate a case is in wrong position.

Common Causes:

- Case positioned too far forward in lane
- Case or debris blocking photocell
- Previous dispense did not complete properly
- Case too tall or irregular
- Sensor malfunction

Solution:

1. Inspect the affected lane
2. Check case position in lane
3. Verify no debris blocking photocell
4. Ensure cases are properly positioned with gaps
5. Check if case dimensions match configuration
6. Adjust case position if needed
7. Clear photocell obstruction
8. Clear alert
9. Monitor subsequent operations on this lane

---

781 - Infeed Motion Control (Galil) Thread Running too Long (Timeout) - Critical Fault

Explanation:

The infeed motion control thread took too long to complete, exceeding the timeout threshold. This indicates a performance issue or system overload.

Common Causes:

- High system load
- Motion controller processing delay
- Communication delay
- Complex motion calculation taking too long
- Controller near processing capacity

Solution:

1. Document what was happening when timeout occurred
2. May be transient - safe to clear if isolated occurrence
3. If repeated, contact system expert
4. Clear alert
5. Monitor for recurrence

---

782 - Sequence Conveyor Motion Control (Galil) Thread Running too Long (Timeout) - Critical Fault

Explanation:

The sequence conveyor motion control thread took too long to complete, exceeding the timeout threshold.

Common Causes & Solution:

See alert 781 above - same troubleshooting applies for sequence conveyor axis.

---

783 - Buffer Motion Control (Galil) Thread Running too Long (Timeout) - Critical Fault

Explanation:

The buffer lane motion control thread took too long to complete, exceeding the timeout threshold.

Common Causes & Solution:

See alert 781 above - same troubleshooting applies for buffer/lane axis.

---

791 - Motion control (Galil) thread crashed while main thread was waiting - critical fault

Explanation:

The motion controller instruction thread has crashed on the indicated level. This is a serious error that should be adequately documented.

Common Causes:

- Software exception
- Motion controller communication failure
- Serious system fault

Solution:

1. Contact System Expert to follow the steps below:
2. **Capture diagnostic data BEFORE clearing the alert:**
    - **IMPORTANT**: DMC-4040 motion controllers do NOT store error history
    - Once alert is cleared, error information is lost forever
    - System expert should capture information from GalilTools immediately:

    ```
    TC1         # Current error code
    TB          # Status byte
    TS          # System status
    SC          # Stop code
    MG _SCA     # Axis A stop code
    MG _SCB     # Axis B stop code
    MG _SCC     # Axis C stop code
    ```

3. **If Alert 791 has already been cleared:**
    - Error information is no longer available
    - Document circumstances and await recurrence
4. **After capturing data, clear the alarm**
5. **Perform quick visual check:**
    - Compare affected level to Inventory Graph
    - Verify case positions match expected
6. **When resuming production:**
    - Pay close attention to affected level
    - An instruction may have been missed or duplicated
    - Sequence conveyor may be offset by 1 position
    - May have extra or missing case
7. **Monitor closely** for several cycles

---

900 - Warning: Back Edge of Case was Adjusted after the Case Stored

Explanation:

After a case was stored in a lane, the system detected the back edge position was not where expected and made an adjustment to the tracking data.

Common Causes:

- Case compressed or shifted during storage
- Case dimensions slightly different than expected
- Normal variation in case positioning
- Gap adjustment in lane

Monitor and Inspect Clutch Assembly if it occurs repeatedly on the same lane

---

901 - Warning: Back Edge of Case was Adjusted Outward after Case Dispense (Back Edge not Seen)

Explanation:

After dispensing a case, the back edge was not detected where expected, and the system adjusted the position outward (further into the lane).

Common Causes:

- Case back edge not clearly detected
- Case compressed in lane
- Photocell timing issue
- Normal variation

Monitor and Inspect Clutch Assembly if it occurs repeatedly on the same lane

---

902 - Warning: Back Edge of Case was Adjusted Outward after Case Dispensed

Explanation:

After dispensing a case, the system adjusted the tracked position of the back edge outward (further into the lane).

Common Causes & Solution:

Similar to alert 901 - normal system adjustment.

Monitor and Inspect Clutch Assembly if it occurs repeatedly on the same lane

---

903 - Warning: Back Edge of Case was Adjusted Inward after Case Dispensed

Explanation:

After dispensing a case, the system adjusted the tracked position of the back edge inward (closer to lane entrance).

Common Causes:

- Case extended further than expected during storage
- Normal variation in case positioning
- Gap adjustment
- System self-correction

Monitor and Inspect Clutch Assembly if it occurs repeatedly on the same lane

---

904 - Warning: Pulled Buffer Lane Cases Inward after Case Store (Photocell was Blocked)

Explanation:

After storing a case, the photocell was blocked (indicating insufficient gap to next case), so the system adjusted by pulling all cases in the lane inward to create proper spacing.

Common Causes:

- Case stored closer to previous case than expected
- Normal gap adjustment
- Case compressed during storage
- System maintaining proper spacing

Monitor and Inspect Clutch Assembly if it occurs repeatedly on the same lane

---

905 - Warning: Pulled Buffer Lane Cases Inward after Case Dispense (Photocell was Blocked)

Explanation:

After dispensing a case, the photocell was still blocked, indicating the next case was too close, so the system adjusted by pulling remaining cases in the lane inward.

Common Causes:

- Next case too close after dispense
- Normal gap adjustment
- System maintaining proper spacing
- Cases compressed in lane

Monitor and Inspect Clutch Assembly if it occurs repeatedly on the same lane

---

Outfeed Alerts

Expecting a case but there is no case present at the outfeed

Explanation:

InnoPick expects a case on the last sequence conveyor position, but no case is detected there.

Common Causes:

- Case never dispensed from lane
- Case fell off somewhere
- Case caught on something earlier in sequence
- Motion controller thread crashed (see alert 791)
- Sequence conveyor offset by one position
- Inventory tracking error

Solution:

1. Review Inventory Graph - what case is supposed to be there?
2. Inspect the outfeed area and upstream positions
3. Determine why the case is not present:
    - **Case is one spot back:** May need to jog conveyor forward
    - **Case never dispensed:** Check source lane, may need to manually dispense
    - **Case caught upstream:** Clear obstruction and position case correctly
    - **Case missing entirely:** Find replacement case
4. **Place needed case on last position:**
5.
    - Must be the correct product
    - Must match what Inventory Graph expects
    - Must block the last sensor on InnoPick
6. Verify Inventory Graph matches physical reality
7. Clear alert
8. Resume operation
9. Monitor closely to ensure tracking is correct

Important: See Alert Guidelines - this usually indicates a tracking issue.

Not expecting a case but there is a case at the outfeed

Explanation:

A case is detected at the last position when InnoPick was not expecting one there.

Common Causes:

- Infeed case not stored properly (see infeed alerts)
- Issue not resolved properly after previous fault
- Case advanced without being tracked
- Manual intervention error

Solution:

1. **Verify the case truly should not be there:**
    - Check Inventory Graph
    - Review recent infeed activity
    - Check if case belongs to current sequence
2. **Remove the unexpected case:**
    - Take case off last position
    - Do not leave it in place
3. **Determine where case should go:**
    - Look at which lanes are receiving current replenishment
    - Compare lane inventory numbers with physical count
    - Check Replenishments Page
4. **Two options for the case:**
    - *Option A - Short the lane:*
    - If lane is inaccessible or case not needed urgently
    - Reduce lane count by 1
    - Update in Lane Inventory Page
    - *Option B - Add to back of lane:*
    - If lane is accessible
    - Case may be needed soon
    - Manually place at back of lane
    - Update inventory count if needed
5. Clear alert
6. Resume operation
7. Monitor to prevent recurrence

Important: See Alert Guidelines - usually caused by improper infeed issue resolution.

Related Topics

- Alert Guidelines - General principles for successful alert resolution
- Alerts Page - Viewing and managing alerts
- Common Issues - Frequently encountered problems
- Case Replacement - Procedures for replacing cases

Navigation: ← Alert Guidelines | Next: Common Issues →

5.5 Deadlocks

Home > Troubleshooting > Deadlocks

Overview

This section provides guidance for resolving deadlock situations.
A deadlock is when InnoPick is not able to fit all the cases that are due to enter the infeed, and it cannot make room by outputting cases because the next case needed for the sequence is coming after those cases which cannot fit.
In other words, a deadlock is when a level of InnoPick is too full and becomes stuck.

Types of Deadlocks

There are two types of deadlocks:

1. **Preview deadlock (forecast)** - warns the operator that a deadlock will occur unless something is done to avert it.
2. **Just-in-Time deadlock** - the system is in a deadlock state and cannot proceed without an operator intervention.

Preview Deadlock

<table>
<tr><th>Level</th><th>Message</th></tr>
<tr><td>5</td><td>Level 5: Caught deadlock 'Level 5: DeadLock detected with given sequence after processing 436 output cases, 451 input cases. The product that is needed to service the next output case '7b546749-be54-4457-9735-ed8ad28454a1||118010574289|145811691' is '49309'. Last input case attempted to process was '7b546749-be54-4457-9735-ed8ad28454a1||118010574289|145811691' with product reference '49309'. If this happens during preview, we still have a chance to avoid the deadlock by manually reassigning this product to a different level than 5. Aborting!': Unable to auto-resolve. This is critical. Halting system. (RunInSimulationMode = True)</td></tr>
</table>

"Level 2. Deadlock detected with given sequence after processing 940 output cases, 850 input cases."

The following is an example alarm message for a preview deadlock:

- *[Level 5] Level 5: Caught deadlock 'Level 5: DeadLock detected with given sequence after processing 111 output cases, 166 input cases. The product that is needed to service the next output case '0aa249a6-d163-4bb2-bfe9-9da964345d38|000000006500247839|119003120537|129563670' is '47692'. Last input case attempted to process was '0aa249a6-d163-4bb2-bfe9-9da964345d38|000000006500247839|119003120537|129563670' with product reference '47692'. If this happens during preview, we still have a chance to avoid the deadlock by manually reassigning this product to a different level than 5. Aborting!': Unable to auto-resolve. This is critical. Halting system. (RunInSimulationMode = False)*

Key Points:

- The deadlock is detected after processing a number of output and input cases
- This forecasts a future deadlock if nothing changes
- Manual reassignment of the key product to a different level can prevent it
- The system automatically enters manual mode to allow operator intervention

Understanding the Causes of Preview Deadlocks

Preview deadlocks can be generated by one or more factors.

- There are too many products assigned to a single level of InnoPick, and the system has not found any way to avoid a deadlock situation.
- A replenishment has come in out of order (before the next priority replen), meaning it has to be stored completely in order for the priority replen can be brought in.
- The number of enabled buffer lanes was reduced since the initial case sequence calculation.
- A replenishment was shorted, but the shorted cases were needed to complete the case sequence. Now a new replen has been generated but the system has to store the active replens that are in the pipeline before it.

Resolving a Preview Deadlock

- Note the product # and use the Products Page to re-assign it to another level.
- After this, the system will re-calculate the case sequence and may re-generate a preview deadlock fault depending on the context.
- Continue re-assigning the problematic product (as listed in the preview deadlock alert message) until no new preview deadlock alerts are generated.

Just-In-Time Deadlocks

A Just-In-Time deadlock occurs when a level of InnoPick is full and cannot make even one move.
This can happen if:

- A preview deadlock is ignored / not resolved and the system is allowed to continue operating.
- Some unexpected change has occurred, and InnoPick can no longer continue on that level, such as the disabling of a lane that InnoPick was expecting to be able to use.

Example of a Just-in-Time deadlock alert message:

- *[Level 5] Warning: Just In Time deadlock detected on Level 5. The product that is needed to service the next output case '0aa249a6-d163-4bb2-bfe9-9da964345d38|000000006500247839|119003120537|129563670' is '47692'. Please fix manually!!!*

Resolving a Just-In-Time Deadlock

The simplest way to resolve a Just-In-Time deadlock is to:

1. Identify some lanes on that level with low inventory (1-10 cases)
2. Makes sure those cases are not immediately going to be needed (using the Case Sequence Page)
3. Empty those lanes, making them available.

Alternatively, another option is to:

1. Identify which product InnoPick needs in order to resume outputting cases (this is listed in the Alert message).
2. Filter the Case Sequence Page to see how many cases of this product are needed.
3. Manually load this number of cases of the required product in an empty lane on that level of InnoPick.
4. Edit the inventory for that lane and save the changes, using the Inventory Page.
5. Upon saving the changes to the inventory change, the system should recalculate the case sequence for that level and production should resume.
6. If needed, use the "Recalculate Level" Recovery Action.

Navigation: ← Common Issues | Case Replacement →

5.6 Case Replacement Procedures

Home > Troubleshooting > Case Replacement

Overview

This page provides procedures for replacing broken, missing, or incorrect cases during InnoPick operation. Proper case replacement is critical to maintaining accurate inventory tracking and preventing cascading faults.

When Case Replacement is Needed

Common scenarios requiring case replacement:

- **Broken case**: Case damaged, collapsed, or torn
- **Missing case**: Case fell off, removed, or never present
- **Wrong product**: Incorrect SKU inducted
- **Quality issue**: Case rejected for damage or defects
- **Operator error**: Case removed during troubleshooting

General Principles

Before Replacing Any Case

1. **Understand why the case needs replacement**
    - What happened to the original case?
    - Is this an isolated incident or pattern?
    - Will the same issue affect the replacement?
2. **Determine the correct product**
    - Check what product is expected
    - Verify SKU from alert message or Inventory Graph
    - Confirm with Replenishments Page or Case Sequence Page
3. **Know where to place the replacement**
    - Sequence conveyor vs. storage lane
    - Specific position on conveyor

Scenario 1: Replace Case Currently Being Inducted

Situation:

- Case is currently part of an active replenishment
- Case is already in InnoPick
- Case is broken, too damaged, or wrong product

Procedure

1. **Obtain replacement case:**
   - Take one from upstream conveyor system
   - Ensure it's the correct product
2. **Update upstream system:**
   - Delete 1 Case Data on upstream conveyor HMI
   - This prevents tracking mismatch
3. **Short the replenishment in InnoPick Manager:**
   - Go to Replenishments Page
   - Find the active replenishment
   - Use **Alter Quantity** action (three dots menu)
   - Reduce quantity by 1
4. **Place replacement case:**
   - Position on the case exactly where the broken was (or was supposed to be)
   - Ensure proper orientation
5. **Resume operation:**
   - Clear any alerts if present
   - Verify the case stores correctly
   - Monitor the transition to the next replenishment

Scenario 2: Damaged Inbound Case on Sequence Conveyor is Not Needed for Client Pallet

Situation:

- Case is on the sequence conveyor and is going to be stored
- No replacement cases are readily available
- Case will NOT be needed for upcoming orders
- This can be determined if: There are no additional replenishment scheduled for this product (filter for this product & level in the Replenishments page to determine this).

If no other replenishment scheduled:

- InnoPick can function without this case
- Case can be shorted rather than replaced
- **Replacement is optional**

Procedure

SHORT THE CASE

1. **Allow the storage move to fault:**
    - **Alert 121 (photocell never detected case)** will occur
    - System expects to store a case but finds none
2. **Jog lane so it is ready to dispense**
    - Use Manual motion control > Lane Operations
    - Jog lane forward so existing cases move up
    - Front case can now be dispensed
3. **Short the lane inventory:**
    - Go to Lane Inventory Page
    - Find the affected lane
    - Reduce quantity by 1
4. **Clear alerts and resume**

Scenario 3: Replace Missing Outfeed Case

Situation:

- Alert: "Expecting a case but there is no case present at the outfeed"
- Case should be at last InnoPick position but is missing
- See Alert Reference for detailed alert resolution

Procedure

1. **Identify the expected case:**
    - Check Inventory Graph
    - Note product SKU and pallet ID
    - Determine from which lane it should have come
2. **Determine why case is missing:**
    - Was it never dispensed from lane?
    - Did it fall off somewhere?
    - Was it incorrectly removed?
    - Is sequence conveyor offset by one position?
3. **Obtain replacement case:**
    - Must be exact product that was expected
    - Check case label/markings
4. **Place case at outfeed position:**
    - Must block the last sensor on InnoPick
    - Position correctly for downstream equipment
    - Verify orientation

5. **Verify Inventory Graph:**
    - Should now show filled square at last position
    - Matches physical case
6. **Clear alert and resume**
7. **Monitor closely:**
    - Verify subsequent cases are tracking correctly
    - Watch for signs of offset sequence conveyor

Special Considerations

Quality Control

If replacing due to quality issue:

- Note the specific problem
- Report pattern of defects to supervision
- May indicate supplier issue

Inventory Accuracy

After any replacement:

- Verify all counts are correct
- Check Lane Inventory
- Confirm Replenishments Page quantities
- Ensure Inventory Graph matches reality

Preventing Case Replacement Needs

Best Practices

Visual Inspections:

- Check case quality of the cases in InnoPick
- Monitor cases as they induct

Proper Handling:

- Ensure upstream handling doesn't damage cases
- Ensure upstream handling properly identifies and rejects damaged or turned cases

Maintain Tracking Accuracy:

- Always follow Alert Guidelines
- Verify Inventory Graph after every alert

Related Topics

- Alert Reference - Specific alerts requiring case replacement
- Alert Guidelines - Proper alert recovery procedures
- Lane Inventory - Updating lane contents
- Replenishments Page - Adjusting quantities

Navigation: ← Deadlocks | Case Centering →

5.7 Case Centering

Home > Troubleshooting > Case Centering Adjustments

Overview

InnoPick requires precise parameterization in order for cases to be accurately centered all the way from the infeed to the last buffer lane.

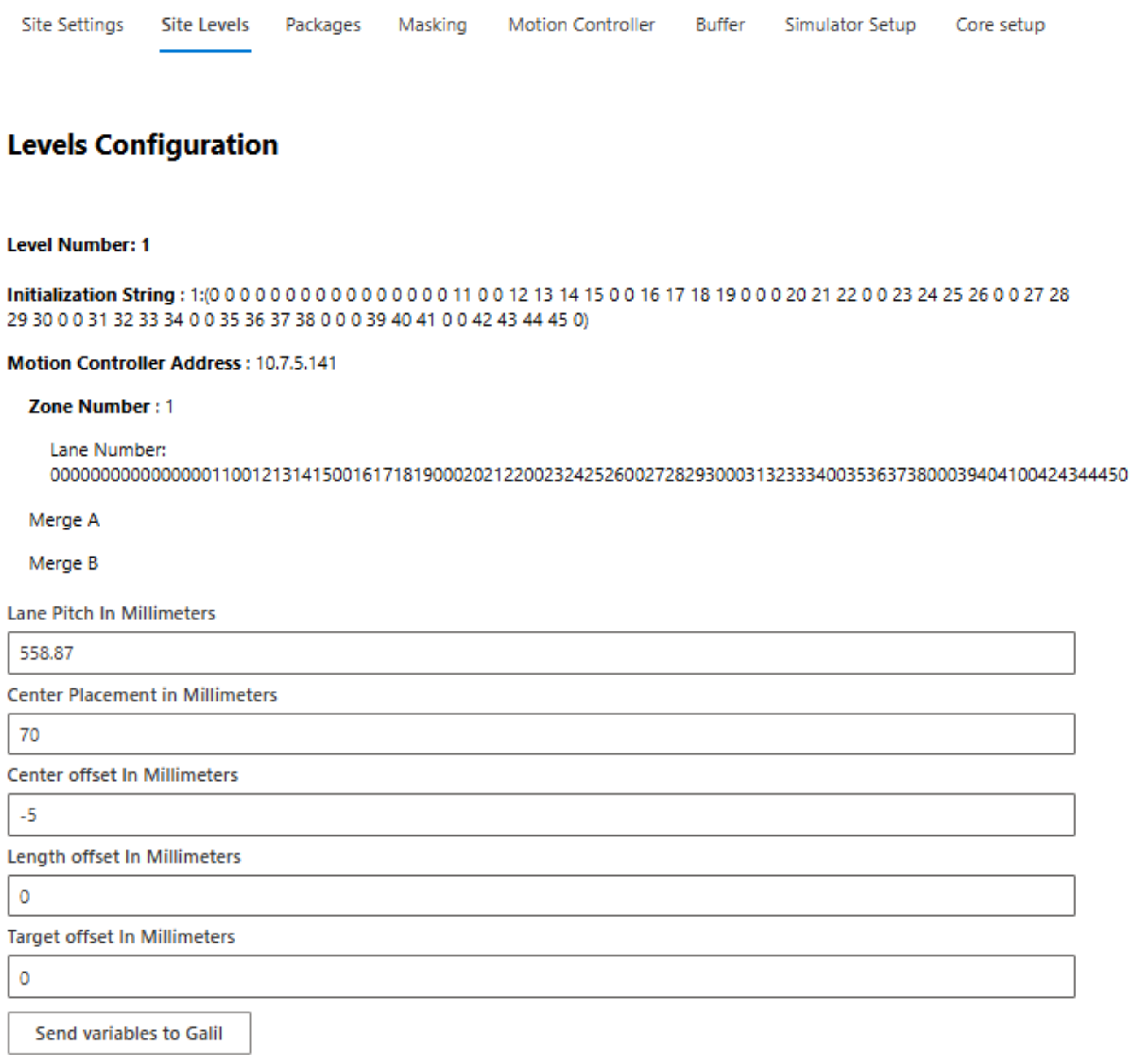
This page walks through the proper process for ensuring good case centering.

Step-by-Step Sequence for Proper Case Centering

1. Validate Sequence Conveyor Chain Tension

- It's not uncommon to have to adjust the case centering parameters after tensionning sequence conveyor chains.
- Therefore, if it is needed, it's best to do that first.

2. Adjust the "Center Placement in Millimeters" Parameter



On the **Administration > Setup InnoPick > Site Levels** page, adjust the "**Center placement in millimeters**" parameter so that cases are properly centered in front of the first buffer lane (the one closest to the infeed).

Procedure:

1. Stop InnoPick when a case is positioned in front of the first lane (closest to the infeed).
2. Calculate the adjustment required to be centered
   - **Example**: The case is too far in the downstream direction, by 10mm
3. Modify the "**Center placement in millimeters**" by **-10**, and let a new case enter to re-validate
   - ⚠ **Note**: The change will only be applied to the next case that is inducted on the infeed section of the upstream conveyor
4. Continue until cases are properly centered in front of the first lane

3. Test Alignment Along the Entire Length of InnoPick

Next, place a perfectly centered case in front of the first lane, and manually jog the sequence conveyor until the case is in front of the last lane (nearest the outfeed).

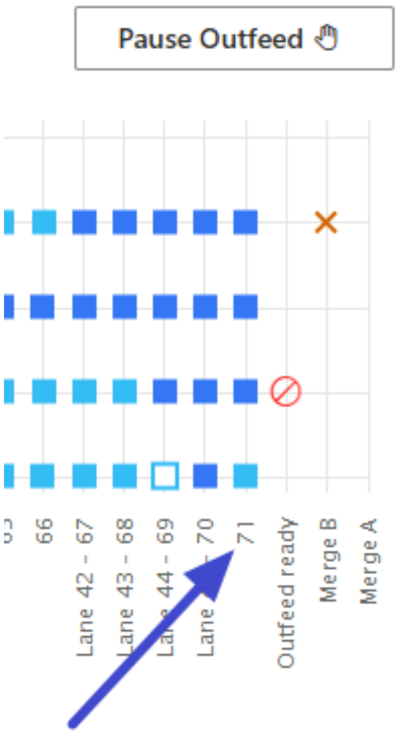
Click here for information on manually jogging InnoPick

Important: When you jog, make sure that the value used for the jogging (on the Manual Motion Controls page) is the same as the **"Lane Pitch in Millimeters"** parameter on the Site Levels page.

- It is usually necessary to input the correct distance from the Lane Pitch paramater for this exercise, in order to recreate the actual movement distances seen in production.

4. Measure the Necessary Adjustment

- Go to the case stopped in front of the last buffer lane.
- Measure the adjustment necessary for it be properly centered, and divide the adjustment by the number of positions on the sequence conveyor (visible on the inventory graph).



- In the example above, there are 71 positions.

5. Calculate the Lane Pitch Correction

Example:

- Need to adjust the position by **12mm toward upstream** (case is too far toward downstream)
- The InnoPick has **71 positions**
- Calculation: **12mm ÷ 71 positions = 0.17mm**

6. Apply the Correction

Adjust the **"Lane Pitch in Millimeters"** parameter by the result of your calculation in Step 5.

Important Note:

- If the adjustment is towards the upstream, subtract the value from the current Lane Pitch distance. This will reduce the size of the standard move and result in the case moving a smaller distance, ending up more upstream.
- If the adjustment is towards the downstream, add the value to the current Lane Pitch distance. This will increase the size of the standard move and result in the case moving a greater distance, ending up further downstream.

Example (continued):

- If our calculation resulted in a value of 0.17mm
- And the Lane Pitch parameter is currently at **558.80mm**
- We will subtract the calculated value from the parameter value:
- **558.80 - 0.17 = 558.63mm**

7. Final Validation

Validate the new value by repeating the test and ensuring that the case ends up properly centered in front of the last lane (near the outfeed).

Parameter Summary

Parameter	Location	Function
Center Placement in Millimeters	Administration > Setup InnoPick > Site Levels	Used to set initial center position, which is validated in front of the **first** lane
Lane Pitch in Millimeters	Administration > Setup InnoPick > Site Settings	Defines the standard sequence conveyor move distance, used for maintaining centering along the **entire length** of InnoPick, and validated in front of the **last** lane

---

Calculation Formula

```
Lane Pitch Adjustment = Total Error (mm) ÷ Number of Positions
```

Direction of adjustments:

- Case too far **downstream** → **Negative** (-) adjustment
- Case too far **upstream** → **Positive** (+) adjustment

Navigation: ← Case Replacement | Disabled Lane →

5.8 Disabled Lane

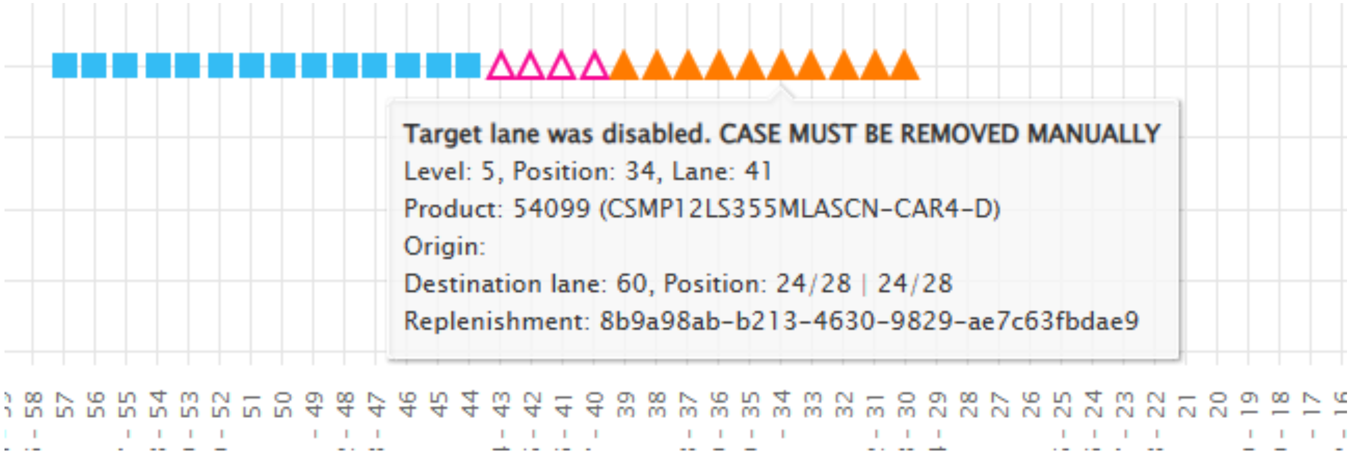
Home > Troubleshooting > Disabled Lane

Overview

If a mechanical issue renders a lane inoperable during production, it may be necessary to disable a lane that InnoPick is expecting to be able to use. This troubleshooting guide covers these scenarios.

Inbound Cases

- Once replenishment cases are on the sequence conveyor, they are assigned to a particular buffer lane, and this destination cannot be changed.
- If that lane suffers a mechanical breakdown and cannot store those cases, then the operator must:
- Disable the lane in question
- Manually remove the cases on the sequence conveyor that are destined for that lane.
- Once the lane is disabled, the affected inbound cases will be highlighted and a message will appear when you hover over them:
- **"Target lane was disabled. CASE MUST BE REMOVED MANUALLY"**
- Failure to remove these cases by hand will certainly cause subsequent issues



Outbound Cases

- Similarly, once an outbound case has been scheduled on the sequence conveyor (appearing as an empty square), the lane cannot be re-assigned.
- If the source lane suffers a mechanical breakdown, the operator must:
- Disable the lane in question
- Manually place the scheduled outbound cases on the correct sequence conveyor positions
- Failure to place the scheduled outbound cases on the correct sequence conveyor positions will result in faults when those cases are not detected as expected on the last position of InnoPick

Navigation: ← Case Centering | Reference →

6. Reference

6.1 Reference

Home > Reference

Overview

This page provides quick reference information for InnoPick Manager, including status indicators, alert code summaries, and other frequently-needed reference data.

Reference Pages

Alert Codes Quick Reference

Condensed summary of all alert codes for quick lookup:

- Alert code number
- Brief description
- Link to detailed resolution in Alert Reference

Error Code	Description
98	ServoDrive Faulted while Moving (Sys_Flt)
99	ServoDrive Time-out Fault occurred during Start-up (Sys_Flt)
109	Photocell above lane was still Blocked after the end of the Storing Move
113	Buffer lane tried to Dispense a Case but the Case was Too Far back
115	Buffer lane tried to Dispense a Case but the Case was Too Close to the front
117	Buffer lane tried to Dispense a Case but the Case was Too Far back
119	Photocell above lane still Blocked after Buffer Lane Dispensed a Case
121	Photocell above lane never Detected the Case during a Storing Move
123	Photocell above lane never Detected the Case during a Dispense Move
124	Photocell above lane never Detected the Case during a Dispense Move
130	Could not Execute a Sequence Conveyor Move. Timed Out Waiting for a previous Dispense Move to Finish
152	The Pop-Up Mechanism stayed down when trying to Store a Case
153	The Pop-Up Mechanism stayed down when trying to Dispense a Case
154	The Pop-Up Mechanism stayed Up after Dispensing a Case
155	The Pop-Up Mechanism never went Down (from a previous Move) before a Storing Move
156	The Pop-Up Mechanism never went Down (from a previous Move) before a Dispense Move
310	Clutch Fault: Mechanical Engagement was not Successfully Completed after 15 Attempts
320	Clutch Fault: Calculation Error in Mechanical Engagement Position
333	Warning: Actual Sensor Transition was Not Seen. System is Assuming Clutch was Engaged Properly
334	Clutch Fault: Prox Sensor Off after Mechanical Transition. Probably Clutch Mechanism Bounce
340	Clutch Fault: Mechanical Disengagement was not Successfully Completed after 15 Attempts
345	Warning: Clutch Disengaged the Mechanism but the Prox Sensor is still Off
350	Warning: Clutch Prox Sensor Changed Status in the Middle of a Buffer Lane Move
371	Auxiliary Encoders (at Outfeed & Infeed) are not Aligned to each other on the Buffer Shaft
372	Auxiliary Encoder C (Infeed side) is not Aligned to Motor Encoder on Buffer Shaft
373	Auxiliary Encoder D (Outfeed side) is not Aligned to Motor Encoder on Buffer Shaft
374	Auxiliary Encoders Offsets have reached the Cumulative threshold
375	Auxiliary Encoder C (Infeed side) Coupling Slip was Detected on Buffer Shaft
376	Auxiliary Encoder D (Outfeed side) Coupling Slip was Detected on Buffer Shaft
377	Buffer Motor Gearbox / Shaft Slip Detected. Critical Fault. Inspect Mechanics.
378	Buffer Motor Gearbox / Shaft and Coupling Slip Detected. Critical Fault. Inspect Mechanics.
400	Communication Loopback TimeOut (Sys_Flt)
401	Communication Loopback TimeOut (Sys_Flt)
402	Timeout Clutch Fault - Waiting for Clutch to be Disengaged to Resume (Sys_Flt)

Error Code	Description
512	Infeed Middle Photocell was not Blocked during an Infeed Case Move
515	Unexpected Case entering onto the Infeed Belts
516	The Length of the Infeed Case (on position 0) is too long: Expected mm. Measured mm.
517	The Length of the Infeed Case (on position 0) is too short: Expected mm. Measured mm
518	Infeed Case (on position 0) is too far downstream by {4}mm
519	Infeed Case (on position 0) is too far upstream by {4}mm
520	Infeed Fault: Previous Length Check was not Completed
525	Pop-up Shaft Reached Final Upper Limit - Probably did not Engage the Pop-Up Mechanism
555	E-Stop Condition Active - Safety is Disarmed (Door Opened or E-Stop Pressed)
556	Open Door Preventing return to RUN Status (Door# 1-4)
570	UPS Power Supply on Battery (Power Loss)
571	Mico Circuit Tripped (24V Short-Circuit)
572	Air Pressure Switch Off - No Air in System
573	Input/Output Murr Module Fault (Read Variable Murr_Flt)
575	12-Volt Power Supply for Auxiliary Encoders Not OK
576	Warning: UPS Battery Faulted
577	Warning: UPS Battery Low
660	Motion Control Following Error - Critical Fault
661	Motion Control Following Error - Axis A (Infeed Conveyor) - Critical Fault
662	Motion Control Following Error - Axis B (Sequence Conveyor) - Critical Fault
663	Motion Control Following Error - Axis C (Buffer Conveyor) - Critical Fault
667	Motion Control Following Error - Axis A B & C (All) - Critical Fault
671	Galil Program Thread Command Error
701	Clutch Prox / Sensor Not On Before Starting Routine I/O Check
702	Clutch Prox Not Off after Clutch Activation - Routine I/O Check
703	Clutch Prox Not On after clutch De-activation - Routine I/O Check
706	Missing Y Pop-Up Flag (H) after Lift - Routine I/O Check
707	Z Pop-Up Flag (HH) Present after Lift - Routine I/O Check
708	Missing W Pop-Up Flag (LL) After Pop-Up De-activation - I/O Check
709	Buffer Lane Photocell Blocked during Routine I/O Check
715	Case Detected at Far Back of Buffer Lane
719	Clutch Prox Sensor Off during Routine I/O Check
725	Photocell above Lane was Blocked on Activated Lane
781	Infeed Motion Control (Galil) Thread Running too Long (Timeout) - Critical Fault

Error Code	Description
782	Sequence Conveyor Motion Control (Galil) Thread Running too Long (Timeout) - Critical Fault
783	Buffer Motion Control (Galil) Thread Running too Long (Timeout) - Critical Fault
791	Motion Control (Galil) Thread Crashed while Main Thread was Waiting - Critical Fault
900	Warning: Back Edge of Case was Adjusted after the Case Stored
901	Warning: Back Edge of Case was Adjusted Outward after Case Dispense (Back Edge not Seen)
902	Warning: Back Edge of Case was Adjusted Outward after Case Dispensed
903	Warning: Back Edge of Case was Adjusted Inward after Case Dispensed
904	Warning: Pulled Buffer Lane Cases Inward after Case Store (Photocell was Blocked)
905	Warning: Pulled Buffer Lane Cases Inward after Case Dispense (Photocell was Blocked)

Quick Reference Tables

Level Status Colors

Color	Status	Meaning	Action Available
Green	Automatic	System operating normally	Can put in Manual
Yellow	Manual	System paused	Can put in Auto
Red	Fault	Active alert	Must resolve alert
Dark Red	E-Stop	Safety system engaged	Clear E-Stop condition
Blue	Manual	Communication Loss	Re-establish communications

Alert Severity Levels

Severity	Description	Typical Response
Warning	Minor issue	Monitor, may not stop production
Error	Standard Alert	Production stopped on that level, buzzer sounds, requires attention.
Critical	Major issue	Production stopped on all levels, buzzer sounds, requires immediate attention

Common Alert Categories

Category	Alert Codes	Description
System	99, 400-401, 570, 572-573, 791	System-level faults
Safety	555	E-Stop and safety interlocks
Storage/Dispense	109, 113, 115, 117, 119, 121, 123	Lane operations
Pop-Up	152-156, 525	Pop-up mechanism
Clutch	310, 333-334, 340	Clutch assembly
Infeed	512, 515, 517	Inbound case issues
Photocell/Sensor	709, 719	Sensor monitoring
Outfeed	(no code)	Output positioning

See Alert Reference for complete list with detailed explanations.

User Roles and Permissions

Role	Access Level	Capabilities
Admin	Full	All features, settings, user management
Manager	Operational	Operations, not site config
Clerk	Production	Monitor and operate, limited config
Guest	View-only	Browse screens, no changes

See User Manager for details.

Common Tasks Quick Reference

Task	Where to Go
Check system status	Home Page
View current alerts	Alerts Page
Check replenishment status	Inventory > Replenishments
View lane contents	Inventory > Lane Inventory
Track specific case	Case Sequence Page
Change system settings	Administration
Resolve an alert	Troubleshooting > Alert Reference
Replace a case	Troubleshooting > Case Replacement

InnoPick Terminology

Term	Definition
Accumulation	Buffer space between InnoPick and merge or palletizer
Case Sequence	Order in which cases are scheduled to exit InnoPick
Clutch	Mechanical assembly that engages lane with pop-up
Dispense	Moving case from storage lane to sequence conveyor
Infeed	Where cases enter InnoPick from upstream conveyor system
Inventory Graph	Visual display of case positions on sequence conveyors
Lane	Storage conveyor that holds buffer inventory
Level	Complete operational unit with sequence conveyor and storage lanes
Outfeed	Where cases exit InnoPick to downstream equipment
Pop-Up	Assembly that lifts cases from the sequence conveyor, used for both storing and dispensing
Replenishment	Quantity of product sent to replenish InnoPick
Sequence Conveyor	Main conveyor that indexes cases position by position
SKU	Stock Keeping Unit (product code)

System Architecture Summary

Case Flow

1. Cases arrive at **Infeed** from depalletizer
2. Cases advance on **Sequence Conveyor**
3. Cases **Store** into **Storage Lanes** via pop-up and clutch
4. Cases remain in lanes until needed
5. Cases **Dispense** back to **Sequence Conveyor**
6. Cases advance to **Outfeed**
7. Cases exit to downstream palletizer

Related Topics

- Getting Started - For new users
- Daily Operations - Routine procedures
- Main Screens - Detailed feature reference
- Troubleshooting - Problem resolution

Navigation: ← Disabled Lane | Home →